

rolling_ball

May 6, 2026

1 Rolling Ball Navigator

Własne środowisko Gymnasium zaimplementowane na potrzeby Projektu 4 z przedmiotu Inteligencja Obliczeniowa.

Kulka tocząca się po planszy musi dotrzeć do wyznaczonego celu, omijając statyczne przeszkody. Agent uczy się nawigacji wyłącznie na podstawie lokalnych obserwacji: prędkości, kierunku do celu i sensorów odległości (raycasting) — bez znajomości swojej pozycji absolutnej.

1.1 1. Środowisko RollingBallEnv

Środowisko spełnia standardowy interfejs Gymnasium (`reset`, `step`, `render`, `close`). Przestrzeń obserwacji jest ciągła (wektor 13 liczb `float32`), przestrzeń akcji jest ciągła (wektor 2 liczb `float32`).

Element	Opis
Obserwacje	[<code>vel_x</code> , <code>vel_y</code> , <code>goal_dx</code> , <code>goal_dy</code> , <code>goal_dist</code> , <code>ray_0</code> ... <code>ray_7</code>]
Akcje	[<code>fx</code> , <code>fy</code>] — siła przykładana do kulki w osi X i Y
Koniec epizodu	Dotarcie do celu (<code>terminated</code>) lub przekroczenie limitu kroków (<code>truncated</code>)

```
[ ]: import math
import numpy as np
import gymnasium as gym
from gymnasium import spaces
import pygame

WORLD_W      = 600.0
WORLD_H      = 600.0
BALL_RADIUS  = 12.0
GOAL_RADIUS  = 18.0
FRICTION     = 0.92
FORCE_SCALE  = 0.4
V_MAX        = 8.0
```

```

MAX_STEPS      = 500
RAY_MAX        = 500.0
DIAGONAL       = math.sqrt(WORLD_W ** 2 + WORLD_H ** 2)
RAY_ANGLES     = [0, 45, 90, 135, 180, 225, 270, 315]

# Static obstacle layout: (left_x, top_y, width, height)
OBSTACLES = [
    ( 80, 80, 80, 200),
    (280, 40, 80, 160),
    (440, 100, 100, 60),
    (140, 340, 160, 60),
    (350, 280, 60, 200),
    (460, 380, 90, 130),
]

class RollingBallEnv(gym.Env):

    metadata = {"render_modes": ["human", "rgb_array"], "render_fps": 30}

    def __init__(self, render_mode=None):
        super().__init__()
        assert render_mode is None or render_mode in self.
        ↪ metadata["render_modes"]
        self.render_mode = render_mode
        self._window = None
        self._clock = None

        low = np.array([-1., -1., -1., -1., 0.] + [0.] * 8, dtype=np.float32)
        high = np.array([ 1., 1., 1., 1., 1.] + [1.] * 8, dtype=np.float32)
        self.observation_space = spaces.Box(low, high, dtype=np.float32)
        self.action_space = spaces.Box(
            np.array([-1.0, -1.0], dtype=np.float32),
            np.array([ 1.0, 1.0], dtype=np.float32),
            dtype=np.float32,
        )

        self._pos = np.zeros(2, dtype=np.float32)
        self._vel = np.zeros(2, dtype=np.float32)
        self._goal = np.zeros(2, dtype=np.float32)
        self._step_count = 0

    def reset(self, seed=None, options=None):
        super().reset(seed=seed)

        while True:
            p = np.array([

```

```

        self.np_random.uniform(BALL_RADIUS, WORLD_W - BALL_RADIUS),
        self.np_random.uniform(BALL_RADIUS, WORLD_H - BALL_RADIUS),
    ], dtype=np.float32)
    if self._is_free(p, BALL_RADIUS):
        self._pos = p
        break

while True:
    g = np.array([
        self.np_random.uniform(GOAL_RADIUS, WORLD_W - GOAL_RADIUS),
        self.np_random.uniform(GOAL_RADIUS, WORLD_H - GOAL_RADIUS),
    ], dtype=np.float32)
    if self._is_free(g, GOAL_RADIUS) and np.linalg.norm(g - self._pos) <
↪ 100.0:
        self._goal = g
        break

self._vel = np.zeros(2, dtype=np.float32)
self._step_count = 0

obs = self._get_obs()
info = self._get_info()
if self.render_mode == "human":
    self._render_frame()
return obs, info

def step(self, action):
    action = np.clip(action, -1.0, 1.0).astype(np.float32)

    prev_dist = float(np.linalg.norm(self._pos - self._goal))

    self._vel = np.clip(self._vel + action * FORCE_SCALE, -V_MAX, V_MAX)
    self._vel *= FRICTION
    self._pos += self._vel

    self._resolve_walls()
    for obs_rect in OBSTACLES:
        self._resolve_obstacle(obs_rect)

    self._step_count += 1

    dist = float(np.linalg.norm(self._pos - self._goal))
    terminated = dist <= GOAL_RADIUS + BALL_RADIUS
    truncated = self._step_count >= MAX_STEPS

    reward = -0.01 + 0.02 * (prev_dist - dist)
    if terminated:

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        reward += 50.0

    obs = self._get_obs()
    info = self._get_info()
    if self.render_mode == "human":
        self._render_frame()
    return obs, float(reward), terminated, truncated, info

def render(self):
    if self.render_mode == "rgb_array":
        return self._render_frame()

def close(self):
    if self._window is not None:
        pygame.display.quit()
        pygame.quit()
        self._window = None
        self._clock = None

def _is_free(self, pos: np.ndarray, radius: float) -> bool:
    if (
        pos[0] - radius < 0 or pos[0] + radius > WORLD_W
        or pos[1] - radius < 0 or pos[1] + radius > WORLD_H
    ):
        return False
    for ox, oy, ow, oh in OBSTACLES:
        cx = float(np.clip(pos[0], ox, ox + ow))
        cy = float(np.clip(pos[1], oy, oy + oh))
        if math.sqrt((pos[0] - cx) ** 2 + (pos[1] - cy) ** 2) < radius + 2.
↪0:
            return False
    return True

def _resolve_walls(self) -> None:
    if self._pos[0] - BALL_RADIUS < 0:
        self._pos[0] = BALL_RADIUS
        self._vel[0] = abs(self._vel[0])
    elif self._pos[0] + BALL_RADIUS > WORLD_W:
        self._pos[0] = WORLD_W - BALL_RADIUS
        self._vel[0] = -abs(self._vel[0])
    if self._pos[1] - BALL_RADIUS < 0:
        self._pos[1] = BALL_RADIUS
        self._vel[1] = abs(self._vel[1])
    elif self._pos[1] + BALL_RADIUS > WORLD_H:
        self._pos[1] = WORLD_H - BALL_RADIUS
        self._vel[1] = -abs(self._vel[1])

```

```

def _resolve_obstacle(self, rect) -> None:
    ox, oy, ow, oh = rect
    cx = float(np.clip(self._pos[0], ox, ox + ow))
    cy = float(np.clip(self._pos[1], oy, oy + oh))
    dx = self._pos[0] - cx
    dy = self._pos[1] - cy
    dist_sq = dx * dx + dy * dy

    if dist_sq >= BALL_RADIUS * BALL_RADIUS:
        return

    dist = math.sqrt(dist_sq)
    nx, ny = (dx / dist, dy / dist) if dist > 1e-6 else (1.0, 0.0)

    overlap = BALL_RADIUS - dist
    self._pos[0] += nx * (overlap + 0.5)
    self._pos[1] += ny * (overlap + 0.5)

    dot = self._vel[0] * nx + self._vel[1] * ny
    if dot < 0:
        self._vel[0] -= 2.0 * dot * nx
        self._vel[1] -= 2.0 * dot * ny

def _cast_ray(self, angle_deg: float) -> float:
    rad = math.radians(angle_deg)
    dx = math.cos(rad)
    dy = -math.sin(rad)
    ox, oy = float(self._pos[0]), float(self._pos[1])

    t = RAY_MAX
    if dx > 1e-9:
        t = min(t, (WORLD_W - ox) / dx)
    elif dx < -1e-9:
        t = min(t, -ox / dx)
    if dy > 1e-9:
        t = min(t, (WORLD_H - oy) / dy)
    elif dy < -1e-9:
        t = min(t, -oy / dy)

    for obs_rect in OBSTACLES:
        t_obs = self._ray_aabb(ox, oy, dx, dy, obs_rect)
        if t_obs < t:
            t = t_obs

    return float(np.clip(t, 0.0, RAY_MAX))

```

@staticmethod

```

def _ray_aabb(ox: float, oy: float, dx: float, dy: float, rect) -> float:
    rx, ry, rw, rh = rect
    t_near, t_far = -1e30, 1e30

    if abs(dx) > 1e-9:
        tx1 = (rx - ox) / dx
        tx2 = (rx + rw - ox) / dx
        t_near = max(t_near, min(tx1, tx2))
        t_far = min(t_far, max(tx1, tx2))
    elif ox < rx or ox > rx + rw:
        return RAY_MAX

    if abs(dy) > 1e-9:
        ty1 = (ry - oy) / dy
        ty2 = (ry + rh - oy) / dy
        t_near = max(t_near, min(ty1, ty2))
        t_far = min(t_far, max(ty1, ty2))
    elif oy < ry or oy > ry + rh:
        return RAY_MAX

    if t_far < t_near or t_near <= 0.0:
        return RAY_MAX
    return float(np.clip(t_near, 0.0, RAY_MAX))

def _get_obs(self) -> np.ndarray:
    ddx = self._goal[0] - self._pos[0]
    ddy = self._goal[1] - self._pos[1]
    dist = math.sqrt(ddx * ddx + ddy * ddy)
    if dist > 1e-9:
        ndx = ddx / dist
        ndy = -ddy / dist
    else:
        ndx = ndy = 0.0

    rays = [self._cast_ray(a) for a in RAY_ANGLES]
    return np.array([
        self._vel[0] / V_MAX,
        self._vel[1] / V_MAX,
        ndx,
        ndy,
        dist / DIAGONAL,
        *[r / RAY_MAX for r in rays],
    ], dtype=np.float32)

def _get_info(self) -> dict:
    return {
        "distance_to_goal": float(np.linalg.norm(self._pos - self._goal)),
    }

```

```

        "step": self._step_count,
    }

    def _render_frame(self):
        if not pygame.get_init():
            pygame.init()
        if self.render_mode == "human":
            if self._window is None:
                pygame.display.init()
                self._window = pygame.display.set_mode((int(WORLD_W),
↪int(WORLD_H)))
                pygame.display.set_caption("Rolling Ball Navigator")
            if self._clock is None:
                self._clock = pygame.time.Clock()

        canvas = pygame.Surface((int(WORLD_W), int(WORLD_H)))
        TILE = 40
        for ty in range(0, int(WORLD_H), TILE):
            for tx in range(0, int(WORLD_W), TILE):
                shade = (42, 33, 24) if (tx // TILE + ty // TILE) % 2 == 0 else
↪(50, 40, 29)
                pygame.draw.rect(canvas, shade, (tx, ty, TILE, TILE))
                pygame.draw.rect(canvas, (28, 20, 14), (tx, ty, TILE, TILE), 1)

        pygame.draw.rect(canvas, (110, 88, 64), (0, 0, int(WORLD_W),
↪int(WORLD_H)), 6)
        pygame.draw.rect(canvas, (70, 54, 38), (3, 3, int(WORLD_W) - 6,
↪int(WORLD_H) - 6), 2)

        for ox, oy, ow, oh in OBSTACLES:
            r = pygame.Rect(int(ox), int(oy), int(ow), int(oh))
            pygame.draw.rect(canvas, (78, 64, 48), r)
            pygame.draw.rect(canvas, (108, 90, 66), r, 2)
            pygame.draw.line(canvas, (48, 38, 28), (r.right - 1, r.top), (r.
↪right - 1, r.bottom), 2)
            pygame.draw.line(canvas, (48, 38, 28), (r.left, r.bottom - 1), (r.
↪right, r.bottom - 1), 2)

        obs_vec = self._get_obs()
        for i, angle_deg in enumerate(RAY_ANGLES):
            rad = math.radians(angle_deg)
            rdx = math.cos(rad)
            rdy = -math.sin(rad)
            ray_len = float(obs_vec[5 + i]) * RAY_MAX
            ex = int(self._pos[0] + rdx * ray_len)

```

```

        ey = int(self._pos[1] + rdy * ray_len)
        pygame.draw.line(
            canvas, (210, 158, 38),
            (int(self._pos[0]), int(self._pos[1])), (ex, ey), 1,
        )

    if not hasattr(self, "_goal_img"):
        def _load_sprite(path, size):
            raw = pygame.image.load(path)
            surf = pygame.Surface(raw.get_size(), pygame.SRCALPHA)
            surf.blit(raw, (0, 0))
            return pygame.transform.scale(surf, size)
        self._goal_img = _load_sprite("sprites/goal.png",
        ↪(int(GOAL_RADIUS * 2), int(GOAL_RADIUS * 2)))
        self._agent_img = _load_sprite("sprites/agent.png",
        ↪(int(BALL_RADIUS * 2), int(BALL_RADIUS * 2)))

        canvas.blit(self._goal_img,
            (int(self._goal[0] - GOAL_RADIUS), int(self._goal[1] -
        ↪GOAL_RADIUS)))
        canvas.blit(self._agent_img,
            (int(self._pos[0] - BALL_RADIUS), int(self._pos[1] - BALL_RADIUS)))

    if self.render_mode == "human":
        self._window.blit(canvas, canvas.get_rect())
        pygame.event.pump()
        pygame.display.update()
        self._clock.tick(self.metadata["render_fps"])
    else:
        return np.transpose(
            np.array(pygame.surfarray.pixels3d(canvas)), axes=(1, 0, 2)
        )

```

1.2 2. Trening agenta (SAC)

Do treningu używamy algorytmu **Soft Actor-Critic (SAC)** z biblioteki `stable-baselines3`. SAC jest dedykowany do ciągłych przestrzeni akcji — maksymalizuje zarówno oczekiwaną nagrodę, jak i entropię polityki, co przekłada się na stabilność uczenia i dobre eksplorowanie środowiska.

Środowisko treningowe nie używa trybu graficznego (`render_mode=None`), co zapewnia maksymalną szybkość. Postęp ewaluacji jest zapisywany do pliku `logs/evaluations.npz` i wizualizowany po zakończeniu treningu.

```

[ ]: import os
from stable_baselines3 import SAC
from stable_baselines3.common.monitor import Monitor
from stable_baselines3.common.callbacks import EvalCallback

```

```

os.makedirs("models", exist_ok=True)
os.makedirs("logs", exist_ok=True)

train_env = RollingBallEnv()
eval_env = Monitor(RollingBallEnv())

model = SAC(
    "MlpPolicy",
    train_env,
    learning_rate=3e-4,
    buffer_size=300_000,
    learning_starts=5_000,
    batch_size=256,
    tau=0.005,
    gamma=0.99,
    ent_coef=0.1,
    verbose=1,
    seed=42,
)

eval_callback = EvalCallback(
    eval_env,
    best_model_save_path="models/",
    log_path="logs/",
    eval_freq=5_000,
    n_eval_episodes=10,
    deterministic=True,
    render=False,
)

model.learn(total_timesteps=500_000, callback=eval_callback)
model.save("models/rolling_ball_final")
train_env.close()
eval_env.close()
print("Trening zakończony. Model zapisany do models/rolling_ball_final.zip")

```

Using cuda device

Wrapping the env with a `Monitor` wrapper

Wrapping the env in a DummyVecEnv.

```

-----
| rollout/          |          |
|   ep_len_mean    | 500      |
|   ep_rew_mean    | -4.58    |
| time/            |          |
|   episodes       | 4        |
|   fps            | 4864     |

```

	time_elapsed		0	
	total_timesteps		2000	

	rollout/			
	ep_len_mean		500	
	ep_rew_mean		-4.59	
	time/			
	episodes		8	
	fps		4847	
	time_elapsed		0	
	total_timesteps		4000	

Eval num_timesteps=5000, episode_reward=1.37 +/- 15.83
 Episode length: 486.60 +/- 40.20

	eval/			
	mean_ep_length		487	
	mean_reward		1.37	
	time/			
	total_timesteps		5000	

New best mean reward!

	rollout/			
	ep_len_mean		500	
	ep_rew_mean		-4.54	
	time/			
	episodes		12	
	fps		373	
	time_elapsed		16	
	total_timesteps		6000	
	train/			
	actor_loss		-0.819	
	critic_loss		0.000282	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		999	

	rollout/			
	ep_len_mean		497	
	ep_rew_mean		-0.569	
	time/			
	episodes		16	
	fps		245	
	time_elapsed		32	
	total_timesteps		7954	

train/		
actor_loss	-1.91	
critic_loss	0.00041	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	2953	

rollout/		
ep_len_mean	479	
ep_rew_mean	7.03	
time/		
episodes	20	
fps	203	
time_elapsed	47	
total_timesteps	9574	
train/		
actor_loss	-2.96	
critic_loss	0.0177	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	4573	

Eval num_timesteps=10000, episode_reward=41.66 +/- 21.24
 Episode length: 261.80 +/- 160.73

eval/		
mean_ep_length	262	
mean_reward	41.7	
time/		
total_timesteps	10000	
train/		
actor_loss	-3.14	
critic_loss	0.0278	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	4999	

New best mean reward!

rollout/		
ep_len_mean	469	
ep_rew_mean	10.1	
time/		
episodes	24	
fps	174	
time_elapsed	64	
total_timesteps	11266	

train/		
actor_loss	-4.21	
critic_loss	0.0687	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	6265	

rollout/		
ep_len_mean	454	
ep_rew_mean	12.6	
time/		
episodes	28	
fps	162	
time_elapsed	78	
total_timesteps	12714	

train/		
actor_loss	-5.56	
critic_loss	0.514	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	7713	

rollout/		
ep_len_mean	426	
ep_rew_mean	15.9	
time/		
episodes	32	
fps	156	
time_elapsed	87	
total_timesteps	13631	

train/		
actor_loss	-6.59	
critic_loss	4.74	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	8630	

rollout/		
ep_len_mean	416	
ep_rew_mean	16.9	
time/		
episodes	36	
fps	152	
time_elapsed	98	
total_timesteps	14977	

train/		
actor_loss	-6.74	
critic_loss	0.274	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	9976	

Eval num_timesteps=15000, episode_reward=30.74 +/- 27.12
 Episode length: 308.50 +/- 194.41

eval/		
mean_ep_length	308	
mean_reward	30.7	
time/		
total_timesteps	15000	
train/		
actor_loss	-7.43	
critic_loss	0.544	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	9999	

rollout/		
ep_len_mean	399	
ep_rew_mean	19.2	
time/		
episodes	40	
fps	143	
time_elapsed	110	
total_timesteps	15961	
train/		
actor_loss	-8.35	
critic_loss	0.373	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	10960	

rollout/		
ep_len_mean	392	
ep_rew_mean	21	
time/		
episodes	44	
fps	141	
time_elapsed	121	
total_timesteps	17268	
train/		

	actor_loss		-9.46	
	critic_loss		0.19	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		12267	

	rollout/			
	ep_len_mean		384	
	ep_rew_mean		21.4	
	time/			
	episodes		48	
	fps		138	
	time_elapsed		132	
	total_timesteps		18414	
	train/			
	actor_loss		-10.3	
	critic_loss		0.241	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		13413	

Eval num_timesteps=20000, episode_reward=13.64 +/- 25.43
 Episode length: 398.60 +/- 165.49

	eval/			
	mean_ep_length		399	
	mean_reward		13.6	
	time/			
	total_timesteps		20000	
	train/			
	actor_loss		-9.76	
	critic_loss		0.135	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		14999	

	rollout/			
	ep_len_mean		385	
	ep_rew_mean		20.7	
	time/			
	episodes		52	
	fps		132	
	time_elapsed		151	
	total_timesteps		20028	
	train/			
	actor_loss		-10.5	

	critic_loss		0.175	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		15027	

	rollout/			
	ep_len_mean		369	
	ep_rew_mean		23	
	time/			
	episodes		56	
	fps		131	
	time_elapsed		157	
	total_timesteps		20671	
	train/			
	actor_loss		-11.7	
	critic_loss		0.261	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		15670	

	rollout/			
	ep_len_mean		368	
	ep_rew_mean		23.1	
	time/			
	episodes		60	
	fps		129	
	time_elapsed		170	
	total_timesteps		22078	
	train/			
	actor_loss		-10.6	
	critic_loss		0.101	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		17077	

	rollout/			
	ep_len_mean		366	
	ep_rew_mean		23.3	
	time/			
	episodes		64	
	fps		127	
	time_elapsed		183	
	total_timesteps		23396	
	train/			
	actor_loss		-12.6	

	critic_loss		0.0837	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		18395	

	rollout/			
	ep_len_mean		354	
	ep_rew_mean		25	
	time/			
	episodes		68	
	fps		128	
	time_elapsed		188	
	total_timesteps		24106	
	train/			
	actor_loss		-12.2	
	critic_loss		0.0786	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		19105	

Eval num_timesteps=25000, episode_reward=36.50 +/- 25.08
Episode length: 289.60 +/- 152.59

	eval/			
	mean_ep_length		290	
	mean_reward		36.5	
	time/			
	total_timesteps		25000	
	train/			
	actor_loss		-12.5	
	critic_loss		0.0722	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		19999	

	rollout/			
	ep_len_mean		352	
	ep_rew_mean		24.9	
	time/			
	episodes		72	
	fps		124	
	time_elapsed		202	
	total_timesteps		25312	
	train/			
	actor_loss		-12.1	
	critic_loss		0.0984	

	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		20311	

	rollout/			
	ep_len_mean		353	
	ep_rew_mean		25.8	
	time/			
	episodes		76	
	fps		124	
	time_elapsed		214	
	total_timesteps		26806	
	train/			
	actor_loss		-12	
	critic_loss		0.104	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		21805	

	rollout/			
	ep_len_mean		360	
	ep_rew_mean		24.4	
	time/			
	episodes		80	
	fps		123	
	time_elapsed		232	
	total_timesteps		28806	
	train/			
	actor_loss		-13.2	
	critic_loss		0.108	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		23805	

Eval num_timesteps=30000, episode_reward=9.75 +/- 22.82
 Episode length: 428.10 +/- 143.95

	eval/			
	mean_ep_length		428	
	mean_reward		9.75	
	time/			
	total_timesteps		30000	
	train/			
	actor_loss		-13.8	
	critic_loss		0.0683	
	ent_coef		0.1	

learning_rate	0.0003
n_updates	24999

rollout/	
ep_len_mean	362
ep_rew_mean	23.8
time/	
episodes	84
fps	121
time_elapsed	250
total_timesteps	30414
train/	
actor_loss	-13.4
critic_loss	0.0588
ent_coef	0.1
learning_rate	0.0003
n_updates	25413

rollout/	
ep_len_mean	359
ep_rew_mean	23.9
time/	
episodes	88
fps	121
time_elapsed	261
total_timesteps	31631
train/	
actor_loss	-14.5
critic_loss	0.169
ent_coef	0.1
learning_rate	0.0003
n_updates	26630

rollout/	
ep_len_mean	357
ep_rew_mean	24
time/	
episodes	92
fps	120
time_elapsed	272
total_timesteps	32845
train/	
actor_loss	-14.5
critic_loss	0.0547
ent_coef	0.1

	learning_rate	0.0003	
	n_updates	27844	

	rollout/		
	ep_len_mean	359	
	ep_rew_mean	24.2	
	time/		
	episodes	96	
	fps	120	
	time_elapsed	285	
	total_timesteps	34490	
	train/		
	actor_loss	-15.6	
	critic_loss	0.0789	
	ent_coef	0.1	
	learning_rate	0.0003	
	n_updates	29489	

Eval num_timesteps=35000, episode_reward=26.49 +/- 28.07
Episode length: 326.90 +/- 191.82

	eval/		
	mean_ep_length	327	
	mean_reward	26.5	
	time/		
	total_timesteps	35000	
	train/		
	actor_loss	-16.9	
	critic_loss	0.216	
	ent_coef	0.1	
	learning_rate	0.0003	
	n_updates	29999	

	rollout/		
	ep_len_mean	356	
	ep_rew_mean	24.2	
	time/		
	episodes	100	
	fps	118	
	time_elapsed	299	
	total_timesteps	35596	
	train/		
	actor_loss	-15	
	critic_loss	0.219	
	ent_coef	0.1	
	learning_rate	0.0003	

	n_updates		30595	

	rollout/			
	ep_len_mean		347	
	ep_rew_mean		25.5	
	time/			
	episodes		104	
	fps		119	
	time_elapsed		308	
	total_timesteps		36700	
	train/			
	actor_loss		-14.7	
	critic_loss		0.181	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		31699	

	rollout/			
	ep_len_mean		337	
	ep_rew_mean		27.2	
	time/			
	episodes		108	
	fps		118	
	time_elapsed		316	
	total_timesteps		37672	
	train/			
	actor_loss		-15.1	
	critic_loss		0.31	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		32671	

	rollout/			
	ep_len_mean		333	
	ep_rew_mean		27.9	
	time/			
	episodes		112	
	fps		118	
	time_elapsed		332	
	total_timesteps		39289	
	train/			
	actor_loss		-15.9	
	critic_loss		0.0961	
	ent_coef		0.1	
	learning_rate		0.0003	

	n_updates		34288	
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	rollout/			
	ep_len_mean		320	
	ep_rew_mean		29.6	
	time/			
	episodes		116	
	fps		118	
	time_elapsed		336	
	total_timesteps		39911	
	train/			
	actor_loss		-15.8	
	critic_loss		0.179	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		34910	

Eval num_timesteps=40000, episode_reward=31.38 +/- 28.88
Episode length: 340.90 +/- 176.16

	eval/			
	mean_ep_length		341	
	mean_reward		31.4	
	time/			
	total_timesteps		40000	
	train/			
	actor_loss		-16.6	
	critic_loss		0.108	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		34999	

	rollout/			
	ep_len_mean		311	
	ep_rew_mean		29.7	
	time/			
	episodes		120	
	fps		117	
	time_elapsed		346	
	total_timesteps		40711	
	train/			
	actor_loss		-17.9	
	critic_loss		0.131	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		35710	

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| rollout/          |          |
|   ep_len_mean    | 310      |
|   ep_rew_mean    | 29.1     |
| time/            |          |
|   episodes       | 124      |
|   fps            | 117      |
|   time_elapsed   | 361      |
|   total_timesteps | 42281    |
| train/           |          |
|   actor_loss     | -15.7    |
|   critic_loss    | 0.147    |
|   ent_coef       | 0.1      |
|   learning_rate  | 0.0003   |
|   n_updates      | 37280    |
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| rollout/          |          |
|   ep_len_mean    | 312      |
|   ep_rew_mean    | 29.1     |
| time/            |          |
|   episodes       | 128      |
|   fps            | 117      |
|   time_elapsed   | 373      |
|   total_timesteps | 43874    |
| train/           |          |
|   actor_loss     | -14.8    |
|   critic_loss    | 0.123    |
|   ent_coef       | 0.1      |
|   learning_rate  | 0.0003   |
|   n_updates      | 38873    |
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| rollout/          |          |
|   ep_len_mean    | 311      |
|   ep_rew_mean    | 29.3     |
| time/            |          |
|   episodes       | 132      |
|   fps            | 117      |
|   time_elapsed   | 381      |
|   total_timesteps | 44772    |
| train/           |          |
|   actor_loss     | -15.7    |
|   critic_loss    | 0.138    |
|   ent_coef       | 0.1      |
|   learning_rate  | 0.0003   |
|   n_updates      | 39771    |
-----

```

 Eval num_timesteps=45000, episode_reward=38.79 +/- 25.67
 Episode length: 257.50 +/- 175.67

eval/		
mean_ep_length	258	
mean_reward	38.8	
time/		
total_timesteps	45000	
train/		
actor_loss	-16.7	
critic_loss	0.0524	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	39999	

rollout/		
ep_len_mean	312	
ep_rew_mean	29.3	
time/		
episodes	136	
fps	117	
time_elapsed	394	
total_timesteps	46165	
train/		
actor_loss	-15.7	
critic_loss	0.0697	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	41164	

rollout/		
ep_len_mean	314	
ep_rew_mean	29.3	
time/		
episodes	140	
fps	116	
time_elapsed	405	
total_timesteps	47363	
train/		
actor_loss	-15.7	
critic_loss	0.118	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	42362	

rollout/		
ep_len_mean	313	
ep_rew_mean	28.8	
time/		
episodes	144	
fps	116	
time_elapsed	417	
total_timesteps	48601	
train/		
actor_loss	-15.7	
critic_loss	0.209	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	43600	

rollout/		
ep_len_mean	310	
ep_rew_mean	29.4	
time/		
episodes	148	
fps	116	
time_elapsed	425	
total_timesteps	49445	
train/		
actor_loss	-15.2	
critic_loss	0.236	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	44444	

Eval num_timesteps=50000, episode_reward=36.91 +/- 25.63
Episode length: 228.70 +/- 185.66

eval/		
mean_ep_length	229	
mean_reward	36.9	
time/		
total_timesteps	50000	
train/		
actor_loss	-15.8	
critic_loss	0.0851	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	44999	

rollout/		
ep_len_mean	310	
ep_rew_mean	29.9	
time/		
episodes	152	
fps	115	
time_elapsed	440	
total_timesteps	50997	
train/		
actor_loss	-15.9	
critic_loss	0.0743	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	45996	

rollout/		
ep_len_mean	312	
ep_rew_mean	29.9	
time/		
episodes	156	
fps	115	
time_elapsed	448	
total_timesteps	51839	
train/		
actor_loss	-16.3	
critic_loss	0.0988	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	46838	

rollout/		
ep_len_mean	300	
ep_rew_mean	31	
time/		
episodes	160	
fps	115	
time_elapsed	450	
total_timesteps	52070	
train/		
actor_loss	-15.7	
critic_loss	0.0322	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	47069	

rollout/		
ep_len_mean	293	
ep_rew_mean	32.2	
time/		
episodes	164	
fps	115	
time_elapsed	454	
total_timesteps	52702	
train/		
actor_loss	-15.6	
critic_loss	0.045	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	47701	

rollout/		
ep_len_mean	295	
ep_rew_mean	31.7	
time/		
episodes	168	
fps	115	
time_elapsed	462	
total_timesteps	53569	
train/		
actor_loss	-16.9	
critic_loss	0.197	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	48568	

Eval num_timesteps=55000, episode_reward=19.83 +/- 27.63
Episode length: 365.40 +/- 193.71

eval/		
mean_ep_length	365	
mean_reward	19.8	
time/		
total_timesteps	55000	
train/		
actor_loss	-16.3	
critic_loss	0.166	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	49999	

rollout/		
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	ep_len_mean		299	
	ep_rew_mean		31.3	
	time/			
	episodes		172	
	fps		114	
	time_elapsed		482	
	total_timesteps		55251	
	train/			
	actor_loss		-16.2	
	critic_loss		0.151	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		50250	

	rollout/			
	ep_len_mean		296	
	ep_rew_mean		30.6	
	time/			
	episodes		176	
	fps		114	
	time_elapsed		490	
	total_timesteps		56393	
	train/			
	actor_loss		-15	
	critic_loss		0.328	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		51392	

	rollout/			
	ep_len_mean		290	
	ep_rew_mean		31.8	
	time/			
	episodes		180	
	fps		114	
	time_elapsed		503	
	total_timesteps		57815	
	train/			
	actor_loss		-15	
	critic_loss		0.109	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		52814	

	rollout/			
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	ep_len_mean		279	
	ep_rew_mean		33.5	
	time/			
	episodes		184	
	fps		114	
	time_elapsed		507	
	total_timesteps		58306	
	train/			
	actor_loss		-16.7	
	critic_loss		0.139	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		53305	

	rollout/			
	ep_len_mean		279	
	ep_rew_mean		34	
	time/			
	episodes		188	
	fps		115	
	time_elapsed		516	
	total_timesteps		59526	
	train/			
	actor_loss		-17	
	critic_loss		0.22	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		54525	

Eval num_timesteps=60000, episode_reward=26.51 +/- 27.53
Episode length: 335.80 +/- 189.81

	eval/			
	mean_ep_length		336	
	mean_reward		26.5	
	time/			
	total_timesteps		60000	
	train/			
	actor_loss		-15.6	
	critic_loss		0.0404	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		54999	

	rollout/			
	ep_len_mean		275	

	ep_rew_mean		34.5	
	time/			
	episodes		192	
	fps		114	
	time_elapsed		526	
	total_timesteps		60365	
	train/			
	actor_loss		-17	
	critic_loss		0.152	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		55364	

	rollout/			
	ep_len_mean		266	
	ep_rew_mean		35.6	
	time/			
	episodes		196	
	fps		114	
	time_elapsed		532	
	total_timesteps		61042	
	train/			
	actor_loss		-16.3	
	critic_loss		0.0841	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		56041	

	rollout/			
	ep_len_mean		262	
	ep_rew_mean		36.2	
	time/			
	episodes		200	
	fps		114	
	time_elapsed		539	
	total_timesteps		61815	
	train/			
	actor_loss		-16.5	
	critic_loss		0.156	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		56814	

	rollout/			
	ep_len_mean		259	

	ep_rew_mean		36.7	
	time/			
	episodes		204	
	fps		114	
	time_elapsed		545	
	total_timesteps		62567	
	train/			
	actor_loss		-16.3	
	critic_loss		0.171	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		57566	

	rollout/			
	ep_len_mean		253	
	ep_rew_mean		37.3	
	time/			
	episodes		208	
	fps		115	
	time_elapsed		547	
	total_timesteps		63019	
	train/			
	actor_loss		-16.4	
	critic_loss		0.162	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		58018	

	rollout/			
	ep_len_mean		246	
	ep_rew_mean		38.4	
	time/			
	episodes		212	
	fps		114	
	time_elapsed		555	
	total_timesteps		63908	
	train/			
	actor_loss		-16.3	
	critic_loss		0.13	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		58907	

	rollout/			
	ep_len_mean		246	

	ep_rew_mean	38.4	
	time/		
	episodes	216	
	fps	114	
	time_elapsed	561	
	total_timesteps	64485	
	train/		
	actor_loss	-16.2	
	critic_loss	0.147	
	ent_coef	0.1	
	learning_rate	0.0003	
	n_updates	59484	

Eval num_timesteps=65000, episode_reward=53.69 +/- 1.85
 Episode length: 180.60 +/- 124.89

	eval/		
	mean_ep_length	181	
	mean_reward	53.7	
	time/		
	total_timesteps	65000	
	train/		
	actor_loss	-16.8	
	critic_loss	0.0798	
	ent_coef	0.1	
	learning_rate	0.0003	
	n_updates	59999	

New best mean reward!

	rollout/		
	ep_len_mean	245	
	ep_rew_mean	39	
	time/		
	episodes	220	
	fps	114	
	time_elapsed	570	
	total_timesteps	65218	
	train/		
	actor_loss	-16.7	
	critic_loss	0.125	
	ent_coef	0.1	
	learning_rate	0.0003	
	n_updates	60217	

	rollout/		
	ep_len_mean	243	

	ep_rew_mean		39.5	
	time/			
	episodes		224	
	fps		114	
	time_elapsed		581	
	total_timesteps		66608	
	train/			
	actor_loss		-16.8	
	critic_loss		0.0756	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		61607	

	rollout/			
	ep_len_mean		236	
	ep_rew_mean		40.5	
	time/			
	episodes		228	
	fps		114	
	time_elapsed		588	
	total_timesteps		67498	
	train/			
	actor_loss		-15.6	
	critic_loss		0.12	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		62497	

	rollout/			
	ep_len_mean		236	
	ep_rew_mean		41	
	time/			
	episodes		232	
	fps		114	
	time_elapsed		596	
	total_timesteps		68339	
	train/			
	actor_loss		-15.9	
	critic_loss		0.116	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		63338	

	rollout/			
	ep_len_mean		227	

	ep_rew_mean		42.1	
	time/			
	episodes		236	
	fps		114	
	time_elapsed		600	
	total_timesteps		68872	
	train/			
	actor_loss		-18.2	
	critic_loss		0.11	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		63871	

	rollout/			
	ep_len_mean		221	
	ep_rew_mean		42.6	
	time/			
	episodes		240	
	fps		114	
	time_elapsed		606	
	total_timesteps		69471	
	train/			
	actor_loss		-17.3	
	critic_loss		0.199	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		64470	

	rollout/			
	ep_len_mean		214	
	ep_rew_mean		43.8	
	time/			
	episodes		244	
	fps		114	
	time_elapsed		608	
	total_timesteps		69979	
	train/			
	actor_loss		-16.3	
	critic_loss		0.0705	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		64978	

Eval num_timesteps=70000, episode_reward=56.39 +/- 1.10
Episode length: 229.90 +/- 123.31

eval/		
mean_ep_length	230	
mean_reward	56.4	
time/		
total_timesteps	70000	
train/		
actor_loss	-16.1	
critic_loss	0.12	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	64999	

New best mean reward!

rollout/		
ep_len_mean	218	
ep_rew_mean	43.8	
time/		
episodes	248	
fps	114	
time_elapsed	621	
total_timesteps	71232	
train/		
actor_loss	-15.4	
critic_loss	0.0981	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	66231	

rollout/		
ep_len_mean	216	
ep_rew_mean	44.3	
time/		
episodes	252	
fps	114	
time_elapsed	634	
total_timesteps	72580	
train/		
actor_loss	-16.1	
critic_loss	0.147	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	67579	

rollout/		
ep_len_mean	214	

	ep_rew_mean		44.4	
	time/			
	episodes		256	
	fps		114	
	time_elapsed		639	
	total_timesteps		73245	
	train/			
	actor_loss		-16.6	
	critic_loss		0.0785	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		68244	

	rollout/			
	ep_len_mean		219	
	ep_rew_mean		44.4	
	time/			
	episodes		260	
	fps		114	
	time_elapsed		645	
	total_timesteps		73927	
	train/			
	actor_loss		-15.8	
	critic_loss		0.115	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		68926	

	rollout/			
	ep_len_mean		221	
	ep_rew_mean		43.9	
	time/			
	episodes		264	
	fps		114	
	time_elapsed		654	
	total_timesteps		74794	
	train/			
	actor_loss		-16.9	
	critic_loss		0.128	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		69793	

Eval num_timesteps=75000, episode_reward=47.97 +/- 17.67
Episode length: 173.50 +/- 139.05

eval/		
mean_ep_length	174	
mean_reward	48	
time/		
total_timesteps	75000	
train/		
actor_loss	-16.2	
critic_loss	0.0755	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	69999	

rollout/		
ep_len_mean	224	
ep_rew_mean	43.9	
time/		
episodes	268	
fps	114	
time_elapsed	666	
total_timesteps	75988	
train/		
actor_loss	-16.3	
critic_loss	0.0661	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	70987	

rollout/		
ep_len_mean	216	
ep_rew_mean	45	
time/		
episodes	272	
fps	114	
time_elapsed	672	
total_timesteps	76811	
train/		
actor_loss	-17.6	
critic_loss	0.0656	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	71810	

rollout/		
ep_len_mean	212	
ep_rew_mean	46.1	

time/		
episodes	276	
fps	114	
time_elapsed	679	
total_timesteps	77544	
train/		
actor_loss	-17.4	
critic_loss	0.0559	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	72543	

rollout/		
ep_len_mean	200	
ep_rew_mean	47.1	
time/		
episodes	280	
fps	114	
time_elapsed	682	
total_timesteps	77786	
train/		
actor_loss	-16.8	
critic_loss	0.0607	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	72785	

rollout/		
ep_len_mean	205	
ep_rew_mean	46.6	
time/		
episodes	284	
fps	113	
time_elapsed	692	
total_timesteps	78779	
train/		
actor_loss	-18.2	
critic_loss	0.125	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	73778	

rollout/		
ep_len_mean	201	
ep_rew_mean	46.5	

time/		
episodes	288	
fps	114	
time_elapsed	698	
total_timesteps	79657	
train/		
actor_loss	-17.9	
critic_loss	0.0379	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	74656	

Eval num_timesteps=80000, episode_reward=37.02 +/- 27.02
Episode length: 281.60 +/- 168.40

eval/		
mean_ep_length	282	
mean_reward	37	
time/		
total_timesteps	80000	
train/		
actor_loss	-18.5	
critic_loss	0.0948	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	74999	

rollout/		
ep_len_mean	197	
ep_rew_mean	47.1	
time/		
episodes	292	
fps	113	
time_elapsed	705	
total_timesteps	80054	
train/		
actor_loss	-16.2	
critic_loss	0.0724	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	75053	

rollout/		
ep_len_mean	201	
ep_rew_mean	46.6	
time/		

	episodes		296	
	fps		113	
	time_elapsed		716	
	total_timesteps		81180	
	train/			
	actor_loss		-18	
	critic_loss		0.156	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		76179	

	rollout/			
	ep_len_mean		202	
	ep_rew_mean		47.1	
	time/			
	episodes		300	
	fps		113	
	time_elapsed		724	
	total_timesteps		81977	
	train/			
	actor_loss		-17.5	
	critic_loss		0.0995	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		76976	

	rollout/			
	ep_len_mean		205	
	ep_rew_mean		47.2	
	time/			
	episodes		304	
	fps		113	
	time_elapsed		732	
	total_timesteps		83036	
	train/			
	actor_loss		-17.7	
	critic_loss		0.187	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		78035	

	rollout/			
	ep_len_mean		205	
	ep_rew_mean		47.2	
	time/			

	episodes		308	
	fps		113	
	time_elapsed		736	
	total_timesteps		83478	
	train/			
	actor_loss		-17.9	
	critic_loss		0.108	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		78477	

	rollout/			
	ep_len_mean		208	
	ep_rew_mean		46.7	
	time/			
	episodes		312	
	fps		113	
	time_elapsed		748	
	total_timesteps		84745	
	train/			
	actor_loss		-16.8	
	critic_loss		0.174	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		79744	

Eval num_timesteps=85000, episode_reward=48.62 +/- 14.55
Episode length: 161.20 +/- 135.79

	eval/			
	mean_ep_length		161	
	mean_reward		48.6	
	time/			
	total_timesteps		85000	
	train/			
	actor_loss		-17.7	
	critic_loss		0.186	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		79999	

	rollout/			
	ep_len_mean		214	
	ep_rew_mean		46.2	
	time/			
	episodes		316	

	fps		113	
	time_elapsed		759	
	total_timesteps		85930	
	train/			
	actor_loss		-17	
	critic_loss		0.195	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		80929	

	rollout/			
	ep_len_mean		216	
	ep_rew_mean		45.6	
	time/			
	episodes		320	
	fps		113	
	time_elapsed		767	
	total_timesteps		86825	
	train/			
	actor_loss		-17.3	
	critic_loss		0.166	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		81824	

	rollout/			
	ep_len_mean		212	
	ep_rew_mean		46.8	
	time/			
	episodes		324	
	fps		113	
	time_elapsed		776	
	total_timesteps		87813	
	train/			
	actor_loss		-17.8	
	critic_loss		0.0494	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		82812	

	rollout/			
	ep_len_mean		211	
	ep_rew_mean		46.3	
	time/			
	episodes		328	

	fps		113	
	time_elapsed		784	
	total_timesteps		88636	
	train/			
	actor_loss		-17.8	
	critic_loss		0.0714	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		83635	

	rollout/			
	ep_len_mean		216	
	ep_rew_mean		45.7	
	time/			
	episodes		332	
	fps		113	
	time_elapsed		794	
	total_timesteps		89928	
	train/			
	actor_loss		-17.7	
	critic_loss		0.0468	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		84927	

Eval num_timesteps=90000, episode_reward=42.51 +/- 23.79
Episode length: 211.70 +/- 157.64

	eval/			
	mean_ep_length		212	
	mean_reward		42.5	
	time/			
	total_timesteps		90000	
	train/			
	actor_loss		-17.3	
	critic_loss		0.0719	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		84999	

	rollout/			
	ep_len_mean		215	
	ep_rew_mean		45.7	
	time/			
	episodes		336	
	fps		112	

	time_elapsed		800	
	total_timesteps		90382	
	train/			
	actor_loss		-17	
	critic_loss		0.0377	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		85381	

	rollout/			
	ep_len_mean		216	
	ep_rew_mean		45.7	
	time/			
	episodes		340	
	fps		112	
	time_elapsed		806	
	total_timesteps		91041	
	train/			
	actor_loss		-18.3	
	critic_loss		0.0872	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		86040	

	rollout/			
	ep_len_mean		222	
	ep_rew_mean		45.2	
	time/			
	episodes		344	
	fps		112	
	time_elapsed		817	
	total_timesteps		92142	
	train/			
	actor_loss		-17	
	critic_loss		0.0584	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		87141	

	rollout/			
	ep_len_mean		219	
	ep_rew_mean		45.2	
	time/			
	episodes		348	
	fps		112	

	time_elapsed		825	
	total_timesteps		93166	
	train/			
	actor_loss		-18.5	
	critic_loss		0.115	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		88165	

	rollout/			
	ep_len_mean		215	
	ep_rew_mean		45.2	
	time/			
	episodes		352	
	fps		112	
	time_elapsed		834	
	total_timesteps		94121	
	train/			
	actor_loss		-18	
	critic_loss		0.0771	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		89120	

Eval num_timesteps=95000, episode_reward=48.31 +/- 17.78
Episode length: 155.80 +/- 129.26

	eval/			
	mean_ep_length		156	
	mean_reward		48.3	
	time/			
	total_timesteps		95000	
	train/			
	actor_loss		-17.9	
	critic_loss		0.0993	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		89999	

	rollout/			
	ep_len_mean		221	
	ep_rew_mean		44.7	
	time/			
	episodes		356	
	fps		112	
	time_elapsed		848	

	total_timesteps		95377	
	train/			
	actor_loss		-16.5	
	critic_loss		0.0805	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		90376	

	rollout/			
	ep_len_mean		219	
	ep_rew_mean		44.7	
	time/			
	episodes		360	
	fps		112	
	time_elapsed		850	
	total_timesteps		95844	
	train/			
	actor_loss		-18.1	
	critic_loss		0.115	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		90843	

	rollout/			
	ep_len_mean		219	
	ep_rew_mean		44.7	
	time/			
	episodes		364	
	fps		112	
	time_elapsed		858	
	total_timesteps		96712	
	train/			
	actor_loss		-20	
	critic_loss		0.161	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		91711	

	rollout/			
	ep_len_mean		216	
	ep_rew_mean		44.7	
	time/			
	episodes		368	
	fps		112	
	time_elapsed		867	

	total_timesteps		97605	
	train/			
	actor_loss		-16.1	
	critic_loss		0.0736	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		92604	

	rollout/			
	ep_len_mean		216	
	ep_rew_mean		44.7	
	time/			
	episodes		372	
	fps		112	
	time_elapsed		875	
	total_timesteps		98433	
	train/			
	actor_loss		-17.9	
	critic_loss		0.156	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		93432	

	rollout/			
	ep_len_mean		217	
	ep_rew_mean		44.1	
	time/			
	episodes		376	
	fps		112	
	time_elapsed		880	
	total_timesteps		99215	
	train/			
	actor_loss		-18.8	
	critic_loss		0.198	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		94214	

Eval num_timesteps=100000, episode_reward=42.26 +/- 23.15
Episode length: 208.90 +/- 168.25

	eval/			
	mean_ep_length		209	
	mean_reward		42.3	
	time/			
	total_timesteps		100000	

train/		
actor_loss	-17.8	
critic_loss	0.0577	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	94999	

rollout/		
ep_len_mean	227	
ep_rew_mean	43.7	
time/		
episodes	380	
fps	112	
time_elapsed	893	
total_timesteps	100505	
train/		
actor_loss	-18.2	
critic_loss	0.0867	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	95504	

rollout/		
ep_len_mean	227	
ep_rew_mean	43.7	
time/		
episodes	384	
fps	112	
time_elapsed	902	
total_timesteps	101437	
train/		
actor_loss	-16.4	
critic_loss	0.129	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	96436	

rollout/		
ep_len_mean	224	
ep_rew_mean	43.7	
time/		
episodes	388	
fps	112	
time_elapsed	909	
total_timesteps	102102	

train/		
actor_loss	-18.2	
critic_loss	0.105	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	97101	

rollout/		
ep_len_mean	232	
ep_rew_mean	43.2	
time/		
episodes	392	
fps	112	
time_elapsed	918	
total_timesteps	103212	
train/		
actor_loss	-17.2	
critic_loss	0.0731	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	98211	

rollout/		
ep_len_mean	229	
ep_rew_mean	43.2	
time/		
episodes	396	
fps	112	
time_elapsed	926	
total_timesteps	104036	
train/		
actor_loss	-17.9	
critic_loss	0.101	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	99035	

rollout/		
ep_len_mean	228	
ep_rew_mean	43.2	
time/		
episodes	400	
fps	112	
time_elapsed	933	
total_timesteps	104758	

train/		
actor_loss	-17.5	
critic_loss	0.0594	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	99757	

Eval num_timesteps=105000, episode_reward=49.09 +/- 16.68
 Episode length: 149.50 +/- 138.27

eval/		
mean_ep_length	150	
mean_reward	49.1	
time/		
total_timesteps	105000	
train/		
actor_loss	-18.3	
critic_loss	0.129	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	99999	

rollout/		
ep_len_mean	222	
ep_rew_mean	43.6	
time/		
episodes	404	
fps	112	
time_elapsed	939	
total_timesteps	105201	
train/		
actor_loss	-16.6	
critic_loss	0.0702	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	100200	

rollout/		
ep_len_mean	222	
ep_rew_mean	43.7	
time/		
episodes	408	
fps	112	
time_elapsed	941	
total_timesteps	105704	
train/		

	actor_loss		-17.8	
	critic_loss		0.0649	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		100703	

	rollout/			
	ep_len_mean		216	
	ep_rew_mean		44.7	
	time/			
	episodes		412	
	fps		112	
	time_elapsed		947	
	total_timesteps		106361	
	train/			
	actor_loss		-17.6	
	critic_loss		0.128	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		101360	

	rollout/			
	ep_len_mean		214	
	ep_rew_mean		44.8	
	time/			
	episodes		416	
	fps		112	
	time_elapsed		956	
	total_timesteps		107319	
	train/			
	actor_loss		-17.8	
	critic_loss		0.0461	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		102318	

	rollout/			
	ep_len_mean		209	
	ep_rew_mean		45.4	
	time/			
	episodes		420	
	fps		112	
	time_elapsed		960	
	total_timesteps		107739	
	train/			

	actor_loss		-17.8	
	critic_loss		0.188	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		102738	

	rollout/			
	ep_len_mean		207	
	ep_rew_mean		44.8	
	time/			
	episodes		424	
	fps		112	
	time_elapsed		967	
	total_timesteps		108496	
	train/			
	actor_loss		-17.7	
	critic_loss		0.152	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		103495	

	rollout/			
	ep_len_mean		205	
	ep_rew_mean		44.6	
	time/			
	episodes		428	
	fps		112	
	time_elapsed		972	
	total_timesteps		109181	
	train/			
	actor_loss		-16.6	
	critic_loss		0.0618	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		104180	

	rollout/			
	ep_len_mean		200	
	ep_rew_mean		45.2	
	time/			
	episodes		432	
	fps		112	
	time_elapsed		979	
	total_timesteps		109884	
	train/			

	actor_loss		-18.1	
	critic_loss		0.0629	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		104883	

Eval num_timesteps=110000, episode_reward=49.06 +/- 17.20
Episode length: 163.20 +/- 132.09

	eval/			
	mean_ep_length		163	
	mean_reward		49.1	
	time/			
	total_timesteps		110000	
	train/			
	actor_loss		-18.5	
	critic_loss		0.164	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		104999	

	rollout/			
	ep_len_mean		200	
	ep_rew_mean		45.2	
	time/			
	episodes		436	
	fps		112	
	time_elapsed		985	
	total_timesteps		110354	
	train/			
	actor_loss		-17.4	
	critic_loss		0.0954	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		105353	

	rollout/			
	ep_len_mean		198	
	ep_rew_mean		45.2	
	time/			
	episodes		440	
	fps		112	
	time_elapsed		989	
	total_timesteps		110851	
	train/			
	actor_loss		-16.9	

	critic_loss		0.122	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		105850	

	rollout/			
	ep_len_mean		198	
	ep_rew_mean		45.2	
	time/			
	episodes		444	
	fps		112	
	time_elapsed		998	
	total_timesteps		111921	
	train/			
	actor_loss		-18.4	
	critic_loss		0.0638	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		106920	

	rollout/			
	ep_len_mean		198	
	ep_rew_mean		45.1	
	time/			
	episodes		448	
	fps		112	
	time_elapsed		1005	
	total_timesteps		112945	
	train/			
	actor_loss		-17.8	
	critic_loss		0.152	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		107944	

	rollout/			
	ep_len_mean		198	
	ep_rew_mean		45.8	
	time/			
	episodes		452	
	fps		112	
	time_elapsed		1014	
	total_timesteps		113907	
	train/			
	actor_loss		-19.2	

	critic_loss		0.0737	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		108906	

	rollout/			
	ep_len_mean		189	
	ep_rew_mean		46.2	
	time/			
	episodes		456	
	fps		112	
	time_elapsed		1018	
	total_timesteps		114286	
	train/			
	actor_loss		-17.7	
	critic_loss		0.11	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		109285	

Eval num_timesteps=115000, episode_reward=54.95 +/- 1.81
 Episode length: 186.70 +/- 103.47

	eval/			
	mean_ep_length		187	
	mean_reward		54.9	
	time/			
	total_timesteps		115000	
	train/			
	actor_loss		-17.7	
	critic_loss		0.153	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		109999	

	rollout/			
	ep_len_mean		192	
	ep_rew_mean		45.7	
	time/			
	episodes		460	
	fps		111	
	time_elapsed		1027	
	total_timesteps		115085	
	train/			
	actor_loss		-17.4	
	critic_loss		0.0779	

	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		110084	

	rollout/			
	ep_len_mean		193	
	ep_rew_mean		46.2	
	time/			
	episodes		464	
	fps		112	
	time_elapsed		1033	
	total_timesteps		115970	
	train/			
	actor_loss		-19	
	critic_loss		0.135	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		110969	

	rollout/			
	ep_len_mean		191	
	ep_rew_mean		46.8	
	time/			
	episodes		468	
	fps		112	
	time_elapsed		1040	
	total_timesteps		116713	
	train/			
	actor_loss		-18.1	
	critic_loss		0.119	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		111712	

	rollout/			
	ep_len_mean		190	
	ep_rew_mean		47.4	
	time/			
	episodes		472	
	fps		112	
	time_elapsed		1046	
	total_timesteps		117400	
	train/			
	actor_loss		-18	
	critic_loss		0.263	

	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		112399	

	rollout/			
	ep_len_mean		190	
	ep_rew_mean		48	
	time/			
	episodes		476	
	fps		112	
	time_elapsed		1053	
	total_timesteps		118216	
	train/			
	actor_loss		-17.1	
	critic_loss		0.102	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		113215	

	rollout/			
	ep_len_mean		185	
	ep_rew_mean		48.5	
	time/			
	episodes		480	
	fps		112	
	time_elapsed		1060	
	total_timesteps		119037	
	train/			
	actor_loss		-18.6	
	critic_loss		0.134	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		114036	

Eval num_timesteps=120000, episode_reward=54.87 +/- 1.93
 Episode length: 117.70 +/- 49.64

	eval/			
	mean_ep_length		118	
	mean_reward		54.9	
	time/			
	total_timesteps		120000	
	train/			
	actor_loss		-18.7	
	critic_loss		0.0738	
	ent_coef		0.1	

	learning_rate		0.0003	
	n_updates		114999	

	rollout/			
	ep_len_mean		187	
	ep_rew_mean		49.2	
	time/			
	episodes		484	
	fps		112	
	time_elapsed		1069	
	total_timesteps		120109	
	train/			
	actor_loss		-17.4	
	critic_loss		0.127	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		115108	

	rollout/			
	ep_len_mean		191	
	ep_rew_mean		49.4	
	time/			
	episodes		488	
	fps		112	
	time_elapsed		1079	
	total_timesteps		121230	
	train/			
	actor_loss		-16.9	
	critic_loss		0.157	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		116229	

	rollout/			
	ep_len_mean		193	
	ep_rew_mean		49.9	
	time/			
	episodes		492	
	fps		112	
	time_elapsed		1091	
	total_timesteps		122528	
	train/			
	actor_loss		-17.6	
	critic_loss		0.1	
	ent_coef		0.1	

learning_rate	0.0003
n_updates	117527

rollout/	
ep_len_mean	196
ep_rew_mean	49.9
time/	
episodes	496
fps	112
time_elapsed	1099
total_timesteps	123590
train/	
actor_loss	-18.5
critic_loss	0.121
ent_coef	0.1
learning_rate	0.0003
n_updates	118589

rollout/	
ep_len_mean	192
ep_rew_mean	49.9
time/	
episodes	500
fps	112
time_elapsed	1102
total_timesteps	123939
train/	
actor_loss	-19.1
critic_loss	0.145
ent_coef	0.1
learning_rate	0.0003
n_updates	118938

rollout/	
ep_len_mean	194
ep_rew_mean	49.9
time/	
episodes	504
fps	112
time_elapsed	1107
total_timesteps	124618
train/	
actor_loss	-18.2
critic_loss	0.317
ent_coef	0.1

learning_rate	0.0003
n_updates	119617

Eval num_timesteps=125000, episode_reward=54.73 +/- 1.54
 Episode length: 119.60 +/- 63.61

eval/	
mean_ep_length	120
mean_reward	54.7
time/	
total_timesteps	125000
train/	
actor_loss	-18.3
critic_loss	0.0497
ent_coef	0.1
learning_rate	0.0003
n_updates	119999

rollout/	
ep_len_mean	198
ep_rew_mean	49.4
time/	
episodes	508
fps	112
time_elapsed	1116
total_timesteps	125474
train/	
actor_loss	-18.3
critic_loss	0.482
ent_coef	0.1
learning_rate	0.0003
n_updates	120473

rollout/	
ep_len_mean	198
ep_rew_mean	49.4
time/	
episodes	512
fps	112
time_elapsed	1122
total_timesteps	126201
train/	
actor_loss	-18.5
critic_loss	0.0516
ent_coef	0.1
learning_rate	0.0003

	n_updates		121200	

	rollout/			
	ep_len_mean		193	
	ep_rew_mean		49.8	
	time/			
	episodes		516	
	fps		112	
	time_elapsed		1124	
	total_timesteps		126633	
	train/			
	actor_loss		-17.1	
	critic_loss		0.0625	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		121632	

	rollout/			
	ep_len_mean		197	
	ep_rew_mean		49.2	
	time/			
	episodes		520	
	fps		112	
	time_elapsed		1131	
	total_timesteps		127403	
	train/			
	actor_loss		-17.6	
	critic_loss		0.0934	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		122402	

	rollout/			
	ep_len_mean		195	
	ep_rew_mean		49.8	
	time/			
	episodes		524	
	fps		112	
	time_elapsed		1137	
	total_timesteps		127983	
	train/			
	actor_loss		-17.4	
	critic_loss		0.0932	
	ent_coef		0.1	
	learning_rate		0.0003	

	n_updates		122982	
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	rollout/			
	ep_len_mean		198	
	ep_rew_mean		49.9	
	time/			
	episodes		528	
	fps		112	
	time_elapsed		1145	
	total_timesteps		128963	
	train/			
	actor_loss		-18.6	
	critic_loss		0.0751	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		123962	

	rollout/			
	ep_len_mean		201	
	ep_rew_mean		49.3	
	time/			
	episodes		532	
	fps		112	
	time_elapsed		1153	
	total_timesteps		129956	
	train/			
	actor_loss		-18.2	
	critic_loss		0.142	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		124955	

Eval num_timesteps=130000, episode_reward=49.81 +/- 15.74

Episode length: 210.40 +/- 149.45

	eval/			
	mean_ep_length		210	
	mean_reward		49.8	
	time/			
	total_timesteps		130000	
	train/			
	actor_loss		-19.2	
	critic_loss		0.0772	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		124999	

```

-----
| rollout/          |          |
|   ep_len_mean    | 203      |
|   ep_rew_mean    | 49.4     |
| time/            |          |
|   episodes       | 536      |
|   fps            | 112      |
|   time_elapsed   | 1161     |
|   total_timesteps | 130644   |
| train/           |          |
|   actor_loss     | -18       |
|   critic_loss    | 0.0445   |
|   ent_coef       | 0.1       |
|   learning_rate  | 0.0003   |
|   n_updates      | 125643   |
-----

```

```

-----
| rollout/          |          |
|   ep_len_mean    | 204      |
|   ep_rew_mean    | 49.4     |
| time/            |          |
|   episodes       | 540      |
|   fps            | 112      |
|   time_elapsed   | 1167     |
|   total_timesteps | 131291   |
| train/           |          |
|   actor_loss     | -17.7     |
|   critic_loss    | 0.0598   |
|   ent_coef       | 0.1       |
|   learning_rate  | 0.0003   |
|   n_updates      | 126290   |
-----

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```

-----
| rollout/          |          |
|   ep_len_mean    | 197      |
|   ep_rew_mean    | 49.9     |
| time/            |          |
|   episodes       | 544      |
|   fps            | 112      |
|   time_elapsed   | 1170     |
|   total_timesteps | 131620   |
| train/           |          |
|   actor_loss     | -18.4     |
|   critic_loss    | 0.119     |
|   ent_coef       | 0.1       |
|   learning_rate  | 0.0003   |
|   n_updates      | 126619   |
-----

```

rollout/		
ep_len_mean	191	
ep_rew_mean	50.5	
time/		
episodes	548	
fps	112	
time_elapsed	1174	
total_timesteps	132068	
train/		
actor_loss	-18.5	
critic_loss	0.104	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	127067	

rollout/		
ep_len_mean	190	
ep_rew_mean	49.9	
time/		
episodes	552	
fps	112	
time_elapsed	1182	
total_timesteps	132953	
train/		
actor_loss	-18.7	
critic_loss	0.145	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	127952	

rollout/		
ep_len_mean	195	
ep_rew_mean	50	
time/		
episodes	556	
fps	112	
time_elapsed	1188	
total_timesteps	133814	
train/		
actor_loss	-18	
critic_loss	0.0736	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	128813	

rollout/		
ep_len_mean	194	
ep_rew_mean	50.5	
time/		
episodes	560	
fps	112	
time_elapsed	1194	
total_timesteps	134447	
train/		
actor_loss	-17.9	
critic_loss	0.145	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	129446	

rollout/		
ep_len_mean	190	
ep_rew_mean	50.6	
time/		
episodes	564	
fps	112	
time_elapsed	1199	
total_timesteps	134986	
train/		
actor_loss	-17.6	
critic_loss	0.0497	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	129985	

Eval num_timesteps=135000, episode_reward=43.26 +/- 22.94
Episode length: 260.10 +/- 177.69

eval/		
mean_ep_length	260	
mean_reward	43.3	
time/		
total_timesteps	135000	
train/		
actor_loss	-17.9	
critic_loss	0.0754	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	129999	

rollout/		
ep_len_mean	187	
ep_rew_mean	50.6	
time/		
episodes	568	
fps	112	
time_elapsed	1206	
total_timesteps	135399	
train/		
actor_loss	-18.7	
critic_loss	0.108	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	130398	

rollout/		
ep_len_mean	188	
ep_rew_mean	50.6	
time/		
episodes	572	
fps	112	
time_elapsed	1212	
total_timesteps	136181	
train/		
actor_loss	-18.4	
critic_loss	0.0457	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	131180	

rollout/		
ep_len_mean	191	
ep_rew_mean	50.6	
time/		
episodes	576	
fps	112	
time_elapsed	1223	
total_timesteps	137267	
train/		
actor_loss	-17.3	
critic_loss	0.0519	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	132266	

rollout/		
ep_len_mean	189	
ep_rew_mean	50.5	
time/		
episodes	580	
fps	112	
time_elapsed	1229	
total_timesteps	137972	
train/		
actor_loss	-18.7	
critic_loss	0.0867	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	132971	

rollout/		
ep_len_mean	183	
ep_rew_mean	50.3	
time/		
episodes	584	
fps	112	
time_elapsed	1233	
total_timesteps	138377	
train/		
actor_loss	-17.6	
critic_loss	0.0816	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	133376	

rollout/		
ep_len_mean	180	
ep_rew_mean	50.8	
time/		
episodes	588	
fps	112	
time_elapsed	1241	
total_timesteps	139189	
train/		
actor_loss	-18.5	
critic_loss	0.17	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	134188	

```

-----
| rollout/          |          |
|   ep_len_mean    | 172      |
|   ep_rew_mean    | 50.9     |
| time/            |          |
|   episodes       | 592      |
|   fps            | 112      |
|   time_elapsed   | 1245     |
|   total_timesteps | 139773   |
| train/           |          |
|   actor_loss     | -17.5     |
|   critic_loss    | 0.0575    |
|   ent_coef       | 0.1       |
|   learning_rate  | 0.0003    |
|   n_updates      | 134772    |
-----

```

Eval num_timesteps=140000, episode_reward=37.50 +/- 26.19
 Episode length: 225.10 +/- 184.59

```

-----
| eval/            |          |
|   mean_ep_length | 225      |
|   mean_reward    | 37.5     |
| time/            |          |
|   total_timesteps | 140000   |
| train/           |          |
|   actor_loss     | -18.6     |
|   critic_loss    | 0.144     |
|   ent_coef       | 0.1       |
|   learning_rate  | 0.0003    |
|   n_updates      | 134999    |
-----

```

```

-----
| rollout/          |          |
|   ep_len_mean    | 165      |
|   ep_rew_mean    | 51.3     |
| time/            |          |
|   episodes       | 596      |
|   fps            | 112      |
|   time_elapsed   | 1250     |
|   total_timesteps | 140093   |
| train/           |          |
|   actor_loss     | -18.6     |
|   critic_loss    | 0.0483    |
|   ent_coef       | 0.1       |
|   learning_rate  | 0.0003    |
|   n_updates      | 135092    |
-----

```

rollout/		
ep_len_mean	167	
ep_rew_mean	51.4	
time/		
episodes	600	
fps	111	
time_elapsed	1255	
total_timesteps	140639	
train/		
actor_loss	-17.8	
critic_loss	0.0534	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	135638	

rollout/		
ep_len_mean	168	
ep_rew_mean	50.8	
time/		
episodes	604	
fps	111	
time_elapsed	1264	
total_timesteps	141468	
train/		
actor_loss	-18.9	
critic_loss	0.145	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	136467	

rollout/		
ep_len_mean	165	
ep_rew_mean	51.3	
time/		
episodes	608	
fps	111	
time_elapsed	1269	
total_timesteps	142014	
train/		
actor_loss	-18.7	
critic_loss	0.0727	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	137013	

rollout/		
ep_len_mean	161	
ep_rew_mean	51.3	
time/		
episodes	612	
fps	111	
time_elapsed	1272	
total_timesteps	142321	
train/		
actor_loss	-16.8	
critic_loss	0.0493	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	137320	

rollout/		
ep_len_mean	164	
ep_rew_mean	51.3	
time/		
episodes	616	
fps	111	
time_elapsed	1276	
total_timesteps	142984	
train/		
actor_loss	-18.5	
critic_loss	0.0742	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	137983	

rollout/		
ep_len_mean	162	
ep_rew_mean	51.9	
time/		
episodes	620	
fps	111	
time_elapsed	1282	
total_timesteps	143590	
train/		
actor_loss	-18.5	
critic_loss	0.109	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	138589	

rollout/		
ep_len_mean	160	
ep_rew_mean	51.9	
time/		
episodes	624	
fps	111	
time_elapsed	1286	
total_timesteps	144005	
train/		
actor_loss	-19.1	
critic_loss	0.0696	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	139004	

rollout/		
ep_len_mean	159	
ep_rew_mean	52.4	
time/		
episodes	628	
fps	111	
time_elapsed	1295	
total_timesteps	144844	
train/		
actor_loss	-19.4	
critic_loss	0.0391	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	139843	

Eval num_timesteps=145000, episode_reward=42.57 +/- 23.11
Episode length: 186.70 +/- 165.92

eval/		
mean_ep_length	187	
mean_reward	42.6	
time/		
total_timesteps	145000	
train/		
actor_loss	-17.7	
critic_loss	0.0628	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	139999	

rollout/		
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	ep_len_mean		153	
	ep_rew_mean		52.9	
	time/			
	episodes		632	
	fps		111	
	time_elapsed		1301	
	total_timesteps		145273	
	train/			
	actor_loss		-19.6	
	critic_loss		0.0747	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		140272	

	rollout/			
	ep_len_mean		150	
	ep_rew_mean		52.9	
	time/			
	episodes		636	
	fps		111	
	time_elapsed		1303	
	total_timesteps		145689	
	train/			
	actor_loss		-19.2	
	critic_loss		0.118	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		140688	

	rollout/			
	ep_len_mean		148	
	ep_rew_mean		52.9	
	time/			
	episodes		640	
	fps		111	
	time_elapsed		1306	
	total_timesteps		146083	
	train/			
	actor_loss		-19.2	
	critic_loss		0.0726	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		141082	

	rollout/			
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	ep_len_mean		153	
	ep_rew_mean		53	
	time/			
	episodes		644	
	fps		111	
	time_elapsed		1315	
	total_timesteps		146946	
	train/			
	actor_loss		-18	
	critic_loss		0.248	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		141945	

	rollout/			
	ep_len_mean		159	
	ep_rew_mean		53.1	
	time/			
	episodes		648	
	fps		111	
	time_elapsed		1324	
	total_timesteps		147976	
	train/			
	actor_loss		-18.7	
	critic_loss		0.068	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		142975	

	rollout/			
	ep_len_mean		159	
	ep_rew_mean		53.7	
	time/			
	episodes		652	
	fps		111	
	time_elapsed		1332	
	total_timesteps		148811	
	train/			
	actor_loss		-19.9	
	critic_loss		0.122	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		143810	

	rollout/			
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	ep_len_mean		161	
	ep_rew_mean		53.7	
	time/			
	episodes		656	
	fps		111	
	time_elapsed		1341	
	total_timesteps		149884	
	train/			
	actor_loss		-18.7	
	critic_loss		0.163	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		144883	

Eval num_timesteps=150000, episode_reward=49.58 +/- 16.86
Episode length: 172.20 +/- 118.89

	eval/			
	mean_ep_length		172	
	mean_reward		49.6	
	time/			
	total_timesteps		150000	
	train/			
	actor_loss		-19.1	
	critic_loss		0.116	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		144999	

	rollout/			
	ep_len_mean		162	
	ep_rew_mean		53.7	
	time/			
	episodes		660	
	fps		111	
	time_elapsed		1350	
	total_timesteps		150639	
	train/			
	actor_loss		-19.2	
	critic_loss		0.0637	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		145638	

	rollout/			
	ep_len_mean		162	

	ep_rew_mean		53.7	
	time/			
	episodes		664	
	fps		111	
	time_elapsed		1356	
	total_timesteps		151160	
	train/			
	actor_loss		-20.2	
	critic_loss		0.0793	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		146159	

	rollout/			
	ep_len_mean		167	
	ep_rew_mean		53.1	
	time/			
	episodes		668	
	fps		111	
	time_elapsed		1363	
	total_timesteps		152089	
	train/			
	actor_loss		-18	
	critic_loss		0.16	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		147088	

	rollout/			
	ep_len_mean		163	
	ep_rew_mean		53.1	
	time/			
	episodes		672	
	fps		111	
	time_elapsed		1367	
	total_timesteps		152495	
	train/			
	actor_loss		-18.2	
	critic_loss		0.133	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		147494	

	rollout/			
	ep_len_mean		163	

	ep_rew_mean	52.6	
	time/		
	episodes	676	
	fps	111	
	time_elapsed	1377	
	total_timesteps	153526	
	train/		
	actor_loss	-19.4	
	critic_loss	0.0614	
	ent_coef	0.1	
	learning_rate	0.0003	
	n_updates	148525	

	rollout/		
	ep_len_mean	164	
	ep_rew_mean	52.6	
	time/		
	episodes	680	
	fps	111	
	time_elapsed	1386	
	total_timesteps	154347	
	train/		
	actor_loss	-18.3	
	critic_loss	0.122	
	ent_coef	0.1	
	learning_rate	0.0003	
	n_updates	149346	

Eval num_timesteps=155000, episode_reward=54.29 +/- 1.64
 Episode length: 128.00 +/- 91.94

	eval/		
	mean_ep_length	128	
	mean_reward	54.3	
	time/		
	total_timesteps	155000	
	train/		
	actor_loss	-18	
	critic_loss	0.152	
	ent_coef	0.1	
	learning_rate	0.0003	
	n_updates	149999	

	rollout/		
	ep_len_mean	169	
	ep_rew_mean	52.1	

time/		
episodes	684	
fps	111	
time_elapsed	1394	
total_timesteps	155293	
train/		
actor_loss	-18.3	
critic_loss	0.0429	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	150292	

rollout/		
ep_len_mean	170	
ep_rew_mean	51.6	
time/		
episodes	688	
fps	111	
time_elapsed	1402	
total_timesteps	156168	
train/		
actor_loss	-19.2	
critic_loss	0.15	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	151167	

rollout/		
ep_len_mean	175	
ep_rew_mean	50.4	
time/		
episodes	692	
fps	111	
time_elapsed	1413	
total_timesteps	157237	
train/		
actor_loss	-19.1	
critic_loss	0.282	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	152236	

rollout/		
ep_len_mean	182	
ep_rew_mean	49.3	

time/		
episodes	696	
fps	111	
time_elapsed	1424	
total_timesteps	158328	
train/		
actor_loss	-19.6	
critic_loss	0.0449	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	153327	

rollout/		
ep_len_mean	183	
ep_rew_mean	49.3	
time/		
episodes	700	
fps	111	
time_elapsed	1428	
total_timesteps	158896	
train/		
actor_loss	-19.1	
critic_loss	0.0814	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	153895	

rollout/		
ep_len_mean	183	
ep_rew_mean	49.4	
time/		
episodes	704	
fps	111	
time_elapsed	1436	
total_timesteps	159768	
train/		
actor_loss	-19	
critic_loss	0.0888	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	154767	

Eval num_timesteps=160000, episode_reward=54.41 +/- 1.67
 Episode length: 144.80 +/- 99.59

eval/		
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	mean_ep_length		145	
	mean_reward		54.4	
	time/			
	total_timesteps		160000	
	train/			
	actor_loss		-19.7	
	critic_loss		0.0899	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		154999	

	rollout/			
	ep_len_mean		187	
	ep_rew_mean		48.7	
	time/			
	episodes		708	
	fps		111	
	time_elapsed		1446	
	total_timesteps		160752	
	train/			
	actor_loss		-18.1	
	critic_loss		0.0553	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		155751	

	rollout/			
	ep_len_mean		192	
	ep_rew_mean		48.3	
	time/			
	episodes		712	
	fps		111	
	time_elapsed		1453	
	total_timesteps		161541	
	train/			
	actor_loss		-19	
	critic_loss		0.0522	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		156540	

	rollout/			
	ep_len_mean		190	
	ep_rew_mean		48.3	
	time/			

	episodes		716	
	fps		111	
	time_elapsed		1456	
	total_timesteps		162024	
	train/			
	actor_loss		-19.4	
	critic_loss		0.0615	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		157023	

	rollout/			
	ep_len_mean		191	
	ep_rew_mean		48.3	
	time/			
	episodes		720	
	fps		111	
	time_elapsed		1463	
	total_timesteps		162673	
	train/			
	actor_loss		-18.8	
	critic_loss		0.117	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		157672	

	rollout/			
	ep_len_mean		192	
	ep_rew_mean		48.3	
	time/			
	episodes		724	
	fps		111	
	time_elapsed		1468	
	total_timesteps		163175	
	train/			
	actor_loss		-18.5	
	critic_loss		0.0967	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		158174	

	rollout/			
	ep_len_mean		192	
	ep_rew_mean		48.3	
	time/			

	episodes		728	
	fps		111	
	time_elapsed		1476	
	total_timesteps		164051	
	train/			
	actor_loss		-18.4	
	critic_loss		0.0685	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		159050	

	rollout/			
	ep_len_mean		193	
	ep_rew_mean		48.3	
	time/			
	episodes		732	
	fps		111	
	time_elapsed		1481	
	total_timesteps		164586	
	train/			
	actor_loss		-18.7	
	critic_loss		0.0444	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		159585	

	rollout/			
	ep_len_mean		192	
	ep_rew_mean		48.3	
	time/			
	episodes		736	
	fps		111	
	time_elapsed		1484	
	total_timesteps		164929	
	train/			
	actor_loss		-19.9	
	critic_loss		0.0716	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		159928	

Eval num_timesteps=165000, episode_reward=48.00 +/- 17.15
Episode length: 155.00 +/- 136.46

	eval/			
	mean_ep_length		155	

	mean_reward		48	
	time/			
	total_timesteps		165000	
	train/			
	actor_loss		-19.9	
	critic_loss		0.124	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		159999	

	rollout/			
	ep_len_mean		191	
	ep_rew_mean		48.3	
	time/			
	episodes		740	
	fps		111	
	time_elapsed		1486	
	total_timesteps		165185	
	train/			
	actor_loss		-18.9	
	critic_loss		0.125	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		160184	

	rollout/			
	ep_len_mean		189	
	ep_rew_mean		48.2	
	time/			
	episodes		744	
	fps		111	
	time_elapsed		1492	
	total_timesteps		165832	
	train/			
	actor_loss		-18.2	
	critic_loss		0.119	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		160831	

	rollout/			
	ep_len_mean		184	
	ep_rew_mean		48	
	time/			
	episodes		748	

	fps		111	
	time_elapsed		1497	
	total_timesteps		166394	
	train/			
	actor_loss		-18.2	
	critic_loss		0.0465	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		161393	

	rollout/			
	ep_len_mean		184	
	ep_rew_mean		48.1	
	time/			
	episodes		752	
	fps		111	
	time_elapsed		1505	
	total_timesteps		167249	
	train/			
	actor_loss		-18.7	
	critic_loss		0.0457	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		162248	

	rollout/			
	ep_len_mean		178	
	ep_rew_mean		48	
	time/			
	episodes		756	
	fps		111	
	time_elapsed		1510	
	total_timesteps		167696	
	train/			
	actor_loss		-18.3	
	critic_loss		0.049	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		162695	

	rollout/			
	ep_len_mean		174	
	ep_rew_mean		47.9	
	time/			
	episodes		760	

	fps		111	
	time_elapsed		1513	
	total_timesteps		168050	
	train/			
	actor_loss		-18.2	
	critic_loss		0.117	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		163049	

	rollout/			
	ep_len_mean		176	
	ep_rew_mean		47.9	
	time/			
	episodes		764	
	fps		111	
	time_elapsed		1518	
	total_timesteps		168806	
	train/			
	actor_loss		-18	
	critic_loss		0.156	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		163805	

	rollout/			
	ep_len_mean		173	
	ep_rew_mean		48.5	
	time/			
	episodes		768	
	fps		111	
	time_elapsed		1524	
	total_timesteps		169361	
	train/			
	actor_loss		-19.2	
	critic_loss		0.184	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		164360	

Eval num_timesteps=170000, episode_reward=54.36 +/- 2.03
Episode length: 164.20 +/- 155.61

	eval/			
	mean_ep_length		164	
	mean_reward		54.4	

time/		
total_timesteps	170000	
train/		
actor_loss	-19.7	
critic_loss	0.0484	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	164999	

rollout/		
ep_len_mean	178	
ep_rew_mean	48.6	
time/		
episodes	772	
fps	110	
time_elapsed	1535	
total_timesteps	170287	
train/		
actor_loss	-19.3	
critic_loss	0.13	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	165286	

rollout/		
ep_len_mean	172	
ep_rew_mean	49.1	
time/		
episodes	776	
fps	110	
time_elapsed	1540	
total_timesteps	170755	
train/		
actor_loss	-17.6	
critic_loss	0.124	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	165754	

rollout/		
ep_len_mean	171	
ep_rew_mean	49.1	
time/		
episodes	780	
fps	110	

	time_elapsed		1546	
	total_timesteps		171424	
	train/			
	actor_loss		-19.9	
	critic_loss		0.0963	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		166423	

	rollout/			
	ep_len_mean		172	
	ep_rew_mean		49.2	
	time/			
	episodes		784	
	fps		110	
	time_elapsed		1555	
	total_timesteps		172464	
	train/			
	actor_loss		-19.5	
	critic_loss		0.0682	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		167463	

	rollout/			
	ep_len_mean		173	
	ep_rew_mean		49.2	
	time/			
	episodes		788	
	fps		110	
	time_elapsed		1565	
	total_timesteps		173492	
	train/			
	actor_loss		-19.4	
	critic_loss		0.0964	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		168491	

	rollout/			
	ep_len_mean		175	
	ep_rew_mean		50.4	
	time/			
	episodes		792	
	fps		110	

	time_elapsed		1577	
	total_timesteps		174704	
	train/			
	actor_loss		-18.5	
	critic_loss		0.094	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		169703	

Eval num_timesteps=175000, episode_reward=55.66 +/- 2.03

Episode length: 147.10 +/- 70.38

	eval/			
	mean_ep_length		147	
	mean_reward		55.7	
	time/			
	total_timesteps		175000	
	train/			
	actor_loss		-19.5	
	critic_loss		0.0773	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		169999	

	rollout/			
	ep_len_mean		169	
	ep_rew_mean		51.5	
	time/			
	episodes		796	
	fps		110	
	time_elapsed		1582	
	total_timesteps		175265	
	train/			
	actor_loss		-18.1	
	critic_loss		0.0633	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		170264	

	rollout/			
	ep_len_mean		173	
	ep_rew_mean		50.9	
	time/			
	episodes		800	
	fps		110	
	time_elapsed		1591	

	total_timesteps		176201	
	train/			
	actor_loss		-17.8	
	critic_loss		0.0915	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		171200	

	rollout/			
	ep_len_mean		175	
	ep_rew_mean		50.9	
	time/			
	episodes		804	
	fps		110	
	time_elapsed		1602	
	total_timesteps		177306	
	train/			
	actor_loss		-18.7	
	critic_loss		0.0623	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		172305	

	rollout/			
	ep_len_mean		170	
	ep_rew_mean		51.5	
	time/			
	episodes		808	
	fps		110	
	time_elapsed		1606	
	total_timesteps		177706	
	train/			
	actor_loss		-19.6	
	critic_loss		0.0509	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		172705	

	rollout/			
	ep_len_mean		166	
	ep_rew_mean		52.1	
	time/			
	episodes		812	
	fps		110	
	time_elapsed		1608	

	total_timesteps		178095	
	train/			
	actor_loss		-18.5	
	critic_loss		0.0498	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		173094	

	rollout/			
	ep_len_mean		168	
	ep_rew_mean		52.1	
	time/			
	episodes		816	
	fps		110	
	time_elapsed		1616	
	total_timesteps		178847	
	train/			
	actor_loss		-20.9	
	critic_loss		0.0836	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		173846	

	rollout/			
	ep_len_mean		171	
	ep_rew_mean		51.6	
	time/			
	episodes		820	
	fps		110	
	time_elapsed		1625	
	total_timesteps		179795	
	train/			
	actor_loss		-19.6	
	critic_loss		0.209	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		174794	

Eval num_timesteps=180000, episode_reward=54.55 +/- 2.05
Episode length: 105.10 +/- 53.09

	eval/			
	mean_ep_length		105	
	mean_reward		54.6	
	time/			
	total_timesteps		180000	

train/		
actor_loss	-18.2	
critic_loss	0.101	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	174999	

rollout/		
ep_len_mean	181	
ep_rew_mean	51.2	
time/		
episodes	824	
fps	110	
time_elapsed	1638	
total_timesteps	181238	
train/		
actor_loss	-18.8	
critic_loss	0.115	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	176237	

rollout/		
ep_len_mean	179	
ep_rew_mean	51.2	
time/		
episodes	828	
fps	110	
time_elapsed	1646	
total_timesteps	181974	
train/		
actor_loss	-18.1	
critic_loss	0.106	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	176973	

rollout/		
ep_len_mean	184	
ep_rew_mean	50.7	
time/		
episodes	832	
fps	110	
time_elapsed	1656	
total_timesteps	182998	

train/		
actor_loss	-19.1	
critic_loss	0.0897	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	177997	

rollout/		
ep_len_mean	187	
ep_rew_mean	50.7	
time/		
episodes	836	
fps	110	
time_elapsed	1662	
total_timesteps	183587	

train/		
actor_loss	-19.3	
critic_loss	0.138	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	178586	

rollout/		
ep_len_mean	190	
ep_rew_mean	50.8	
time/		
episodes	840	
fps	110	
time_elapsed	1666	
total_timesteps	184172	

train/		
actor_loss	-19.3	
critic_loss	0.128	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	179171	

rollout/		
ep_len_mean	190	
ep_rew_mean	50.8	
time/		
episodes	844	
fps	110	
time_elapsed	1672	
total_timesteps	184803	

train/		
actor_loss	-19.3	
critic_loss	0.0829	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	179802	

Eval num_timesteps=185000, episode_reward=49.94 +/- 18.14
 Episode length: 187.80 +/- 116.87

eval/		
mean_ep_length	188	
mean_reward	49.9	
time/		
total_timesteps	185000	
train/		
actor_loss	-19.1	
critic_loss	0.15	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	179999	

rollout/		
ep_len_mean	189	
ep_rew_mean	50.8	
time/		
episodes	848	
fps	110	
time_elapsed	1679	
total_timesteps	185276	
train/		
actor_loss	-19.4	
critic_loss	0.111	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	180275	

rollout/		
ep_len_mean	186	
ep_rew_mean	50.7	
time/		
episodes	852	
fps	110	
time_elapsed	1685	
total_timesteps	185814	
train/		

	actor_loss		-19.3	
	critic_loss		0.23	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		180813	

	rollout/			
	ep_len_mean		185	
	ep_rew_mean		50.8	
	time/			
	episodes		856	
	fps		110	
	time_elapsed		1688	
	total_timesteps		186156	
	train/			
	actor_loss		-18.7	
	critic_loss		0.156	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		181155	

	rollout/			
	ep_len_mean		190	
	ep_rew_mean		50.8	
	time/			
	episodes		860	
	fps		110	
	time_elapsed		1697	
	total_timesteps		187004	
	train/			
	actor_loss		-18.7	
	critic_loss		0.111	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		182003	

	rollout/			
	ep_len_mean		193	
	ep_rew_mean		50.9	
	time/			
	episodes		864	
	fps		110	
	time_elapsed		1706	
	total_timesteps		188131	
	train/			

	actor_loss		-19	
	critic_loss		0.0699	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		183130	

	rollout/			
	ep_len_mean		192	
	ep_rew_mean		50.8	
	time/			
	episodes		868	
	fps		110	
	time_elapsed		1710	
	total_timesteps		188519	
	train/			
	actor_loss		-18.2	
	critic_loss		0.074	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		183518	

	rollout/			
	ep_len_mean		191	
	ep_rew_mean		50.1	
	time/			
	episodes		872	
	fps		110	
	time_elapsed		1719	
	total_timesteps		189364	
	train/			
	actor_loss		-18.7	
	critic_loss		0.117	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		184363	

	rollout/			
	ep_len_mean		189	
	ep_rew_mean		50	
	time/			
	episodes		876	
	fps		110	
	time_elapsed		1722	
	total_timesteps		189651	
	train/			

	actor_loss		-19.7	
	critic_loss		0.178	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		184650	

	rollout/			
	ep_len_mean		186	
	ep_rew_mean		50	
	time/			
	episodes		880	
	fps		110	
	time_elapsed		1725	
	total_timesteps		189975	
	train/			
	actor_loss		-17.8	
	critic_loss		0.0575	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		184974	

Eval num_timesteps=190000, episode_reward=54.40 +/- 1.90
 Episode length: 92.90 +/- 51.61

	eval/			
	mean_ep_length		92.9	
	mean_reward		54.4	
	time/			
	total_timesteps		190000	
	train/			
	actor_loss		-18.3	
	critic_loss		0.0886	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		184999	

	rollout/			
	ep_len_mean		188	
	ep_rew_mean		50.1	
	time/			
	episodes		884	
	fps		110	
	time_elapsed		1736	
	total_timesteps		191240	
	train/			
	actor_loss		-18.3	

	critic_loss		0.0535	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		186239	

	rollout/			
	ep_len_mean		190	
	ep_rew_mean		49.5	
	time/			
	episodes		888	
	fps		109	
	time_elapsed		1749	
	total_timesteps		192470	
	train/			
	actor_loss		-19.3	
	critic_loss		0.0961	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		187469	

	rollout/			
	ep_len_mean		182	
	ep_rew_mean		49.4	
	time/			
	episodes		892	
	fps		109	
	time_elapsed		1754	
	total_timesteps		192932	
	train/			
	actor_loss		-18.9	
	critic_loss		0.155	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		187931	

	rollout/			
	ep_len_mean		182	
	ep_rew_mean		49.4	
	time/			
	episodes		896	
	fps		110	
	time_elapsed		1757	
	total_timesteps		193425	
	train/			
	actor_loss		-18.1	

	critic_loss		0.138	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		188424	

	rollout/			
	ep_len_mean		178	
	ep_rew_mean		50	
	time/			
	episodes		900	
	fps		110	
	time_elapsed		1763	
	total_timesteps		194025	
	train/			
	actor_loss		-20.2	
	critic_loss		0.0529	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		189024	

	rollout/			
	ep_len_mean		173	
	ep_rew_mean		50.6	
	time/			
	episodes		904	
	fps		109	
	time_elapsed		1769	
	total_timesteps		194563	
	train/			
	actor_loss		-20.1	
	critic_loss		0.0889	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		189562	

	rollout/			
	ep_len_mean		172	
	ep_rew_mean		50.6	
	time/			
	episodes		908	
	fps		109	
	time_elapsed		1772	
	total_timesteps		194929	
	train/			
	actor_loss		-18.2	

	critic_loss		0.0779	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		189928	

Eval num_timesteps=195000, episode_reward=54.15 +/- 1.62
Episode length: 140.10 +/- 74.17

	eval/			
	mean_ep_length		140	
	mean_reward		54.2	
	time/			
	total_timesteps		195000	
	train/			
	actor_loss		-18.7	
	critic_loss		0.084	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		189999	

	rollout/			
	ep_len_mean		175	
	ep_rew_mean		50.7	
	time/			
	episodes		912	
	fps		109	
	time_elapsed		1780	
	total_timesteps		195562	
	train/			
	actor_loss		-20	
	critic_loss		0.163	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		190561	

	rollout/			
	ep_len_mean		174	
	ep_rew_mean		50.6	
	time/			
	episodes		916	
	fps		109	
	time_elapsed		1787	
	total_timesteps		196206	
	train/			
	actor_loss		-20.4	
	critic_loss		0.1	

	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		191205	

	rollout/			
	ep_len_mean		175	
	ep_rew_mean		51.2	
	time/			
	episodes		920	
	fps		109	
	time_elapsed		1796	
	total_timesteps		197276	
	train/			
	actor_loss		-19.7	
	critic_loss		0.0685	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		192275	

	rollout/			
	ep_len_mean		167	
	ep_rew_mean		51.7	
	time/			
	episodes		924	
	fps		109	
	time_elapsed		1803	
	total_timesteps		197962	
	train/			
	actor_loss		-18.6	
	critic_loss		0.113	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		192961	

	rollout/			
	ep_len_mean		170	
	ep_rew_mean		51.6	
	time/			
	episodes		928	
	fps		109	
	time_elapsed		1813	
	total_timesteps		198938	
	train/			
	actor_loss		-18.7	
	critic_loss		0.194	

	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		193937	

	rollout/			
	ep_len_mean		163	
	ep_rew_mean		52.2	
	time/			
	episodes		932	
	fps		109	
	time_elapsed		1817	
	total_timesteps		199317	
	train/			
	actor_loss		-18.9	
	critic_loss		0.101	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		194316	

	rollout/			
	ep_len_mean		163	
	ep_rew_mean		52.2	
	time/			
	episodes		936	
	fps		109	
	time_elapsed		1821	
	total_timesteps		199905	
	train/			
	actor_loss		-18.8	
	critic_loss		0.118	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		194904	

Eval num_timesteps=200000, episode_reward=54.84 +/- 2.15
Episode length: 162.70 +/- 106.31

	eval/			
	mean_ep_length		163	
	mean_reward		54.8	
	time/			
	total_timesteps		200000	
	train/			
	actor_loss		-19.4	
	critic_loss		0.077	
	ent_coef		0.1	

learning_rate	0.0003
n_updates	194999

rollout/	
ep_len_mean	168
ep_rew_mean	51
time/	
episodes	940
fps	109
time_elapsed	1833
total_timesteps	201009
train/	
actor_loss	-18.3
critic_loss	0.101
ent_coef	0.1
learning_rate	0.0003
n_updates	196008

rollout/	
ep_len_mean	171
ep_rew_mean	50.4
time/	
episodes	944
fps	109
time_elapsed	1842
total_timesteps	201857
train/	
actor_loss	-18.6
critic_loss	0.0971
ent_coef	0.1
learning_rate	0.0003
n_updates	196856

rollout/	
ep_len_mean	169
ep_rew_mean	50.4
time/	
episodes	948
fps	109
time_elapsed	1845
total_timesteps	202162
train/	
actor_loss	-20.2
critic_loss	0.124
ent_coef	0.1

learning_rate	0.0003
n_updates	197161

rollout/	
ep_len_mean	171
ep_rew_mean	50.4
time/	
episodes	952
fps	109
time_elapsed	1851
total_timesteps	202931
train/	
actor_loss	-20.4
critic_loss	0.157
ent_coef	0.1
learning_rate	0.0003
n_updates	197930

rollout/	
ep_len_mean	173
ep_rew_mean	50.4
time/	
episodes	956
fps	109
time_elapsed	1857
total_timesteps	203457
train/	
actor_loss	-19.9
critic_loss	0.0753
ent_coef	0.1
learning_rate	0.0003
n_updates	198456

rollout/	
ep_len_mean	170
ep_rew_mean	50.4
time/	
episodes	960
fps	109
time_elapsed	1862
total_timesteps	204027
train/	
actor_loss	-20.7
critic_loss	0.15
ent_coef	0.1

learning_rate	0.0003
n_updates	199026

Eval num_timesteps=205000, episode_reward=54.71 +/- 2.03
 Episode length: 119.50 +/- 61.30

eval/	
mean_ep_length	120
mean_reward	54.7
time/	
total_timesteps	205000
train/	
actor_loss	-19.9
critic_loss	0.0498
ent_coef	0.1
learning_rate	0.0003
n_updates	199999

rollout/	
ep_len_mean	170
ep_rew_mean	49.9
time/	
episodes	964
fps	109
time_elapsed	1874
total_timesteps	205169
train/	
actor_loss	-19
critic_loss	0.0412
ent_coef	0.1
learning_rate	0.0003
n_updates	200168

rollout/	
ep_len_mean	176
ep_rew_mean	49.9
time/	
episodes	968
fps	109
time_elapsed	1881
total_timesteps	206109
train/	
actor_loss	-19
critic_loss	0.0829
ent_coef	0.1
learning_rate	0.0003

	n_updates		201108	

	rollout/			
	ep_len_mean		172	
	ep_rew_mean		50.5	
	time/			
	episodes		972	
	fps		109	
	time_elapsed		1885	
	total_timesteps		206536	
	train/			
	actor_loss		-19.5	
	critic_loss		0.117	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		201535	

	rollout/			
	ep_len_mean		179	
	ep_rew_mean		50.6	
	time/			
	episodes		976	
	fps		109	
	time_elapsed		1895	
	total_timesteps		207600	
	train/			
	actor_loss		-19.4	
	critic_loss		0.0658	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		202599	

	rollout/			
	ep_len_mean		184	
	ep_rew_mean		50.7	
	time/			
	episodes		980	
	fps		109	
	time_elapsed		1903	
	total_timesteps		208359	
	train/			
	actor_loss		-19.8	
	critic_loss		0.0804	
	ent_coef		0.1	
	learning_rate		0.0003	

n_updates	203358
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rollout/	
ep_len_mean	180
ep_rew_mean	51.1
time/	
episodes	984
fps	109
time_elapsed	1909
total_timesteps	209212
train/	
actor_loss	-19.3
critic_loss	0.0831
ent_coef	0.1
learning_rate	0.0003
n_updates	204211

rollout/	
ep_len_mean	171
ep_rew_mean	52.2
time/	
episodes	988
fps	109
time_elapsed	1913
total_timesteps	209547
train/	
actor_loss	-19.2
critic_loss	0.0938
ent_coef	0.1
learning_rate	0.0003
n_updates	204546

Eval num_timesteps=210000, episode_reward=53.89 +/- 1.31

Episode length: 101.10 +/- 50.49

eval/	
mean_ep_length	101
mean_reward	53.9
time/	
total_timesteps	210000
train/	
actor_loss	-17.7
critic_loss	0.494
ent_coef	0.1
learning_rate	0.0003
n_updates	204999

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-----
| rollout/          |          |
|   ep_len_mean    | 173      |
|   ep_rew_mean    | 52.2     |
| time/            |          |
|   episodes       | 992      |
|   fps            | 109      |
|   time_elapsed   | 1921     |
|   total_timesteps | 210271   |
| train/           |          |
|   actor_loss     | -19.3     |
|   critic_loss    | 0.105    |
|   ent_coef       | 0.1       |
|   learning_rate  | 0.0003   |
|   n_updates      | 205270   |
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| rollout/          |          |
|   ep_len_mean    | 177      |
|   ep_rew_mean    | 51.7     |
| time/            |          |
|   episodes       | 996      |
|   fps            | 109      |
|   time_elapsed   | 1931     |
|   total_timesteps | 211157   |
| train/           |          |
|   actor_loss     | -19.1     |
|   critic_loss    | 0.112    |
|   ent_coef       | 0.1       |
|   learning_rate  | 0.0003   |
|   n_updates      | 206156   |
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| rollout/          |          |
|   ep_len_mean    | 180      |
|   ep_rew_mean    | 51.7     |
| time/            |          |
|   episodes       | 1000     |
|   fps            | 109      |
|   time_elapsed   | 1940     |
|   total_timesteps | 212058   |
| train/           |          |
|   actor_loss     | -18.9     |
|   critic_loss    | 0.143    |
|   ent_coef       | 0.1       |
|   learning_rate  | 0.0003   |
|   n_updates      | 207057   |
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rollout/		
ep_len_mean	179	
ep_rew_mean	51.6	
time/		
episodes	1004	
fps	109	
time_elapsed	1942	
total_timesteps	212475	
train/		
actor_loss	-19.8	
critic_loss	0.136	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	207474	

rollout/		
ep_len_mean	180	
ep_rew_mean	51.7	
time/		
episodes	1008	
fps	109	
time_elapsed	1947	
total_timesteps	212953	
train/		
actor_loss	-20.2	
critic_loss	0.147	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	207952	

rollout/		
ep_len_mean	179	
ep_rew_mean	51.6	
time/		
episodes	1012	
fps	109	
time_elapsed	1952	
total_timesteps	213414	
train/		
actor_loss	-17.8	
critic_loss	0.0773	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	208413	

rollout/		
ep_len_mean	182	
ep_rew_mean	51	
time/		
episodes	1016	
fps	109	
time_elapsed	1962	
total_timesteps	214416	
train/		
actor_loss	-17.4	
critic_loss	0.0435	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	209415	

Eval num_timesteps=215000, episode_reward=54.54 +/- 1.82
Episode length: 129.50 +/- 74.97

eval/		
mean_ep_length	130	
mean_reward	54.5	
time/		
total_timesteps	215000	
train/		
actor_loss	-18.6	
critic_loss	0.0858	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	209999	

rollout/		
ep_len_mean	178	
ep_rew_mean	50.9	
time/		
episodes	1020	
fps	109	
time_elapsed	1970	
total_timesteps	215040	
train/		
actor_loss	-19.8	
critic_loss	0.0928	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	210039	

rollout/		
ep_len_mean	178	
ep_rew_mean	51	
time/		
episodes	1024	
fps	109	
time_elapsed	1975	
total_timesteps	215809	
train/		
actor_loss	-19.3	
critic_loss	0.0781	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	210808	

rollout/		
ep_len_mean	180	
ep_rew_mean	51.1	
time/		
episodes	1028	
fps	109	
time_elapsed	1986	
total_timesteps	216981	
train/		
actor_loss	-19.7	
critic_loss	0.104	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	211980	

rollout/		
ep_len_mean	183	
ep_rew_mean	51	
time/		
episodes	1032	
fps	109	
time_elapsed	1992	
total_timesteps	217658	
train/		
actor_loss	-18.5	
critic_loss	0.0872	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	212657	

rollout/		
ep_len_mean	185	
ep_rew_mean	50.4	
time/		
episodes	1036	
fps	109	
time_elapsed	1999	
total_timesteps	218435	
train/		
actor_loss	-20	
critic_loss	0.103	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	213434	

rollout/		
ep_len_mean	179	
ep_rew_mean	51.6	
time/		
episodes	1040	
fps	109	
time_elapsed	2001	
total_timesteps	218861	
train/		
actor_loss	-18.4	
critic_loss	0.0667	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	213860	

rollout/		
ep_len_mean	174	
ep_rew_mean	52.1	
time/		
episodes	1044	
fps	109	
time_elapsed	2004	
total_timesteps	219296	
train/		
actor_loss	-18.9	
critic_loss	0.103	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	214295	

rollout/		
ep_len_mean	175	
ep_rew_mean	52.1	
time/		
episodes	1048	
fps	109	
time_elapsed	2008	
total_timesteps	219687	
train/		
actor_loss	-18.9	
critic_loss	0.0728	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	214686	

Eval num_timesteps=220000, episode_reward=49.10 +/- 17.86

Episode length: 176.30 +/- 122.54

eval/		
mean_ep_length	176	
mean_reward	49.1	
time/		
total_timesteps	220000	
train/		
actor_loss	-19	
critic_loss	0.0977	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	214999	

rollout/		
ep_len_mean	176	
ep_rew_mean	51.6	
time/		
episodes	1052	
fps	109	
time_elapsed	2017	
total_timesteps	220504	
train/		
actor_loss	-19.7	
critic_loss	0.124	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	215503	

rollout/		
ep_len_mean	179	
ep_rew_mean	51.1	
time/		
episodes	1056	
fps	109	
time_elapsed	2025	
total_timesteps	221389	
train/		
actor_loss	-19.1	
critic_loss	0.0803	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	216388	

rollout/		
ep_len_mean	182	
ep_rew_mean	51.1	
time/		
episodes	1060	
fps	109	
time_elapsed	2031	
total_timesteps	222190	
train/		
actor_loss	-18.3	
critic_loss	0.0723	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	217189	

rollout/		
ep_len_mean	176	
ep_rew_mean	51.6	
time/		
episodes	1064	
fps	109	
time_elapsed	2036	
total_timesteps	222771	
train/		
actor_loss	-19.3	
critic_loss	0.0752	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	217770	

rollout/		
ep_len_mean	176	
ep_rew_mean	51.7	
time/		
episodes	1068	
fps	109	
time_elapsed	2045	
total_timesteps	223742	
train/		
actor_loss	-18.5	
critic_loss	0.0549	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	218741	

rollout/		
ep_len_mean	179	
ep_rew_mean	51.7	
time/		
episodes	1072	
fps	109	
time_elapsed	2052	
total_timesteps	224414	
train/		
actor_loss	-18.8	
critic_loss	0.145	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	219413	

Eval num_timesteps=225000, episode_reward=54.79 +/- 1.90
Episode length: 119.60 +/- 64.17

eval/		
mean_ep_length	120	
mean_reward	54.8	
time/		
total_timesteps	225000	
train/		
actor_loss	-19.3	
critic_loss	0.073	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	219999	

rollout/		
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	ep_len_mean		175	
	ep_rew_mean		51.6	
	time/			
	episodes		1076	
	fps		109	
	time_elapsed		2059	
	total_timesteps		225056	
	train/			
	actor_loss		-18.6	
	critic_loss		0.0761	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		220055	

	rollout/			
	ep_len_mean		173	
	ep_rew_mean		51.6	
	time/			
	episodes		1080	
	fps		109	
	time_elapsed		2062	
	total_timesteps		225615	
	train/			
	actor_loss		-19.8	
	critic_loss		0.169	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		220614	

	rollout/			
	ep_len_mean		172	
	ep_rew_mean		51.6	
	time/			
	episodes		1084	
	fps		109	
	time_elapsed		2068	
	total_timesteps		226386	
	train/			
	actor_loss		-18.6	
	critic_loss		0.0689	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		221385	

	rollout/			
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	ep_len_mean		176	
	ep_rew_mean		51.7	
	time/			
	episodes		1088	
	fps		109	
	time_elapsed		2076	
	total_timesteps		227190	
	train/			
	actor_loss		-20.6	
	critic_loss		0.0773	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		222189	

	rollout/			
	ep_len_mean		175	
	ep_rew_mean		51.6	
	time/			
	episodes		1092	
	fps		109	
	time_elapsed		2082	
	total_timesteps		227797	
	train/			
	actor_loss		-19.4	
	critic_loss		0.113	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		222796	

	rollout/			
	ep_len_mean		172	
	ep_rew_mean		52.2	
	time/			
	episodes		1096	
	fps		109	
	time_elapsed		2087	
	total_timesteps		228398	
	train/			
	actor_loss		-19.6	
	critic_loss		0.104	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		223397	

	rollout/			
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	ep_len_mean		172	
	ep_rew_mean		51.6	
	time/			
	episodes		1100	
	fps		109	
	time_elapsed		2092	
	total_timesteps		229245	
	train/			
	actor_loss		-20.7	
	critic_loss		0.219	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		224244	

	rollout/			
	ep_len_mean		173	
	ep_rew_mean		51.7	
	time/			
	episodes		1104	
	fps		109	
	time_elapsed		2097	
	total_timesteps		229791	
	train/			
	actor_loss		-18.6	
	critic_loss		0.0836	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		224790	

Eval num_timesteps=230000, episode_reward=53.61 +/- 1.10
 Episode length: 79.70 +/- 30.72

	eval/			
	mean_ep_length		79.7	
	mean_reward		53.6	
	time/			
	total_timesteps		230000	
	train/			
	actor_loss		-18.9	
	critic_loss		0.0763	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		224999	

	rollout/			
	ep_len_mean		178	

	ep_rew_mean		51.7	
	time/			
	episodes		1108	
	fps		109	
	time_elapsed		2106	
	total_timesteps		230760	
	train/			
	actor_loss		-18.5	
	critic_loss		0.0851	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		225759	

	rollout/			
	ep_len_mean		179	
	ep_rew_mean		51.8	
	time/			
	episodes		1112	
	fps		109	
	time_elapsed		2112	
	total_timesteps		231306	
	train/			
	actor_loss		-18.9	
	critic_loss		0.176	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		226305	

	rollout/			
	ep_len_mean		174	
	ep_rew_mean		52.4	
	time/			
	episodes		1116	
	fps		109	
	time_elapsed		2116	
	total_timesteps		231778	
	train/			
	actor_loss		-18.8	
	critic_loss		0.159	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		226777	

	rollout/			
	ep_len_mean		171	

	ep_rew_mean		52.3	
	time/			
	episodes		1120	
	fps		109	
	time_elapsed		2119	
	total_timesteps		232136	
	train/			
	actor_loss		-18.8	
	critic_loss		0.0839	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		227135	

	rollout/			
	ep_len_mean		172	
	ep_rew_mean		51.8	
	time/			
	episodes		1124	
	fps		109	
	time_elapsed		2125	
	total_timesteps		233015	
	train/			
	actor_loss		-18.5	
	critic_loss		0.0664	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		228014	

	rollout/			
	ep_len_mean		164	
	ep_rew_mean		51.7	
	time/			
	episodes		1128	
	fps		109	
	time_elapsed		2129	
	total_timesteps		233410	
	train/			
	actor_loss		-20.5	
	critic_loss		0.0999	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		228409	

	rollout/			
	ep_len_mean		162	

	ep_rew_mean		51.7	
	time/			
	episodes		1132	
	fps		109	
	time_elapsed		2133	
	total_timesteps		233860	
	train/			
	actor_loss		-18.4	
	critic_loss		0.126	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		228859	

	rollout/			
	ep_len_mean		158	
	ep_rew_mean		52.3	
	time/			
	episodes		1136	
	fps		109	
	time_elapsed		2137	
	total_timesteps		234243	
	train/			
	actor_loss		-20.4	
	critic_loss		0.102	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		229242	

	rollout/			
	ep_len_mean		159	
	ep_rew_mean		52.3	
	time/			
	episodes		1140	
	fps		109	
	time_elapsed		2143	
	total_timesteps		234794	
	train/			
	actor_loss		-19.6	
	critic_loss		0.124	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		229793	

Eval num_timesteps=235000, episode_reward=44.00 +/- 21.15
Episode length: 205.30 +/- 158.33

eval/		
mean_ep_length	205	
mean_reward	44	
time/		
total_timesteps	235000	
train/		
actor_loss	-19.2	
critic_loss	0.174	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	229999	

rollout/		
ep_len_mean	165	
ep_rew_mean	51.8	
time/		
episodes	1144	
fps	109	
time_elapsed	2153	
total_timesteps	235780	
train/		
actor_loss	-18.4	
critic_loss	0.0643	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	230779	

rollout/		
ep_len_mean	169	
ep_rew_mean	51.9	
time/		
episodes	1148	
fps	109	
time_elapsed	2161	
total_timesteps	236568	
train/		
actor_loss	-19.3	
critic_loss	0.166	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	231567	

rollout/		
ep_len_mean	165	
ep_rew_mean	52.4	

time/		
episodes	1152	
fps	109	
time_elapsed	2165	
total_timesteps	236983	
train/		
actor_loss	-20	
critic_loss	0.0798	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	231982	

rollout/		
ep_len_mean	159	
ep_rew_mean	52.9	
time/		
episodes	1156	
fps	109	
time_elapsed	2168	
total_timesteps	237289	
train/		
actor_loss	-20.1	
critic_loss	0.0489	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	232288	

rollout/		
ep_len_mean	156	
ep_rew_mean	52.8	
time/		
episodes	1160	
fps	109	
time_elapsed	2173	
total_timesteps	237769	
train/		
actor_loss	-19.3	
critic_loss	0.0879	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	232768	

rollout/		
ep_len_mean	153	
ep_rew_mean	52.8	

time/		
episodes	1164	
fps	109	
time_elapsed	2177	
total_timesteps	238100	
train/		
actor_loss	-20.1	
critic_loss	0.187	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	233099	

rollout/		
ep_len_mean	149	
ep_rew_mean	52.7	
time/		
episodes	1168	
fps	109	
time_elapsed	2182	
total_timesteps	238640	
train/		
actor_loss	-19.6	
critic_loss	0.0733	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	233639	

rollout/		
ep_len_mean	152	
ep_rew_mean	52.8	
time/		
episodes	1172	
fps	109	
time_elapsed	2190	
total_timesteps	239622	
train/		
actor_loss	-19.3	
critic_loss	0.073	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	234621	

Eval num_timesteps=240000, episode_reward=54.30 +/- 2.17
Episode length: 129.10 +/- 82.54

eval/		
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	mean_ep_length		129	
	mean_reward		54.3	
	time/			
	total_timesteps		240000	
	train/			
	actor_loss		-19.6	
	critic_loss		0.074	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		234999	

	rollout/			
	ep_len_mean		151	
	ep_rew_mean		52.8	
	time/			
	episodes		1176	
	fps		109	
	time_elapsed		2196	
	total_timesteps		240134	
	train/			
	actor_loss		-19.5	
	critic_loss		0.103	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		235133	

	rollout/			
	ep_len_mean		156	
	ep_rew_mean		52.8	
	time/			
	episodes		1180	
	fps		109	
	time_elapsed		2207	
	total_timesteps		241217	
	train/			
	actor_loss		-19.7	
	critic_loss		0.117	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		236216	

	rollout/			
	ep_len_mean		157	
	ep_rew_mean		52.2	
	time/			

	episodes		1184	
	fps		109	
	time_elapsed		2213	
	total_timesteps		242071	
	train/			
	actor_loss		-19.3	
	critic_loss		0.17	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		237070	

	rollout/			
	ep_len_mean		159	
	ep_rew_mean		51.6	
	time/			
	episodes		1188	
	fps		109	
	time_elapsed		2223	
	total_timesteps		243108	
	train/			
	actor_loss		-18.6	
	critic_loss		0.211	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		238107	

	rollout/			
	ep_len_mean		166	
	ep_rew_mean		50.5	
	time/			
	episodes		1192	
	fps		109	
	time_elapsed		2235	
	total_timesteps		244369	
	train/			
	actor_loss		-19	
	critic_loss		0.136	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		239368	

Eval num_timesteps=245000, episode_reward=54.97 +/- 1.37
Episode length: 159.30 +/- 83.49

	eval/			
	mean_ep_length		159	

	mean_reward		55	
	time/			
	total_timesteps		245000	
	train/			
	actor_loss		-18.9	
	critic_loss		0.143	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		239999	

	rollout/			
	ep_len_mean		169	
	ep_rew_mean		49.9	
	time/			
	episodes		1196	
	fps		109	
	time_elapsed		2243	
	total_timesteps		245254	
	train/			
	actor_loss		-18.1	
	critic_loss		0.0547	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		240253	

	rollout/			
	ep_len_mean		165	
	ep_rew_mean		50.4	
	time/			
	episodes		1200	
	fps		109	
	time_elapsed		2247	
	total_timesteps		245728	
	train/			
	actor_loss		-18.5	
	critic_loss		0.13	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		240727	

	rollout/			
	ep_len_mean		170	
	ep_rew_mean		49.8	
	time/			
	episodes		1204	

	fps		109	
	time_elapsed		2258	
	total_timesteps		246804	
	train/			
	actor_loss		-19.3	
	critic_loss		0.0448	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		241803	

	rollout/			
	ep_len_mean		166	
	ep_rew_mean		49.8	
	time/			
	episodes		1208	
	fps		109	
	time_elapsed		2264	
	total_timesteps		247400	
	train/			
	actor_loss		-19.5	
	critic_loss		0.126	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		242399	

	rollout/			
	ep_len_mean		169	
	ep_rew_mean		49.3	
	time/			
	episodes		1212	
	fps		109	
	time_elapsed		2272	
	total_timesteps		248202	
	train/			
	actor_loss		-21.1	
	critic_loss		0.08	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		243201	

	rollout/			
	ep_len_mean		171	
	ep_rew_mean		49.3	
	time/			
	episodes		1216	

	fps		109	
	time_elapsed		2276	
	total_timesteps		248868	
	train/			
	actor_loss		-18	
	critic_loss		0.111	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		243867	

	rollout/			
	ep_len_mean		171	
	ep_rew_mean		49.3	
	time/			
	episodes		1220	
	fps		109	
	time_elapsed		2280	
	total_timesteps		249214	
	train/			
	actor_loss		-20.2	
	critic_loss		0.0459	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		244213	

	rollout/			
	ep_len_mean		165	
	ep_rew_mean		49.7	
	time/			
	episodes		1224	
	fps		109	
	time_elapsed		2283	
	total_timesteps		249561	
	train/			
	actor_loss		-21	
	critic_loss		0.0761	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		244560	

Eval num_timesteps=250000, episode_reward=54.96 +/- 1.76
Episode length: 116.20 +/- 47.16

	eval/			
	mean_ep_length		116	
	mean_reward		55	

time/		
total_timesteps	250000	
train/		
actor_loss	-19.7	
critic_loss	0.0908	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	244999	

rollout/		
ep_len_mean	169	
ep_rew_mean	49.8	
time/		
episodes	1228	
fps	109	
time_elapsed	2291	
total_timesteps	250320	
train/		
actor_loss	-19.3	
critic_loss	0.0932	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	245319	

rollout/		
ep_len_mean	172	
ep_rew_mean	49.8	
time/		
episodes	1232	
fps	109	
time_elapsed	2299	
total_timesteps	251055	
train/		
actor_loss	-19.1	
critic_loss	0.0651	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	246054	

rollout/		
ep_len_mean	177	
ep_rew_mean	49.9	
time/		
episodes	1236	
fps	109	

	time_elapsed		2306	
	total_timesteps		251975	
	train/			
	actor_loss		-20.3	
	critic_loss		0.13	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		246974	

	rollout/			
	ep_len_mean		179	
	ep_rew_mean		50	
	time/			
	episodes		1240	
	fps		109	
	time_elapsed		2313	
	total_timesteps		252704	
	train/			
	actor_loss		-19.1	
	critic_loss		0.0736	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		247703	

	rollout/			
	ep_len_mean		176	
	ep_rew_mean		50.5	
	time/			
	episodes		1244	
	fps		109	
	time_elapsed		2320	
	total_timesteps		253363	
	train/			
	actor_loss		-18.6	
	critic_loss		0.0897	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		248362	

	rollout/			
	ep_len_mean		177	
	ep_rew_mean		50	
	time/			
	episodes		1248	
	fps		109	

	time_elapsed		2330	
	total_timesteps		254310	
	train/			
	actor_loss		-19.5	
	critic_loss		0.121	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		249309	

Eval num_timesteps=255000, episode_reward=47.96 +/- 17.20

Episode length: 122.00 +/- 130.68

	eval/			
	mean_ep_length		122	
	mean_reward		48	
	time/			
	total_timesteps		255000	
	train/			
	actor_loss		-18.2	
	critic_loss		0.0475	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		249999	

	rollout/			
	ep_len_mean		182	
	ep_rew_mean		50	
	time/			
	episodes		1252	
	fps		109	
	time_elapsed		2338	
	total_timesteps		255143	
	train/			
	actor_loss		-21.3	
	critic_loss		0.0731	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		250142	

	rollout/			
	ep_len_mean		183	
	ep_rew_mean		50.1	
	time/			
	episodes		1256	
	fps		109	
	time_elapsed		2343	

	total_timesteps		255636	
	train/			
	actor_loss		-19.6	
	critic_loss		0.159	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		250635	

	rollout/			
	ep_len_mean		187	
	ep_rew_mean		50.2	
	time/			
	episodes		1260	
	fps		109	
	time_elapsed		2351	
	total_timesteps		256459	
	train/			
	actor_loss		-19.4	
	critic_loss		0.0672	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		251458	

	rollout/			
	ep_len_mean		192	
	ep_rew_mean		49.6	
	time/			
	episodes		1264	
	fps		109	
	time_elapsed		2360	
	total_timesteps		257333	
	train/			
	actor_loss		-19.7	
	critic_loss		0.125	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		252332	

	rollout/			
	ep_len_mean		191	
	ep_rew_mean		49.6	
	time/			
	episodes		1268	
	fps		109	
	time_elapsed		2364	

	total_timesteps		257768	
	train/			
	actor_loss		-18.7	
	critic_loss		0.0941	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		252767	

	rollout/			
	ep_len_mean		185	
	ep_rew_mean		49.5	
	time/			
	episodes		1272	
	fps		109	
	time_elapsed		2365	
	total_timesteps		258082	
	train/			
	actor_loss		-20.5	
	critic_loss		0.0601	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		253081	

	rollout/			
	ep_len_mean		191	
	ep_rew_mean		49	
	time/			
	episodes		1276	
	fps		109	
	time_elapsed		2377	
	total_timesteps		259243	
	train/			
	actor_loss		-19	
	critic_loss		0.135	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		254242	

	rollout/			
	ep_len_mean		188	
	ep_rew_mean		49	
	time/			
	episodes		1280	
	fps		108	
	time_elapsed		2385	

	total_timesteps		259980	
	train/			
	actor_loss		-19.3	
	critic_loss		0.0919	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		254979	

Eval num_timesteps=260000, episode_reward=44.60 +/- 20.67
 Episode length: 195.60 +/- 159.81

	eval/			
	mean_ep_length		196	
	mean_reward		44.6	
	time/			
	total_timesteps		260000	
	train/			
	actor_loss		-20.2	
	critic_loss		0.078	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		254999	

	rollout/			
	ep_len_mean		187	
	ep_rew_mean		49.6	
	time/			
	episodes		1284	
	fps		108	
	time_elapsed		2393	
	total_timesteps		260816	
	train/			
	actor_loss		-19.3	
	critic_loss		0.0781	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		255815	

	rollout/			
	ep_len_mean		183	
	ep_rew_mean		50.1	
	time/			
	episodes		1288	
	fps		108	
	time_elapsed		2399	
	total_timesteps		261415	

train/		
actor_loss	-19.5	
critic_loss	0.254	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	256414	

rollout/		
ep_len_mean	176	
ep_rew_mean	51.3	
time/		
episodes	1292	
fps	108	
time_elapsed	2405	
total_timesteps	262008	

train/		
actor_loss	-20	
critic_loss	0.0934	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	257007	

rollout/		
ep_len_mean	175	
ep_rew_mean	51.9	
time/		
episodes	1296	
fps	108	
time_elapsed	2413	
total_timesteps	262766	

train/		
actor_loss	-19.2	
critic_loss	0.161	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	257765	

rollout/		
ep_len_mean	178	
ep_rew_mean	51.3	
time/		
episodes	1300	
fps	108	
time_elapsed	2421	
total_timesteps	263568	

train/		
actor_loss	-18.4	
critic_loss	0.131	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	258567	

rollout/		
ep_len_mean	172	
ep_rew_mean	51.9	
time/		
episodes	1304	
fps	108	
time_elapsed	2423	
total_timesteps	263955	
train/		
actor_loss	-20.7	
critic_loss	0.0405	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	258954	

rollout/		
ep_len_mean	172	
ep_rew_mean	51.9	
time/		
episodes	1308	
fps	108	
time_elapsed	2429	
total_timesteps	264583	
train/		
actor_loss	-18.5	
critic_loss	0.0663	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	259582	

Eval num_timesteps=265000, episode_reward=49.36 +/- 17.06
 Episode length: 196.40 +/- 138.17

eval/		
mean_ep_length	196	
mean_reward	49.4	
time/		
total_timesteps	265000	
train/		

	actor_loss		-18.9	
	critic_loss		0.179	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		259999	

	rollout/			
	ep_len_mean		170	
	ep_rew_mean		52.5	
	time/			
	episodes		1312	
	fps		108	
	time_elapsed		2436	
	total_timesteps		265236	
	train/			
	actor_loss		-20.7	
	critic_loss		0.113	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		260235	

	rollout/			
	ep_len_mean		169	
	ep_rew_mean		52.4	
	time/			
	episodes		1316	
	fps		108	
	time_elapsed		2441	
	total_timesteps		265812	
	train/			
	actor_loss		-20.7	
	critic_loss		0.119	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		260811	

	rollout/			
	ep_len_mean		170	
	ep_rew_mean		52.5	
	time/			
	episodes		1320	
	fps		108	
	time_elapsed		2445	
	total_timesteps		266263	
	train/			

	actor_loss		-19.5	
	critic_loss		0.0808	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		261262	

	rollout/			
	ep_len_mean		176	
	ep_rew_mean		52.5	
	time/			
	episodes		1324	
	fps		108	
	time_elapsed		2454	
	total_timesteps		267181	
	train/			
	actor_loss		-19.2	
	critic_loss		0.0847	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		262180	

	rollout/			
	ep_len_mean		174	
	ep_rew_mean		52.5	
	time/			
	episodes		1328	
	fps		108	
	time_elapsed		2457	
	total_timesteps		267719	
	train/			
	actor_loss		-19.9	
	critic_loss		0.267	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		262718	

	rollout/			
	ep_len_mean		174	
	ep_rew_mean		52.5	
	time/			
	episodes		1332	
	fps		108	
	time_elapsed		2465	
	total_timesteps		268479	
	train/			

	actor_loss		-19.1	
	critic_loss		0.123	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		263478	

	rollout/			
	ep_len_mean		171	
	ep_rew_mean		52.4	
	time/			
	episodes		1336	
	fps		108	
	time_elapsed		2471	
	total_timesteps		269100	
	train/			
	actor_loss		-19.7	
	critic_loss		0.0573	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		264099	

	rollout/			
	ep_len_mean		171	
	ep_rew_mean		52.5	
	time/			
	episodes		1340	
	fps		108	
	time_elapsed		2478	
	total_timesteps		269785	
	train/			
	actor_loss		-20.6	
	critic_loss		0.0802	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		264784	

Eval num_timesteps=270000, episode_reward=54.91 +/- 1.83
 Episode length: 113.10 +/- 59.68

	eval/			
	mean_ep_length		113	
	mean_reward		54.9	
	time/			
	total_timesteps		270000	
	train/			
	actor_loss		-19.4	

	critic_loss		0.116	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		264999	

	rollout/			
	ep_len_mean		171	
	ep_rew_mean		52.5	
	time/			
	episodes		1344	
	fps		108	
	time_elapsed		2484	
	total_timesteps		270459	
	train/			
	actor_loss		-18.8	
	critic_loss		0.0627	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		265458	

	rollout/			
	ep_len_mean		167	
	ep_rew_mean		53	
	time/			
	episodes		1348	
	fps		108	
	time_elapsed		2490	
	total_timesteps		270994	
	train/			
	actor_loss		-18.4	
	critic_loss		0.0804	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		265993	

	rollout/			
	ep_len_mean		164	
	ep_rew_mean		53	
	time/			
	episodes		1352	
	fps		108	
	time_elapsed		2495	
	total_timesteps		271540	
	train/			
	actor_loss		-19.6	

	critic_loss		0.11	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		266539	

	rollout/			
	ep_len_mean		162	
	ep_rew_mean		52.9	
	time/			
	episodes		1356	
	fps		108	
	time_elapsed		2498	
	total_timesteps		271867	
	train/			
	actor_loss		-19.4	
	critic_loss		0.0952	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		266866	

	rollout/			
	ep_len_mean		159	
	ep_rew_mean		52.9	
	time/			
	episodes		1360	
	fps		108	
	time_elapsed		2503	
	total_timesteps		272375	
	train/			
	actor_loss		-18.8	
	critic_loss		0.132	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		267374	

	rollout/			
	ep_len_mean		153	
	ep_rew_mean		53.4	
	time/			
	episodes		1364	
	fps		108	
	time_elapsed		2506	
	total_timesteps		272612	
	train/			
	actor_loss		-20.5	

	critic_loss		0.0655	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		267611	

	rollout/			
	ep_len_mean		158	
	ep_rew_mean		53.5	
	time/			
	episodes		1368	
	fps		108	
	time_elapsed		2516	
	total_timesteps		273596	
	train/			
	actor_loss		-21	
	critic_loss		0.116	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		268595	

	rollout/			
	ep_len_mean		162	
	ep_rew_mean		53.6	
	time/			
	episodes		1372	
	fps		108	
	time_elapsed		2521	
	total_timesteps		274301	
	train/			
	actor_loss		-19.4	
	critic_loss		0.0673	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		269300	

Eval num_timesteps=275000, episode_reward=54.10 +/- 1.64
Episode length: 96.90 +/- 63.72

	eval/			
	mean_ep_length		96.9	
	mean_reward		54.1	
	time/			
	total_timesteps		275000	
	train/			
	actor_loss		-19.2	
	critic_loss		0.112	

	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		269999	

	rollout/			
	ep_len_mean		158	
	ep_rew_mean		54.1	
	time/			
	episodes		1376	
	fps		108	
	time_elapsed		2530	
	total_timesteps		275072	
	train/			
	actor_loss		-20.2	
	critic_loss		0.0743	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		270071	

	rollout/			
	ep_len_mean		155	
	ep_rew_mean		54.1	
	time/			
	episodes		1380	
	fps		108	
	time_elapsed		2535	
	total_timesteps		275497	
	train/			
	actor_loss		-20.9	
	critic_loss		0.0527	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		270496	

	rollout/			
	ep_len_mean		154	
	ep_rew_mean		54.1	
	time/			
	episodes		1384	
	fps		108	
	time_elapsed		2543	
	total_timesteps		276242	
	train/			
	actor_loss		-19.2	
	critic_loss		0.135	

	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		271241	

	rollout/			
	ep_len_mean		158	
	ep_rew_mean		54.2	
	time/			
	episodes		1388	
	fps		108	
	time_elapsed		2551	
	total_timesteps		277265	
	train/			
	actor_loss		-20.6	
	critic_loss		0.0906	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		272264	

	rollout/			
	ep_len_mean		159	
	ep_rew_mean		54.2	
	time/			
	episodes		1392	
	fps		108	
	time_elapsed		2558	
	total_timesteps		277890	
	train/			
	actor_loss		-18.8	
	critic_loss		0.138	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		272889	

	rollout/			
	ep_len_mean		157	
	ep_rew_mean		54.2	
	time/			
	episodes		1396	
	fps		108	
	time_elapsed		2564	
	total_timesteps		278424	
	train/			
	actor_loss		-20.8	
	critic_loss		0.078	

	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		273423	

	rollout/			
	ep_len_mean		156	
	ep_rew_mean		54.2	
	time/			
	episodes		1400	
	fps		108	
	time_elapsed		2572	
	total_timesteps		279207	
	train/			
	actor_loss		-19	
	critic_loss		0.12	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		274206	

Eval num_timesteps=280000, episode_reward=49.28 +/- 17.46
Episode length: 180.10 +/- 150.41

	eval/			
	mean_ep_length		180	
	mean_reward		49.3	
	time/			
	total_timesteps		280000	
	train/			
	actor_loss		-19.5	
	critic_loss		0.104	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		274999	

	rollout/			
	ep_len_mean		162	
	ep_rew_mean		54.2	
	time/			
	episodes		1404	
	fps		108	
	time_elapsed		2579	
	total_timesteps		280139	
	train/			
	actor_loss		-19.3	
	critic_loss		0.136	
	ent_coef		0.1	

learning_rate	0.0003
n_updates	275138

rollout/	
ep_len_mean	159
ep_rew_mean	54.2
time/	
episodes	1408
fps	108
time_elapsed	2582
total_timesteps	280527
train/	
actor_loss	-17.9
critic_loss	0.0583
ent_coef	0.1
learning_rate	0.0003
n_updates	275526

rollout/	
ep_len_mean	159
ep_rew_mean	54.1
time/	
episodes	1412
fps	108
time_elapsed	2588
total_timesteps	281139
train/	
actor_loss	-19.8
critic_loss	0.0935
ent_coef	0.1
learning_rate	0.0003
n_updates	276138

rollout/	
ep_len_mean	161
ep_rew_mean	54.2
time/	
episodes	1416
fps	108
time_elapsed	2595
total_timesteps	281918
train/	
actor_loss	-19.1
critic_loss	0.108
ent_coef	0.1

learning_rate	0.0003
n_updates	276917

rollout/	
ep_len_mean	164
ep_rew_mean	54.2
time/	
episodes	1420
fps	108
time_elapsed	2604
total_timesteps	282647
train/	
actor_loss	-19
critic_loss	0.0999
ent_coef	0.1
learning_rate	0.0003
n_updates	277646

rollout/	
ep_len_mean	159
ep_rew_mean	54.2
time/	
episodes	1424
fps	108
time_elapsed	2607
total_timesteps	283085
train/	
actor_loss	-19.4
critic_loss	0.0631
ent_coef	0.1
learning_rate	0.0003
n_updates	278084

rollout/	
ep_len_mean	158
ep_rew_mean	54.3
time/	
episodes	1428
fps	108
time_elapsed	2612
total_timesteps	283547
train/	
actor_loss	-19.4
critic_loss	0.085
ent_coef	0.1

	learning_rate		0.0003	
	n_updates		278546	

	rollout/			
	ep_len_mean		159	
	ep_rew_mean		53.6	
	time/			
	episodes		1432	
	fps		108	
	time_elapsed		2621	
	total_timesteps		284359	
	train/			
	actor_loss		-19.1	
	critic_loss		0.0446	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		279358	

	rollout/			
	ep_len_mean		158	
	ep_rew_mean		53.6	
	time/			
	episodes		1436	
	fps		108	
	time_elapsed		2626	
	total_timesteps		284852	
	train/			
	actor_loss		-20.1	
	critic_loss		0.0656	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		279851	

Eval num_timesteps=285000, episode_reward=53.22 +/- 1.16
Episode length: 69.30 +/- 21.63

	eval/			
	mean_ep_length		69.3	
	mean_reward		53.2	
	time/			
	total_timesteps		285000	
	train/			
	actor_loss		-19	
	critic_loss		0.0615	
	ent_coef		0.1	
	learning_rate		0.0003	

	n_updates		279999	

	rollout/			
	ep_len_mean		156	
	ep_rew_mean		53.6	
	time/			
	episodes		1440	
	fps		108	
	time_elapsed		2633	
	total_timesteps		285402	
	train/			
	actor_loss		-20.2	
	critic_loss		0.0658	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		280401	

	rollout/			
	ep_len_mean		157	
	ep_rew_mean		53.6	
	time/			
	episodes		1444	
	fps		108	
	time_elapsed		2639	
	total_timesteps		286179	
	train/			
	actor_loss		-19.6	
	critic_loss		0.083	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		281178	

	rollout/			
	ep_len_mean		162	
	ep_rew_mean		53.6	
	time/			
	episodes		1448	
	fps		108	
	time_elapsed		2649	
	total_timesteps		287170	
	train/			
	actor_loss		-18.7	
	critic_loss		0.0805	
	ent_coef		0.1	
	learning_rate		0.0003	

	n_updates		282169	

	rollout/			
	ep_len_mean		161	
	ep_rew_mean		53.6	
	time/			
	episodes		1452	
	fps		108	
	time_elapsed		2654	
	total_timesteps		287679	
	train/			
	actor_loss		-20.5	
	critic_loss		0.122	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		282678	

	rollout/			
	ep_len_mean		164	
	ep_rew_mean		53.6	
	time/			
	episodes		1456	
	fps		108	
	time_elapsed		2661	
	total_timesteps		288313	
	train/			
	actor_loss		-19.7	
	critic_loss		0.147	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		283312	

	rollout/			
	ep_len_mean		165	
	ep_rew_mean		53.6	
	time/			
	episodes		1460	
	fps		108	
	time_elapsed		2666	
	total_timesteps		288881	
	train/			
	actor_loss		-18.4	
	critic_loss		0.0701	
	ent_coef		0.1	
	learning_rate		0.0003	

	n_updates		283880	
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	rollout/			
	ep_len_mean		168	
	ep_rew_mean		53.7	
	time/			
	episodes		1464	
	fps		108	
	time_elapsed		2670	
	total_timesteps		289409	
	train/			
	actor_loss		-20.2	
	critic_loss		0.152	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		284408	

	rollout/			
	ep_len_mean		164	
	ep_rew_mean		53.7	
	time/			
	episodes		1468	
	fps		108	
	time_elapsed		2675	
	total_timesteps		289959	
	train/			
	actor_loss		-20.3	
	critic_loss		0.0812	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		284958	

Eval num_timesteps=290000, episode_reward=54.89 +/- 2.09

Episode length: 114.80 +/- 72.77

	eval/			
	mean_ep_length		115	
	mean_reward		54.9	
	time/			
	total_timesteps		290000	
	train/			
	actor_loss		-18.3	
	critic_loss		0.085	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		284999	

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| rollout/          |          |
|   ep_len_mean    | 160      |
|   ep_rew_mean    | 53.6     |
| time/            |          |
|   episodes       | 1472     |
|   fps            | 108      |
|   time_elapsed   | 2680     |
|   total_timesteps | 290279   |
| train/           |          |
|   actor_loss     | -18.8    |
|   critic_loss    | 0.0913   |
|   ent_coef       | 0.1      |
|   learning_rate  | 0.0003   |
|   n_updates      | 285278   |
-----

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-----
| rollout/          |          |
|   ep_len_mean    | 159      |
|   ep_rew_mean    | 53.6     |
| time/            |          |
|   episodes       | 1476     |
|   fps            | 108      |
|   time_elapsed   | 2687     |
|   total_timesteps | 291009   |
| train/           |          |
|   actor_loss     | -19.9    |
|   critic_loss    | 0.126    |
|   ent_coef       | 0.1      |
|   learning_rate  | 0.0003   |
|   n_updates      | 286008   |
-----

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| rollout/          |          |
|   ep_len_mean    | 162      |
|   ep_rew_mean    | 53.6     |
| time/            |          |
|   episodes       | 1480     |
|   fps            | 108      |
|   time_elapsed   | 2694     |
|   total_timesteps | 291677   |
| train/           |          |
|   actor_loss     | -19.1    |
|   critic_loss    | 0.103    |
|   ent_coef       | 0.1      |
|   learning_rate  | 0.0003   |
|   n_updates      | 286676   |
-----

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| rollout/          |          |
|   ep_len_mean    | 163      |
|   ep_rew_mean    | 53       |
| time/            |          |
|   episodes       | 1484     |
|   fps            | 108      |
|   time_elapsed   | 2701     |
|   total_timesteps | 292545   |
| train/           |          |
|   actor_loss     | -19.7    |
|   critic_loss    | 0.133    |
|   ent_coef       | 0.1      |
|   learning_rate  | 0.0003   |
|   n_updates      | 287544   |
-----

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-----
| rollout/          |          |
|   ep_len_mean    | 157      |
|   ep_rew_mean    | 52.9     |
| time/            |          |
|   episodes       | 1488     |
|   fps            | 108      |
|   time_elapsed   | 2705     |
|   total_timesteps | 292933   |
| train/           |          |
|   actor_loss     | -19.5    |
|   critic_loss    | 0.124    |
|   ent_coef       | 0.1      |
|   learning_rate  | 0.0003   |
|   n_updates      | 287932   |
-----

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-----
| rollout/          |          |
|   ep_len_mean    | 163      |
|   ep_rew_mean    | 52.9     |
| time/            |          |
|   episodes       | 1492     |
|   fps            | 108      |
|   time_elapsed   | 2718     |
|   total_timesteps | 294175   |
| train/           |          |
|   actor_loss     | -19.8    |
|   critic_loss    | 0.0893   |
|   ent_coef       | 0.1      |
|   learning_rate  | 0.0003   |
|   n_updates      | 289174   |
-----

```

rollout/		
ep_len_mean	163	
ep_rew_mean	52.9	
time/		
episodes	1496	
fps	108	
time_elapsed	2723	
total_timesteps	294677	
train/		
actor_loss	-19.1	
critic_loss	0.0592	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	289676	

Eval num_timesteps=295000, episode_reward=54.36 +/- 1.80
Episode length: 101.50 +/- 75.01

eval/		
mean_ep_length	102	
mean_reward	54.4	
time/		
total_timesteps	295000	
train/		
actor_loss	-19.4	
critic_loss	0.0982	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	289999	

rollout/		
ep_len_mean	160	
ep_rew_mean	53.4	
time/		
episodes	1500	
fps	108	
time_elapsed	2727	
total_timesteps	295183	
train/		
actor_loss	-19.1	
critic_loss	0.0507	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	290182	

rollout/		
ep_len_mean	157	
ep_rew_mean	53.4	
time/		
episodes	1504	
fps	108	
time_elapsed	2733	
total_timesteps	295808	
train/		
actor_loss	-18.6	
critic_loss	0.11	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	290807	

rollout/		
ep_len_mean	161	
ep_rew_mean	53.5	
time/		
episodes	1508	
fps	108	
time_elapsed	2742	
total_timesteps	296646	
train/		
actor_loss	-20.5	
critic_loss	0.114	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	291645	

rollout/		
ep_len_mean	163	
ep_rew_mean	53.6	
time/		
episodes	1512	
fps	108	
time_elapsed	2749	
total_timesteps	297422	
train/		
actor_loss	-19	
critic_loss	0.104	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	292421	

rollout/		
ep_len_mean	159	
ep_rew_mean	53.5	
time/		
episodes	1516	
fps	108	
time_elapsed	2753	
total_timesteps	297866	
train/		
actor_loss	-19.6	
critic_loss	0.125	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	292865	

rollout/		
ep_len_mean	159	
ep_rew_mean	53.5	
time/		
episodes	1520	
fps	108	
time_elapsed	2757	
total_timesteps	298523	
train/		
actor_loss	-19.3	
critic_loss	0.0831	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	293522	

rollout/		
ep_len_mean	159	
ep_rew_mean	53.5	
time/		
episodes	1524	
fps	108	
time_elapsed	2762	
total_timesteps	299010	
train/		
actor_loss	-20.6	
critic_loss	0.0863	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	294009	

rollout/		
ep_len_mean	162	
ep_rew_mean	53.5	
time/		
episodes	1528	
fps	108	
time_elapsed	2768	
total_timesteps	299736	
train/		
actor_loss	-19.7	
critic_loss	0.0954	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	294735	

Eval num_timesteps=300000, episode_reward=54.24 +/- 2.02

Episode length: 161.10 +/- 123.42

eval/		
mean_ep_length	161	
mean_reward	54.2	
time/		
total_timesteps	300000	
train/		
actor_loss	-19.4	
critic_loss	0.205	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	294999	

rollout/		
ep_len_mean	158	
ep_rew_mean	54.1	
time/		
episodes	1532	
fps	108	
time_elapsed	2773	
total_timesteps	300182	
train/		
actor_loss	-19.7	
critic_loss	0.111	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	295181	

rollout/		
ep_len_mean	159	
ep_rew_mean	54.2	
time/		
episodes	1536	
fps	108	
time_elapsed	2778	
total_timesteps	300726	
train/		
actor_loss	-19.7	
critic_loss	0.153	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	295725	

rollout/		
ep_len_mean	158	
ep_rew_mean	54.1	
time/		
episodes	1540	
fps	108	
time_elapsed	2782	
total_timesteps	301208	
train/		
actor_loss	-18.3	
critic_loss	0.111	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	296207	

rollout/		
ep_len_mean	155	
ep_rew_mean	54	
time/		
episodes	1544	
fps	108	
time_elapsed	2787	
total_timesteps	301709	
train/		
actor_loss	-18.6	
critic_loss	0.174	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	296708	

rollout/		
ep_len_mean	153	
ep_rew_mean	54.1	
time/		
episodes	1548	
fps	108	
time_elapsed	2792	
total_timesteps	302517	
train/		
actor_loss	-18.8	
critic_loss	0.0922	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	297516	

rollout/		
ep_len_mean	152	
ep_rew_mean	54	
time/		
episodes	1552	
fps	108	
time_elapsed	2796	
total_timesteps	302888	
train/		
actor_loss	-20.1	
critic_loss	0.128	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	297887	

rollout/		
ep_len_mean	153	
ep_rew_mean	54.1	
time/		
episodes	1556	
fps	108	
time_elapsed	2803	
total_timesteps	303629	
train/		
actor_loss	-19.9	
critic_loss	0.0907	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	298628	

rollout/		
ep_len_mean	159	
ep_rew_mean	53.5	
time/		
episodes	1560	
fps	108	
time_elapsed	2813	
total_timesteps	304749	
train/		
actor_loss	-18.9	
critic_loss	0.212	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	299748	

Eval num_timesteps=305000, episode_reward=55.94 +/- 2.15
 Episode length: 162.60 +/- 69.17

eval/		
mean_ep_length	163	
mean_reward	55.9	
time/		
total_timesteps	305000	
train/		
actor_loss	-20.1	
critic_loss	0.136	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	299999	

rollout/		
ep_len_mean	158	
ep_rew_mean	53.5	
time/		
episodes	1564	
fps	108	
time_elapsed	2817	
total_timesteps	305180	
train/		
actor_loss	-20.8	
critic_loss	0.188	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	300179	

rollout/		
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	ep_len_mean		159	
	ep_rew_mean		53.6	
	time/			
	episodes		1568	
	fps		108	
	time_elapsed		2822	
	total_timesteps		305817	
	train/			
	actor_loss		-19.6	
	critic_loss		0.0635	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		300816	

	rollout/			
	ep_len_mean		162	
	ep_rew_mean		53.6	
	time/			
	episodes		1572	
	fps		108	
	time_elapsed		2828	
	total_timesteps		306452	
	train/			
	actor_loss		-19.9	
	critic_loss		0.0717	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		301451	

	rollout/			
	ep_len_mean		160	
	ep_rew_mean		53.6	
	time/			
	episodes		1576	
	fps		108	
	time_elapsed		2834	
	total_timesteps		307043	
	train/			
	actor_loss		-19	
	critic_loss		0.0636	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		302042	

	rollout/			
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	ep_len_mean		160	
	ep_rew_mean		53.6	
	time/			
	episodes		1580	
	fps		108	
	time_elapsed		2839	
	total_timesteps		307673	
	train/			
	actor_loss		-19.2	
	critic_loss		0.0499	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		302672	

	rollout/			
	ep_len_mean		159	
	ep_rew_mean		54.2	
	time/			
	episodes		1584	
	fps		108	
	time_elapsed		2846	
	total_timesteps		308407	
	train/			
	actor_loss		-20.3	
	critic_loss		0.248	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		303406	

	rollout/			
	ep_len_mean		163	
	ep_rew_mean		53.7	
	time/			
	episodes		1588	
	fps		108	
	time_elapsed		2851	
	total_timesteps		309279	
	train/			
	actor_loss		-20.8	
	critic_loss		0.0522	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		304278	

	rollout/			
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	ep_len_mean		156	
	ep_rew_mean		53.6	
	time/			
	episodes		1592	
	fps		108	
	time_elapsed		2856	
	total_timesteps		309770	
	train/			
	actor_loss		-18.6	
	critic_loss		0.125	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		304769	

Eval num_timesteps=310000, episode_reward=55.28 +/- 2.33
 Episode length: 128.00 +/- 80.07

	eval/			
	mean_ep_length		128	
	mean_reward		55.3	
	time/			
	total_timesteps		310000	
	train/			
	actor_loss		-19.7	
	critic_loss		0.0906	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		304999	

	rollout/			
	ep_len_mean		157	
	ep_rew_mean		53.6	
	time/			
	episodes		1596	
	fps		108	
	time_elapsed		2863	
	total_timesteps		310329	
	train/			
	actor_loss		-18.6	
	critic_loss		0.0822	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		305328	

	rollout/			
	ep_len_mean		158	

	ep_rew_mean		53.7	
	time/			
	episodes		1600	
	fps		108	
	time_elapsed		2869	
	total_timesteps		311018	
	train/			
	actor_loss		-18.8	
	critic_loss		0.0579	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		306017	

	rollout/			
	ep_len_mean		157	
	ep_rew_mean		53.7	
	time/			
	episodes		1604	
	fps		108	
	time_elapsed		2874	
	total_timesteps		311498	
	train/			
	actor_loss		-19.8	
	critic_loss		0.0775	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		306497	

	rollout/			
	ep_len_mean		154	
	ep_rew_mean		53.6	
	time/			
	episodes		1608	
	fps		108	
	time_elapsed		2877	
	total_timesteps		312065	
	train/			
	actor_loss		-19.7	
	critic_loss		0.212	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		307064	

	rollout/			
	ep_len_mean		156	

	ep_rew_mean		53.6	
	time/			
	episodes		1612	
	fps		108	
	time_elapsed		2887	
	total_timesteps		313072	
	train/			
	actor_loss		-20.2	
	critic_loss		0.0508	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		308071	

	rollout/			
	ep_len_mean		156	
	ep_rew_mean		53.6	
	time/			
	episodes		1616	
	fps		108	
	time_elapsed		2890	
	total_timesteps		313421	
	train/			
	actor_loss		-21.1	
	critic_loss		0.118	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		308420	

	rollout/			
	ep_len_mean		158	
	ep_rew_mean		53.7	
	time/			
	episodes		1620	
	fps		108	
	time_elapsed		2899	
	total_timesteps		314362	
	train/			
	actor_loss		-20.3	
	critic_loss		0.0862	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		309361	

	rollout/			
	ep_len_mean		159	

	ep_rew_mean		53.7	
	time/			
	episodes		1624	
	fps		108	
	time_elapsed		2905	
	total_timesteps		314930	
	train/			
	actor_loss		-20.2	
	critic_loss		0.227	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		309929	

Eval num_timesteps=315000, episode_reward=55.22 +/- 2.10
Episode length: 130.00 +/- 83.82

	eval/			
	mean_ep_length		130	
	mean_reward		55.2	
	time/			
	total_timesteps		315000	
	train/			
	actor_loss		-20.1	
	critic_loss		0.192	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		309999	

	rollout/			
	ep_len_mean		157	
	ep_rew_mean		53.6	
	time/			
	episodes		1628	
	fps		108	
	time_elapsed		2909	
	total_timesteps		315428	
	train/			
	actor_loss		-18.8	
	critic_loss		0.0977	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		310427	

	rollout/			
	ep_len_mean		159	
	ep_rew_mean		53.6	

time/		
episodes	1632	
fps	108	
time_elapsed	2914	
total_timesteps	316051	
train/		
actor_loss	-19.1	
critic_loss	0.186	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	311050	

rollout/		
ep_len_mean	163	
ep_rew_mean	53.6	
time/		
episodes	1636	
fps	108	
time_elapsed	2923	
total_timesteps	317019	
train/		
actor_loss	-19.9	
critic_loss	0.0469	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	312018	

rollout/		
ep_len_mean	163	
ep_rew_mean	53.7	
time/		
episodes	1640	
fps	108	
time_elapsed	2927	
total_timesteps	317480	
train/		
actor_loss	-19.3	
critic_loss	0.187	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	312479	

rollout/		
ep_len_mean	163	
ep_rew_mean	53.7	

time/		
episodes	1644	
fps	108	
time_elapsed	2931	
total_timesteps	317974	
train/		
actor_loss	-20.7	
critic_loss	0.0991	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	312973	

rollout/		
ep_len_mean	163	
ep_rew_mean	53.1	
time/		
episodes	1648	
fps	108	
time_elapsed	2938	
total_timesteps	318809	
train/		
actor_loss	-19.6	
critic_loss	0.288	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	313808	

rollout/		
ep_len_mean	166	
ep_rew_mean	52.6	
time/		
episodes	1652	
fps	108	
time_elapsed	2941	
total_timesteps	319453	
train/		
actor_loss	-19.2	
critic_loss	0.0869	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	314452	

rollout/		
ep_len_mean	162	
ep_rew_mean	52.5	

time/		
episodes	1656	
fps	108	
time_elapsed	2944	
total_timesteps	319867	
train/		
actor_loss	-18.7	
critic_loss	0.114	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	314866	

Eval num_timesteps=320000, episode_reward=55.43 +/- 1.98
Episode length: 156.60 +/- 86.79

eval/		
mean_ep_length	157	
mean_reward	55.4	
time/		
total_timesteps	320000	
train/		
actor_loss	-20.7	
critic_loss	0.0681	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	314999	

rollout/		
ep_len_mean	162	
ep_rew_mean	52.5	
time/		
episodes	1660	
fps	108	
time_elapsed	2955	
total_timesteps	320978	
train/		
actor_loss	-17.9	
critic_loss	0.0935	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	315977	

rollout/		
ep_len_mean	164	
ep_rew_mean	52.5	
time/		

	episodes		1664	
	fps		108	
	time_elapsed		2959	
	total_timesteps		321536	
	train/			
	actor_loss		-19.7	
	critic_loss		0.138	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		316535	

	rollout/			
	ep_len_mean		164	
	ep_rew_mean		52.5	
	time/			
	episodes		1668	
	fps		108	
	time_elapsed		2965	
	total_timesteps		322256	
	train/			
	actor_loss		-19.1	
	critic_loss		0.0463	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		317255	

	rollout/			
	ep_len_mean		164	
	ep_rew_mean		52.5	
	time/			
	episodes		1672	
	fps		108	
	time_elapsed		2968	
	total_timesteps		322827	
	train/			
	actor_loss		-20.2	
	critic_loss		0.0814	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		317826	

	rollout/			
	ep_len_mean		165	
	ep_rew_mean		52.5	
	time/			

	episodes		1676	
	fps		108	
	time_elapsed		2974	
	total_timesteps		323528	
	train/			
	actor_loss		-19.4	
	critic_loss		0.114	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		318527	

	rollout/			
	ep_len_mean		163	
	ep_rew_mean		52.4	
	time/			
	episodes		1680	
	fps		108	
	time_elapsed		2977	
	total_timesteps		323932	
	train/			
	actor_loss		-19.7	
	critic_loss		0.105	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		318931	

	rollout/			
	ep_len_mean		159	
	ep_rew_mean		52.4	
	time/			
	episodes		1684	
	fps		108	
	time_elapsed		2980	
	total_timesteps		324297	
	train/			
	actor_loss		-20.9	
	critic_loss		0.0705	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		319296	

	rollout/			
	ep_len_mean		157	
	ep_rew_mean		53	
	time/			

	episodes		1688	
	fps		108	
	time_elapsed		2986	
	total_timesteps		324983	
	train/			
	actor_loss		-19.9	
	critic_loss		0.0631	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		319982	

Eval num_timesteps=325000, episode_reward=56.14 +/- 1.23
Episode length: 167.30 +/- 52.50

	eval/			
	mean_ep_length		167	
	mean_reward		56.1	
	time/			
	total_timesteps		325000	
	train/			
	actor_loss		-20.3	
	critic_loss		0.119	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		319999	

	rollout/			
	ep_len_mean		157	
	ep_rew_mean		53	
	time/			
	episodes		1692	
	fps		108	
	time_elapsed		2991	
	total_timesteps		325461	
	train/			
	actor_loss		-20.1	
	critic_loss		0.0753	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		320460	

	rollout/			
	ep_len_mean		155	
	ep_rew_mean		53	
	time/			
	episodes		1696	

	fps		108	
	time_elapsed		2994	
	total_timesteps		325862	
	train/			
	actor_loss		-20.5	
	critic_loss		0.0494	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		320861	

	rollout/			
	ep_len_mean		152	
	ep_rew_mean		52.9	
	time/			
	episodes		1700	
	fps		108	
	time_elapsed		2997	
	total_timesteps		326209	
	train/			
	actor_loss		-20.1	
	critic_loss		0.235	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		321208	

	rollout/			
	ep_len_mean		154	
	ep_rew_mean		52.9	
	time/			
	episodes		1704	
	fps		108	
	time_elapsed		3000	
	total_timesteps		326916	
	train/			
	actor_loss		-19.2	
	critic_loss		0.0927	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		321915	

	rollout/			
	ep_len_mean		156	
	ep_rew_mean		52.4	
	time/			
	episodes		1708	

	fps		108	
	time_elapsed		3006	
	total_timesteps		327701	
	train/			
	actor_loss		-20.5	
	critic_loss		0.0781	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		322700	

	rollout/			
	ep_len_mean		150	
	ep_rew_mean		52.3	
	time/			
	episodes		1712	
	fps		109	
	time_elapsed		3009	
	total_timesteps		328047	
	train/			
	actor_loss		-19	
	critic_loss		0.0597	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		323046	

	rollout/			
	ep_len_mean		153	
	ep_rew_mean		52.3	
	time/			
	episodes		1716	
	fps		109	
	time_elapsed		3014	
	total_timesteps		328744	
	train/			
	actor_loss		-20.1	
	critic_loss		0.0686	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		323743	

	rollout/			
	ep_len_mean		154	
	ep_rew_mean		51.8	
	time/			
	episodes		1720	

	fps		109	
	time_elapsed		3023	
	total_timesteps		329743	
	train/			
	actor_loss		-19.8	
	critic_loss		0.303	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		324742	

Eval num_timesteps=330000, episode_reward=50.12 +/- 17.41
Episode length: 197.70 +/- 131.52

	eval/			
	mean_ep_length		198	
	mean_reward		50.1	
	time/			
	total_timesteps		330000	
	train/			
	actor_loss		-18.6	
	critic_loss		0.0926	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		324999	

	rollout/			
	ep_len_mean		155	
	ep_rew_mean		51.8	
	time/			
	episodes		1724	
	fps		109	
	time_elapsed		3029	
	total_timesteps		330464	
	train/			
	actor_loss		-19.7	
	critic_loss		0.126	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		325463	

	rollout/			
	ep_len_mean		156	
	ep_rew_mean		51.8	
	time/			
	episodes		1728	
	fps		109	

	time_elapsed		3034	
	total_timesteps		331014	
	train/			
	actor_loss		-20.3	
	critic_loss		0.0681	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		326013	

	rollout/			
	ep_len_mean		155	
	ep_rew_mean		51.7	
	time/			
	episodes		1732	
	fps		109	
	time_elapsed		3038	
	total_timesteps		331555	
	train/			
	actor_loss		-21.5	
	critic_loss		0.0805	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		326554	

	rollout/			
	ep_len_mean		153	
	ep_rew_mean		51.8	
	time/			
	episodes		1736	
	fps		109	
	time_elapsed		3044	
	total_timesteps		332302	
	train/			
	actor_loss		-20.3	
	critic_loss		0.159	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		327301	

	rollout/			
	ep_len_mean		152	
	ep_rew_mean		51.7	
	time/			
	episodes		1740	
	fps		109	

	time_elapsed		3048	
	total_timesteps		332729	
	train/			
	actor_loss		-19.5	
	critic_loss		0.115	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		327728	

	rollout/			
	ep_len_mean		153	
	ep_rew_mean		51.8	
	time/			
	episodes		1744	
	fps		109	
	time_elapsed		3052	
	total_timesteps		333272	
	train/			
	actor_loss		-20.1	
	critic_loss		0.228	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		328271	

	rollout/			
	ep_len_mean		149	
	ep_rew_mean		52.2	
	time/			
	episodes		1748	
	fps		109	
	time_elapsed		3055	
	total_timesteps		333688	
	train/			
	actor_loss		-18.4	
	critic_loss		0.0883	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		328687	

	rollout/			
	ep_len_mean		148	
	ep_rew_mean		52.8	
	time/			
	episodes		1752	
	fps		109	

	time_elapsed		3059	
	total_timesteps		334219	
	train/			
	actor_loss		-21	
	critic_loss		0.104	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		329218	

	rollout/			
	ep_len_mean		150	
	ep_rew_mean		52.8	
	time/			
	episodes		1756	
	fps		109	
	time_elapsed		3062	
	total_timesteps		334845	
	train/			
	actor_loss		-19.7	
	critic_loss		0.0678	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		329844	

Eval num_timesteps=335000, episode_reward=54.83 +/- 1.62
Episode length: 120.30 +/- 41.36

	eval/			
	mean_ep_length		120	
	mean_reward		54.8	
	time/			
	total_timesteps		335000	
	train/			
	actor_loss		-20.2	
	critic_loss		0.0539	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		329999	

	rollout/			
	ep_len_mean		145	
	ep_rew_mean		53.4	
	time/			
	episodes		1760	
	fps		109	
	time_elapsed		3068	

	total_timesteps		335456	
	train/			
	actor_loss		-21.5	
	critic_loss		0.059	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		330455	

	rollout/			
	ep_len_mean		146	
	ep_rew_mean		53.4	
	time/			
	episodes		1764	
	fps		109	
	time_elapsed		3074	
	total_timesteps		336125	
	train/			
	actor_loss		-20.2	
	critic_loss		0.12	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		331124	

	rollout/			
	ep_len_mean		147	
	ep_rew_mean		53.4	
	time/			
	episodes		1768	
	fps		109	
	time_elapsed		3081	
	total_timesteps		336956	
	train/			
	actor_loss		-19.8	
	critic_loss		0.119	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		331955	

	rollout/			
	ep_len_mean		148	
	ep_rew_mean		53.5	
	time/			
	episodes		1772	
	fps		109	
	time_elapsed		3086	

	total_timesteps		337584	
	train/			
	actor_loss		-22.3	
	critic_loss		0.119	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		332583	

	rollout/			
	ep_len_mean		144	
	ep_rew_mean		53.4	
	time/			
	episodes		1776	
	fps		109	
	time_elapsed		3089	
	total_timesteps		337922	
	train/			
	actor_loss		-20.9	
	critic_loss		0.0925	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		332921	

	rollout/			
	ep_len_mean		148	
	ep_rew_mean		53.4	
	time/			
	episodes		1780	
	fps		109	
	time_elapsed		3093	
	total_timesteps		338698	
	train/			
	actor_loss		-19.4	
	critic_loss		0.201	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		333697	

	rollout/			
	ep_len_mean		149	
	ep_rew_mean		53.5	
	time/			
	episodes		1784	
	fps		109	
	time_elapsed		3098	

	total_timesteps		339246	
	train/			
	actor_loss		-20.5	
	critic_loss		0.11	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		334245	

Eval num_timesteps=340000, episode_reward=53.48 +/- 1.55
 Episode length: 75.60 +/- 30.40

	eval/			
	mean_ep_length		75.6	
	mean_reward		53.5	
	time/			
	total_timesteps		340000	
	train/			
	actor_loss		-19.1	
	critic_loss		0.142	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		334999	

	rollout/			
	ep_len_mean		153	
	ep_rew_mean		53	
	time/			
	episodes		1788	
	fps		109	
	time_elapsed		3107	
	total_timesteps		340252	
	train/			
	actor_loss		-20.1	
	critic_loss		0.14	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		335251	

	rollout/			
	ep_len_mean		155	
	ep_rew_mean		53	
	time/			
	episodes		1792	
	fps		109	
	time_elapsed		3113	
	total_timesteps		340965	

train/		
actor_loss	-20	
critic_loss	0.13	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	335964	

rollout/		
ep_len_mean	155	
ep_rew_mean	53	
time/		
episodes	1796	
fps	109	
time_elapsed	3117	
total_timesteps	341382	
train/		
actor_loss	-19.5	
critic_loss	0.105	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	336381	

rollout/		
ep_len_mean	154	
ep_rew_mean	53	
time/		
episodes	1800	
fps	109	
time_elapsed	3119	
total_timesteps	341653	
train/		
actor_loss	-20.1	
critic_loss	0.114	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	336652	

rollout/		
ep_len_mean	151	
ep_rew_mean	52.9	
time/		
episodes	1804	
fps	109	
time_elapsed	3120	
total_timesteps	342032	

train/		
actor_loss	-20.5	
critic_loss	0.0639	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	337031	

rollout/		
ep_len_mean	149	
ep_rew_mean	53.5	
time/		
episodes	1808	
fps	109	
time_elapsed	3125	
total_timesteps	342615	
train/		
actor_loss	-19.1	
critic_loss	0.0501	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	337614	

rollout/		
ep_len_mean	149	
ep_rew_mean	53.5	
time/		
episodes	1812	
fps	109	
time_elapsed	3128	
total_timesteps	342926	
train/		
actor_loss	-20.6	
critic_loss	0.0649	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	337925	

rollout/		
ep_len_mean	148	
ep_rew_mean	53.5	
time/		
episodes	1816	
fps	109	
time_elapsed	3133	
total_timesteps	343552	

train/		
actor_loss	-21	
critic_loss	0.137	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	338551	

rollout/		
ep_len_mean	143	
ep_rew_mean	54	
time/		
episodes	1820	
fps	109	
time_elapsed	3137	
total_timesteps	344052	
train/		
actor_loss	-19.5	
critic_loss	0.0572	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	339051	

rollout/		
ep_len_mean	144	
ep_rew_mean	54.1	
time/		
episodes	1824	
fps	109	
time_elapsed	3144	
total_timesteps	344895	
train/		
actor_loss	-21.6	
critic_loss	0.0504	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	339894	

Eval num_timesteps=345000, episode_reward=55.82 +/- 1.87
 Episode length: 165.00 +/- 84.40

eval/		
mean_ep_length	165	
mean_reward	55.8	
time/		
total_timesteps	345000	
train/		

	actor_loss		-20.8	
	critic_loss		0.082	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		339999	

	rollout/			
	ep_len_mean		147	
	ep_rew_mean		54.1	
	time/			
	episodes		1828	
	fps		109	
	time_elapsed		3150	
	total_timesteps		345721	
	train/			
	actor_loss		-22.4	
	critic_loss		0.0463	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		340720	

	rollout/			
	ep_len_mean		149	
	ep_rew_mean		54.1	
	time/			
	episodes		1832	
	fps		109	
	time_elapsed		3156	
	total_timesteps		346440	
	train/			
	actor_loss		-20.2	
	critic_loss		0.0824	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		341439	

	rollout/			
	ep_len_mean		144	
	ep_rew_mean		54	
	time/			
	episodes		1836	
	fps		109	
	time_elapsed		3158	
	total_timesteps		346694	
	train/			

	actor_loss		-20.3	
	critic_loss		0.0532	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		341693	

	rollout/			
	ep_len_mean		144	
	ep_rew_mean		53.9	
	time/			
	episodes		1840	
	fps		109	
	time_elapsed		3161	
	total_timesteps		347088	
	train/			
	actor_loss		-21.3	
	critic_loss		0.0848	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		342087	

	rollout/			
	ep_len_mean		146	
	ep_rew_mean		54	
	time/			
	episodes		1844	
	fps		109	
	time_elapsed		3167	
	total_timesteps		347881	
	train/			
	actor_loss		-21.9	
	critic_loss		0.0735	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		342880	

	rollout/			
	ep_len_mean		146	
	ep_rew_mean		54	
	time/			
	episodes		1848	
	fps		109	
	time_elapsed		3171	
	total_timesteps		348302	
	train/			

	actor_loss		-21	
	critic_loss		0.262	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		343301	

	rollout/			
	ep_len_mean		146	
	ep_rew_mean		54	
	time/			
	episodes		1852	
	fps		109	
	time_elapsed		3176	
	total_timesteps		348846	
	train/			
	actor_loss		-20.8	
	critic_loss		0.0556	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		343845	

	rollout/			
	ep_len_mean		143	
	ep_rew_mean		53.9	
	time/			
	episodes		1856	
	fps		109	
	time_elapsed		3178	
	total_timesteps		349132	
	train/			
	actor_loss		-19.5	
	critic_loss		0.111	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		344131	

	rollout/			
	ep_len_mean		141	
	ep_rew_mean		53.9	
	time/			
	episodes		1860	
	fps		109	
	time_elapsed		3180	
	total_timesteps		349597	
	train/			

	actor_loss		-20.7	
	critic_loss		0.155	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		344596	

Eval num_timesteps=350000, episode_reward=54.38 +/- 2.08
Episode length: 105.70 +/- 87.17

	eval/			
	mean_ep_length		106	
	mean_reward		54.4	
	time/			
	total_timesteps		350000	
	train/			
	actor_loss		-20.4	
	critic_loss		0.127	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		344999	

	rollout/			
	ep_len_mean		139	
	ep_rew_mean		53.8	
	time/			
	episodes		1864	
	fps		109	
	time_elapsed		3184	
	total_timesteps		350009	
	train/			
	actor_loss		-19.9	
	critic_loss		0.0631	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		345008	

	rollout/			
	ep_len_mean		134	
	ep_rew_mean		53.7	
	time/			
	episodes		1868	
	fps		109	
	time_elapsed		3187	
	total_timesteps		350370	
	train/			
	actor_loss		-22.1	

	critic_loss		0.0996	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		345369	

	rollout/			
	ep_len_mean		132	
	ep_rew_mean		53.7	
	time/			
	episodes		1872	
	fps		109	
	time_elapsed		3191	
	total_timesteps		350783	
	train/			
	actor_loss		-20.9	
	critic_loss		0.0816	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		345782	

	rollout/			
	ep_len_mean		132	
	ep_rew_mean		53.7	
	time/			
	episodes		1876	
	fps		109	
	time_elapsed		3193	
	total_timesteps		351103	
	train/			
	actor_loss		-20.1	
	critic_loss		0.105	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		346102	

	rollout/			
	ep_len_mean		129	
	ep_rew_mean		53.6	
	time/			
	episodes		1880	
	fps		109	
	time_elapsed		3197	
	total_timesteps		351579	
	train/			
	actor_loss		-20.8	

	critic_loss		0.083	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		346578	

	rollout/			
	ep_len_mean		132	
	ep_rew_mean		53.6	
	time/			
	episodes		1884	
	fps		109	
	time_elapsed		3205	
	total_timesteps		352447	
	train/			
	actor_loss		-20.4	
	critic_loss		0.0839	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		347446	

	rollout/			
	ep_len_mean		128	
	ep_rew_mean		54.1	
	time/			
	episodes		1888	
	fps		109	
	time_elapsed		3210	
	total_timesteps		353054	
	train/			
	actor_loss		-20.6	
	critic_loss		0.108	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		348053	

	rollout/			
	ep_len_mean		126	
	ep_rew_mean		54.1	
	time/			
	episodes		1892	
	fps		110	
	time_elapsed		3212	
	total_timesteps		353601	
	train/			
	actor_loss		-19	

	critic_loss		0.0946	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		348600	

	rollout/			
	ep_len_mean		131	
	ep_rew_mean		54.2	
	time/			
	episodes		1896	
	fps		110	
	time_elapsed		3219	
	total_timesteps		354478	
	train/			
	actor_loss		-20.9	
	critic_loss		0.0698	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		349477	

Eval num_timesteps=355000, episode_reward=54.17 +/- 1.67
Episode length: 93.40 +/- 52.50

	eval/			
	mean_ep_length		93.4	
	mean_reward		54.2	
	time/			
	total_timesteps		355000	
	train/			
	actor_loss		-21.8	
	critic_loss		0.0764	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		349999	

	rollout/			
	ep_len_mean		134	
	ep_rew_mean		54.2	
	time/			
	episodes		1900	
	fps		110	
	time_elapsed		3226	
	total_timesteps		355100	
	train/			
	actor_loss		-20.6	
	critic_loss		0.198	

	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		350099	

	rollout/			
	ep_len_mean		136	
	ep_rew_mean		54.3	
	time/			
	episodes		1904	
	fps		110	
	time_elapsed		3232	
	total_timesteps		355677	
	train/			
	actor_loss		-21.6	
	critic_loss		0.0663	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		350676	

	rollout/			
	ep_len_mean		141	
	ep_rew_mean		53.7	
	time/			
	episodes		1908	
	fps		110	
	time_elapsed		3241	
	total_timesteps		356675	
	train/			
	actor_loss		-21	
	critic_loss		0.18	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		351674	

	rollout/			
	ep_len_mean		144	
	ep_rew_mean		53.7	
	time/			
	episodes		1912	
	fps		110	
	time_elapsed		3244	
	total_timesteps		357347	
	train/			
	actor_loss		-21.8	
	critic_loss		0.136	

	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		352346	

	rollout/			
	ep_len_mean		141	
	ep_rew_mean		53.7	
	time/			
	episodes		1916	
	fps		110	
	time_elapsed		3247	
	total_timesteps		357648	
	train/			
	actor_loss		-21	
	critic_loss		0.0855	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		352647	

	rollout/			
	ep_len_mean		141	
	ep_rew_mean		53.7	
	time/			
	episodes		1920	
	fps		110	
	time_elapsed		3252	
	total_timesteps		358198	
	train/			
	actor_loss		-19.6	
	critic_loss		0.102	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		353197	

	rollout/			
	ep_len_mean		142	
	ep_rew_mean		53.7	
	time/			
	episodes		1924	
	fps		110	
	time_elapsed		3259	
	total_timesteps		359057	
	train/			
	actor_loss		-21	
	critic_loss		0.123	

	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		354056	

	rollout/			
	ep_len_mean		137	
	ep_rew_mean		53.7	
	time/			
	episodes		1928	
	fps		110	
	time_elapsed		3262	
	total_timesteps		359464	
	train/			
	actor_loss		-22	
	critic_loss		0.106	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		354463	

	rollout/			
	ep_len_mean		135	
	ep_rew_mean		53.6	
	time/			
	episodes		1932	
	fps		110	
	time_elapsed		3266	
	total_timesteps		359979	
	train/			
	actor_loss		-21.5	
	critic_loss		0.0969	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		354978	

Eval num_timesteps=360000, episode_reward=54.52 +/- 2.05
 Episode length: 114.40 +/- 80.76

	eval/			
	mean_ep_length		114	
	mean_reward		54.5	
	time/			
	total_timesteps		360000	
	train/			
	actor_loss		-21	
	critic_loss		0.152	
	ent_coef		0.1	

learning_rate	0.0003
n_updates	354999

rollout/	
ep_len_mean	136
ep_rew_mean	53.6
time/	
episodes	1936
fps	110
time_elapsed	3270
total_timesteps	360336
train/	
actor_loss	-20.8
critic_loss	0.173
ent_coef	0.1
learning_rate	0.0003
n_updates	355335

rollout/	
ep_len_mean	140
ep_rew_mean	53.7
time/	
episodes	1940
fps	110
time_elapsed	3275
total_timesteps	361118
train/	
actor_loss	-20.8
critic_loss	0.0524
ent_coef	0.1
learning_rate	0.0003
n_updates	356117

rollout/	
ep_len_mean	138
ep_rew_mean	53.6
time/	
episodes	1944
fps	110
time_elapsed	3280
total_timesteps	361725
train/	
actor_loss	-20.6
critic_loss	0.135
ent_coef	0.1

learning_rate	0.0003
n_updates	356724

rollout/	
ep_len_mean	139
ep_rew_mean	53.6
time/	
episodes	1948
fps	110
time_elapsed	3284
total_timesteps	362226
train/	
actor_loss	-21.1
critic_loss	0.0823
ent_coef	0.1
learning_rate	0.0003
n_updates	357225

rollout/	
ep_len_mean	140
ep_rew_mean	53.7
time/	
episodes	1952
fps	110
time_elapsed	3290
total_timesteps	362867
train/	
actor_loss	-20.7
critic_loss	0.0872
ent_coef	0.1
learning_rate	0.0003
n_updates	357866

rollout/	
ep_len_mean	141
ep_rew_mean	53.7
time/	
episodes	1956
fps	110
time_elapsed	3293
total_timesteps	363247
train/	
actor_loss	-21.4
critic_loss	0.066
ent_coef	0.1

	learning_rate		0.0003	
	n_updates		358246	

	rollout/			
	ep_len_mean		144	
	ep_rew_mean		53.8	
	time/			
	episodes		1960	
	fps		110	
	time_elapsed		3299	
	total_timesteps		363999	
	train/			
	actor_loss		-20.8	
	critic_loss		0.0874	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		358998	

	rollout/			
	ep_len_mean		149	
	ep_rew_mean		53.2	
	time/			
	episodes		1964	
	fps		110	
	time_elapsed		3305	
	total_timesteps		364956	
	train/			
	actor_loss		-20.1	
	critic_loss		0.0898	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		359955	

Eval num_timesteps=365000, episode_reward=55.40 +/- 1.76
 Episode length: 179.50 +/- 115.11

	eval/			
	mean_ep_length		180	
	mean_reward		55.4	
	time/			
	total_timesteps		365000	
	train/			
	actor_loss		-20.4	
	critic_loss		0.125	
	ent_coef		0.1	
	learning_rate		0.0003	

	n_updates		359999	

	rollout/			
	ep_len_mean		153	
	ep_rew_mean		53.3	
	time/			
	episodes		1968	
	fps		110	
	time_elapsed		3313	
	total_timesteps		365677	
	train/			
	actor_loss		-20.4	
	critic_loss		0.0687	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		360676	

	rollout/			
	ep_len_mean		160	
	ep_rew_mean		52.7	
	time/			
	episodes		1972	
	fps		110	
	time_elapsed		3322	
	total_timesteps		366749	
	train/			
	actor_loss		-20.7	
	critic_loss		0.153	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		361748	

	rollout/			
	ep_len_mean		163	
	ep_rew_mean		52.8	
	time/			
	episodes		1976	
	fps		110	
	time_elapsed		3327	
	total_timesteps		367366	
	train/			
	actor_loss		-21.1	
	critic_loss		0.0673	
	ent_coef		0.1	
	learning_rate		0.0003	

	n_updates		362365	

	rollout/			
	ep_len_mean		165	
	ep_rew_mean		52.9	
	time/			
	episodes		1980	
	fps		110	
	time_elapsed		3331	
	total_timesteps		368099	
	train/			
	actor_loss		-19.2	
	critic_loss		0.0826	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		363098	

	rollout/			
	ep_len_mean		166	
	ep_rew_mean		52.9	
	time/			
	episodes		1984	
	fps		110	
	time_elapsed		3339	
	total_timesteps		369005	
	train/			
	actor_loss		-19.9	
	critic_loss		0.0961	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		364004	

	rollout/			
	ep_len_mean		166	
	ep_rew_mean		52.9	
	time/			
	episodes		1988	
	fps		110	
	time_elapsed		3344	
	total_timesteps		369661	
	train/			
	actor_loss		-21.5	
	critic_loss		0.0693	
	ent_coef		0.1	
	learning_rate		0.0003	

	n_updates		364660	
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Eval num_timesteps=370000, episode_reward=56.31 +/- 1.69

Episode length: 150.70 +/- 56.14

	eval/			
	mean_ep_length		151	
	mean_reward		56.3	
	time/			
	total_timesteps		370000	
	train/			
	actor_loss		-20.3	
	critic_loss		0.0756	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		364999	

	rollout/			
	ep_len_mean		166	
	ep_rew_mean		52.9	
	time/			
	episodes		1992	
	fps		110	
	time_elapsed		3350	
	total_timesteps		370245	
	train/			
	actor_loss		-20.4	
	critic_loss		0.0668	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		365244	

	rollout/			
	ep_len_mean		166	
	ep_rew_mean		52.9	
	time/			
	episodes		1996	
	fps		110	
	time_elapsed		3356	
	total_timesteps		371056	
	train/			
	actor_loss		-20.5	
	critic_loss		0.062	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		366055	

rollout/		
ep_len_mean	167	
ep_rew_mean	52.3	
time/		
episodes	2000	
fps	110	
time_elapsed	3363	
total_timesteps	371831	
train/		
actor_loss	-20.4	
critic_loss	0.139	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	366830	

rollout/		
ep_len_mean	167	
ep_rew_mean	52.3	
time/		
episodes	2004	
fps	110	
time_elapsed	3366	
total_timesteps	372385	
train/		
actor_loss	-20.9	
critic_loss	0.113	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	367384	

rollout/		
ep_len_mean	168	
ep_rew_mean	53	
time/		
episodes	2008	
fps	110	
time_elapsed	3376	
total_timesteps	373465	
train/		
actor_loss	-21	
critic_loss	0.0853	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	368464	

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| rollout/          |          |
|   ep_len_mean    | 167      |
|   ep_rew_mean    | 53       |
| time/            |          |
|   episodes       | 2012     |
|   fps            | 110      |
|   time_elapsed   | 3381     |
|   total_timesteps | 374042   |
| train/           |          |
|   actor_loss     | -19.8    |
|   critic_loss    | 0.0581   |
|   ent_coef       | 0.1      |
|   learning_rate  | 0.0003   |
|   n_updates      | 369041   |
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| rollout/          |          |
|   ep_len_mean    | 169      |
|   ep_rew_mean    | 53       |
| time/            |          |
|   episodes       | 2016     |
|   fps            | 110      |
|   time_elapsed   | 3385     |
|   total_timesteps | 374528   |
| train/           |          |
|   actor_loss     | -20.9    |
|   critic_loss    | 0.105    |
|   ent_coef       | 0.1      |
|   learning_rate  | 0.0003   |
|   n_updates      | 369527   |
-----

```

Eval num_timesteps=375000, episode_reward=53.60 +/- 1.98
Episode length: 89.10 +/- 51.58

```

-----
| eval/            |          |
|   mean_ep_length | 89.1     |
|   mean_reward    | 53.6     |
| time/            |          |
|   total_timesteps | 375000   |
| train/           |          |
|   actor_loss     | -20.1    |
|   critic_loss    | 0.0536   |
|   ent_coef       | 0.1      |
|   learning_rate  | 0.0003   |
|   n_updates      | 369999   |
-----

```

rollout/		
ep_len_mean	172	
ep_rew_mean	52.4	
time/		
episodes	2020	
fps	110	
time_elapsed	3393	
total_timesteps	375406	
train/		
actor_loss	-19.9	
critic_loss	0.0386	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	370405	

rollout/		
ep_len_mean	169	
ep_rew_mean	52.4	
time/		
episodes	2024	
fps	110	
time_elapsed	3396	
total_timesteps	375920	
train/		
actor_loss	-21.9	
critic_loss	0.117	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	370919	

rollout/		
ep_len_mean	171	
ep_rew_mean	52.5	
time/		
episodes	2028	
fps	110	
time_elapsed	3401	
total_timesteps	376591	
train/		
actor_loss	-21.5	
critic_loss	0.16	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	371590	

rollout/		
ep_len_mean	174	
ep_rew_mean	52	
time/		
episodes	2032	
fps	110	
time_elapsed	3406	
total_timesteps	377330	
train/		
actor_loss	-20.1	
critic_loss	0.0594	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	372329	

rollout/		
ep_len_mean	174	
ep_rew_mean	52.1	
time/		
episodes	2036	
fps	110	
time_elapsed	3410	
total_timesteps	377717	
train/		
actor_loss	-20.8	
critic_loss	0.0722	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	372716	

rollout/		
ep_len_mean	170	
ep_rew_mean	52.1	
time/		
episodes	2040	
fps	110	
time_elapsed	3413	
total_timesteps	378150	
train/		
actor_loss	-20.5	
critic_loss	0.0684	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	373149	

rollout/		
ep_len_mean	171	
ep_rew_mean	52.1	
time/		
episodes	2044	
fps	110	
time_elapsed	3419	
total_timesteps	378870	
train/		
actor_loss	-21.4	
critic_loss	0.122	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	373869	

rollout/		
ep_len_mean	173	
ep_rew_mean	52.1	
time/		
episodes	2048	
fps	110	
time_elapsed	3423	
total_timesteps	379504	
train/		
actor_loss	-20.4	
critic_loss	0.0767	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	374503	

rollout/		
ep_len_mean	170	
ep_rew_mean	52.1	
time/		
episodes	2052	
fps	110	
time_elapsed	3426	
total_timesteps	379894	
train/		
actor_loss	-20.9	
critic_loss	0.123	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	374893	

Eval num_timesteps=380000, episode_reward=54.18 +/- 2.39
Episode length: 129.50 +/- 134.24

```
-----  
| eval/                |          |  
|   mean_ep_length    | 130      |  
|   mean_reward       | 54.2     |  
| time/               |          |  
|   total_timesteps   | 380000   |  
| train/              |          |  
|   actor_loss        | -20.6    |  
|   critic_loss       | 0.0486   |  
|   ent_coef          | 0.1      |  
|   learning_rate     | 0.0003   |  
|   n_updates         | 374999   |  
-----
```

```
-----  
| rollout/             |          |  
|   ep_len_mean       | 171      |  
|   ep_rew_mean       | 52.1     |  
| time/               |          |  
|   episodes          | 2056     |  
|   fps               | 110      |  
|   time_elapsed      | 3431     |  
|   total_timesteps   | 380348   |  
| train/              |          |  
|   actor_loss        | -21.3    |  
|   critic_loss       | 0.0688   |  
|   ent_coef          | 0.1      |  
|   learning_rate     | 0.0003   |  
|   n_updates         | 375347   |  
-----
```

```
-----  
| rollout/             |          |  
|   ep_len_mean       | 167      |  
|   ep_rew_mean       | 52.1     |  
| time/               |          |  
|   episodes          | 2060     |  
|   fps               | 110      |  
|   time_elapsed      | 3435     |  
|   total_timesteps   | 380743   |  
| train/              |          |  
|   actor_loss        | -21      |  
|   critic_loss       | 0.0809   |  
|   ent_coef          | 0.1      |  
|   learning_rate     | 0.0003   |  
|   n_updates         | 375742   |  
-----  
-----
```

rollout/		
ep_len_mean	163	
ep_rew_mean	52.6	
time/		
episodes	2064	
fps	110	
time_elapsed	3439	
total_timesteps	381255	
train/		
actor_loss	-21.3	
critic_loss	0.161	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	376254	

rollout/		
ep_len_mean	161	
ep_rew_mean	52.6	
time/		
episodes	2068	
fps	110	
time_elapsed	3443	
total_timesteps	381742	
train/		
actor_loss	-21.4	
critic_loss	0.0858	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	376741	

rollout/		
ep_len_mean	155	
ep_rew_mean	53.2	
time/		
episodes	2072	
fps	110	
time_elapsed	3447	
total_timesteps	382261	
train/		
actor_loss	-21.7	
critic_loss	0.0946	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	377260	

rollout/		
ep_len_mean	153	
ep_rew_mean	53.2	
time/		
episodes	2076	
fps	110	
time_elapsed	3450	
total_timesteps	382667	
train/		
actor_loss	-19.9	
critic_loss	0.143	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	377666	

rollout/		
ep_len_mean	150	
ep_rew_mean	53.1	
time/		
episodes	2080	
fps	110	
time_elapsed	3454	
total_timesteps	383120	
train/		
actor_loss	-21.7	
critic_loss	0.0861	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	378119	

rollout/		
ep_len_mean	146	
ep_rew_mean	53.1	
time/		
episodes	2084	
fps	110	
time_elapsed	3456	
total_timesteps	383590	
train/		
actor_loss	-20.5	
critic_loss	0.0697	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	378589	

rollout/		
ep_len_mean	147	
ep_rew_mean	53.1	
time/		
episodes	2088	
fps	110	
time_elapsed	3462	
total_timesteps	384317	
train/		
actor_loss	-20.5	
critic_loss	0.085	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	379316	

rollout/		
ep_len_mean	144	
ep_rew_mean	53	
time/		
episodes	2092	
fps	110	
time_elapsed	3465	
total_timesteps	384662	
train/		
actor_loss	-19.7	
critic_loss	0.142	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	379661	

Eval num_timesteps=385000, episode_reward=54.65 +/- 2.02
Episode length: 102.80 +/- 52.67

eval/		
mean_ep_length	103	
mean_reward	54.6	
time/		
total_timesteps	385000	
train/		
actor_loss	-21.1	
critic_loss	0.0884	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	379999	

rollout/		
----------	--	--

	ep_len_mean		141	
	ep_rew_mean		53	
	time/			
	episodes		2096	
	fps		110	
	time_elapsed		3470	
	total_timesteps		385134	
	train/			
	actor_loss		-21.1	
	critic_loss		0.157	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		380133	

	rollout/			
	ep_len_mean		135	
	ep_rew_mean		53.5	
	time/			
	episodes		2100	
	fps		110	
	time_elapsed		3472	
	total_timesteps		385363	
	train/			
	actor_loss		-21.8	
	critic_loss		0.198	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		380362	

	rollout/			
	ep_len_mean		134	
	ep_rew_mean		53.4	
	time/			
	episodes		2104	
	fps		111	
	time_elapsed		3475	
	total_timesteps		385812	
	train/			
	actor_loss		-20.3	
	critic_loss		0.0669	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		380811	

	rollout/			
--	----------	--	--	--

	ep_len_mean		130	
	ep_rew_mean		53.3	
	time/			
	episodes		2108	
	fps		111	
	time_elapsed		3481	
	total_timesteps		386463	
	train/			
	actor_loss		-20.8	
	critic_loss		0.0788	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		381462	

	rollout/			
	ep_len_mean		128	
	ep_rew_mean		53.3	
	time/			
	episodes		2112	
	fps		111	
	time_elapsed		3484	
	total_timesteps		386840	
	train/			
	actor_loss		-21.5	
	critic_loss		0.063	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		381839	

	rollout/			
	ep_len_mean		130	
	ep_rew_mean		53.2	
	time/			
	episodes		2116	
	fps		111	
	time_elapsed		3488	
	total_timesteps		387539	
	train/			
	actor_loss		-21.4	
	critic_loss		0.0774	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		382538	

	rollout/			
--	----------	--	--	--

	ep_len_mean		127	
	ep_rew_mean		53.8	
	time/			
	episodes		2120	
	fps		111	
	time_elapsed		3493	
	total_timesteps		388081	
	train/			
	actor_loss		-19.9	
	critic_loss		0.072	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		383080	

	rollout/			
	ep_len_mean		132	
	ep_rew_mean		53.3	
	time/			
	episodes		2124	
	fps		111	
	time_elapsed		3501	
	total_timesteps		389114	
	train/			
	actor_loss		-21.3	
	critic_loss		0.0685	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		384113	

	rollout/			
	ep_len_mean		129	
	ep_rew_mean		53.2	
	time/			
	episodes		2128	
	fps		111	
	time_elapsed		3505	
	total_timesteps		389526	
	train/			
	actor_loss		-21.1	
	critic_loss		0.108	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		384525	

Eval num_timesteps=390000, episode_reward=54.98 +/- 2.05
Episode length: 108.30 +/- 52.73

eval/		
mean_ep_length	108	
mean_reward	55	
time/		
total_timesteps	390000	
train/		
actor_loss	-20.7	
critic_loss	0.0851	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	384999	

rollout/		
ep_len_mean	130	
ep_rew_mean	53.7	
time/		
episodes	2132	
fps	111	
time_elapsed	3513	
total_timesteps	390318	
train/		
actor_loss	-21.9	
critic_loss	0.0883	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	385317	

rollout/		
ep_len_mean	133	
ep_rew_mean	53.8	
time/		
episodes	2136	
fps	111	
time_elapsed	3517	
total_timesteps	391005	
train/		
actor_loss	-21.9	
critic_loss	0.103	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	386004	

rollout/		
ep_len_mean	135	

	ep_rew_mean		53.9	
	time/			
	episodes		2140	
	fps		111	
	time_elapsed		3523	
	total_timesteps		391660	
	train/			
	actor_loss		-22	
	critic_loss		0.092	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		386659	

	rollout/			
	ep_len_mean		133	
	ep_rew_mean		53.8	
	time/			
	episodes		2144	
	fps		111	
	time_elapsed		3528	
	total_timesteps		392142	
	train/			
	actor_loss		-21.9	
	critic_loss		0.159	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		387141	

	rollout/			
	ep_len_mean		130	
	ep_rew_mean		53.8	
	time/			
	episodes		2148	
	fps		111	
	time_elapsed		3531	
	total_timesteps		392510	
	train/			
	actor_loss		-20.6	
	critic_loss		0.0962	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		387509	

	rollout/			
	ep_len_mean		130	

	ep_rew_mean		53.8	
	time/			
	episodes		2152	
	fps		111	
	time_elapsed		3535	
	total_timesteps		392930	
	train/			
	actor_loss		-20.8	
	critic_loss		0.0765	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		387929	

	rollout/			
	ep_len_mean		130	
	ep_rew_mean		53.8	
	time/			
	episodes		2156	
	fps		111	
	time_elapsed		3539	
	total_timesteps		393375	
	train/			
	actor_loss		-20.4	
	critic_loss		0.0873	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		388374	

	rollout/			
	ep_len_mean		132	
	ep_rew_mean		53.8	
	time/			
	episodes		2160	
	fps		111	
	time_elapsed		3544	
	total_timesteps		393905	
	train/			
	actor_loss		-20	
	critic_loss		0.0814	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		388904	

	rollout/			
	ep_len_mean		133	

	ep_rew_mean		53.8	
	time/			
	episodes		2164	
	fps		111	
	time_elapsed		3547	
	total_timesteps		394520	
	train/			
	actor_loss		-20.9	
	critic_loss		0.114	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		389519	

Eval num_timesteps=395000, episode_reward=48.68 +/- 17.09
Episode length: 145.60 +/- 129.74

	eval/			
	mean_ep_length		146	
	mean_reward		48.7	
	time/			
	total_timesteps		395000	
	train/			
	actor_loss		-20.7	
	critic_loss		0.0795	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		389999	

	rollout/			
	ep_len_mean		134	
	ep_rew_mean		53.8	
	time/			
	episodes		2168	
	fps		111	
	time_elapsed		3553	
	total_timesteps		395125	
	train/			
	actor_loss		-21.4	
	critic_loss		0.209	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		390124	

	rollout/			
	ep_len_mean		133	
	ep_rew_mean		53.8	

time/		
episodes	2172	
fps	111	
time_elapsed	3557	
total_timesteps	395527	
train/		
actor_loss	-20.8	
critic_loss	0.0458	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	390526	

rollout/		
ep_len_mean	135	
ep_rew_mean	53.9	
time/		
episodes	2176	
fps	111	
time_elapsed	3562	
total_timesteps	396135	
train/		
actor_loss	-20	
critic_loss	0.042	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	391134	

rollout/		
ep_len_mean	137	
ep_rew_mean	53.9	
time/		
episodes	2180	
fps	111	
time_elapsed	3568	
total_timesteps	396869	
train/		
actor_loss	-20.4	
critic_loss	0.0964	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	391868	

rollout/		
ep_len_mean	139	
ep_rew_mean	53.9	

time/		
episodes	2184	
fps	111	
time_elapsed	3574	
total_timesteps	397494	
train/		
actor_loss	-21.9	
critic_loss	0.0856	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	392493	

rollout/		
ep_len_mean	136	
ep_rew_mean	53.9	
time/		
episodes	2188	
fps	111	
time_elapsed	3575	
total_timesteps	397886	
train/		
actor_loss	-21.1	
critic_loss	0.155	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	392885	

rollout/		
ep_len_mean	138	
ep_rew_mean	53.9	
time/		
episodes	2192	
fps	111	
time_elapsed	3580	
total_timesteps	398470	
train/		
actor_loss	-20.9	
critic_loss	0.0802	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	393469	

rollout/		
ep_len_mean	139	
ep_rew_mean	53.9	

time/		
episodes	2196	
fps	111	
time_elapsed	3584	
total_timesteps	399003	
train/		
actor_loss	-20.9	
critic_loss	0.112	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	394002	

rollout/		
ep_len_mean	140	
ep_rew_mean	54	
time/		
episodes	2200	
fps	111	
time_elapsed	3587	
total_timesteps	399381	
train/		
actor_loss	-21.1	
critic_loss	0.113	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	394380	

Eval num_timesteps=400000, episode_reward=54.83 +/- 2.40
 Episode length: 113.90 +/- 64.07

eval/		
mean_ep_length	114	
mean_reward	54.8	
time/		
total_timesteps	400000	
train/		
actor_loss	-20.5	
critic_loss	0.105	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	394999	

rollout/		
ep_len_mean	144	
ep_rew_mean	54.1	
time/		

	episodes		2204	
	fps		111	
	time_elapsed		3596	
	total_timesteps		400183	
	train/			
	actor_loss		-21.3	
	critic_loss		0.0698	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		395182	

	rollout/			
	ep_len_mean		147	
	ep_rew_mean		54.2	
	time/			
	episodes		2208	
	fps		111	
	time_elapsed		3605	
	total_timesteps		401206	
	train/			
	actor_loss		-21.2	
	critic_loss		0.0861	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		396205	

	rollout/			
	ep_len_mean		153	
	ep_rew_mean		54.3	
	time/			
	episodes		2212	
	fps		111	
	time_elapsed		3611	
	total_timesteps		402092	
	train/			
	actor_loss		-21.8	
	critic_loss		0.0628	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		397091	

	rollout/			
	ep_len_mean		150	
	ep_rew_mean		54.3	
	time/			

	episodes		2216	
	fps		111	
	time_elapsed		3615	
	total_timesteps		402523	
	train/			
	actor_loss		-21.3	
	critic_loss		0.0514	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		397522	

	rollout/			
	ep_len_mean		153	
	ep_rew_mean		54.3	
	time/			
	episodes		2220	
	fps		111	
	time_elapsed		3622	
	total_timesteps		403405	
	train/			
	actor_loss		-19.7	
	critic_loss		0.0659	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		398404	

	rollout/			
	ep_len_mean		148	
	ep_rew_mean		54.8	
	time/			
	episodes		2224	
	fps		111	
	time_elapsed		3626	
	total_timesteps		403875	
	train/			
	actor_loss		-22.4	
	critic_loss		0.0866	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		398874	

	rollout/			
	ep_len_mean		149	
	ep_rew_mean		54.8	
	time/			

	episodes		2228	
	fps		111	
	time_elapsed		3631	
	total_timesteps		404466	
	train/			
	actor_loss		-22.2	
	critic_loss		0.0718	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		399465	

Eval num_timesteps=405000, episode_reward=54.58 +/- 2.46
Episode length: 161.50 +/- 145.63

	eval/			
	mean_ep_length		162	
	mean_reward		54.6	
	time/			
	total_timesteps		405000	
	train/			
	actor_loss		-21.2	
	critic_loss		0.0989	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		399999	

	rollout/			
	ep_len_mean		147	
	ep_rew_mean		54.8	
	time/			
	episodes		2232	
	fps		111	
	time_elapsed		3635	
	total_timesteps		405028	
	train/			
	actor_loss		-20.9	
	critic_loss		0.0732	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		400027	

	rollout/			
	ep_len_mean		142	
	ep_rew_mean		54.6	
	time/			
	episodes		2236	

	fps		111	
	time_elapsed		3637	
	total_timesteps		405228	
	train/			
	actor_loss		-21	
	critic_loss		0.0742	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		400227	

	rollout/			
	ep_len_mean		144	
	ep_rew_mean		54.7	
	time/			
	episodes		2240	
	fps		111	
	time_elapsed		3645	
	total_timesteps		406074	
	train/			
	actor_loss		-22.5	
	critic_loss		0.124	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		401073	

	rollout/			
	ep_len_mean		148	
	ep_rew_mean		54.7	
	time/			
	episodes		2244	
	fps		111	
	time_elapsed		3653	
	total_timesteps		406933	
	train/			
	actor_loss		-20.1	
	critic_loss		0.0691	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		401932	

	rollout/			
	ep_len_mean		148	
	ep_rew_mean		54.7	
	time/			
	episodes		2248	

	fps		111	
	time_elapsed		3656	
	total_timesteps		407280	
	train/			
	actor_loss		-21.1	
	critic_loss		0.0969	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		402279	

	rollout/			
	ep_len_mean		152	
	ep_rew_mean		54.7	
	time/			
	episodes		2252	
	fps		111	
	time_elapsed		3663	
	total_timesteps		408129	
	train/			
	actor_loss		-21	
	critic_loss		0.0733	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		403128	

	rollout/			
	ep_len_mean		153	
	ep_rew_mean		54.7	
	time/			
	episodes		2256	
	fps		111	
	time_elapsed		3666	
	total_timesteps		408692	
	train/			
	actor_loss		-21.4	
	critic_loss		0.245	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		403691	

	rollout/			
	ep_len_mean		154	
	ep_rew_mean		54.7	
	time/			
	episodes		2260	

	fps		111	
	time_elapsed		3671	
	total_timesteps		409264	
	train/			
	actor_loss		-20.2	
	critic_loss		0.0708	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		404263	

	rollout/			
	ep_len_mean		151	
	ep_rew_mean		54.7	
	time/			
	episodes		2264	
	fps		111	
	time_elapsed		3674	
	total_timesteps		409652	
	train/			
	actor_loss		-20.6	
	critic_loss		0.123	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		404651	

Eval num_timesteps=410000, episode_reward=54.75 +/- 2.27
Episode length: 126.10 +/- 77.47

	eval/			
	mean_ep_length		126	
	mean_reward		54.7	
	time/			
	total_timesteps		410000	
	train/			
	actor_loss		-21.5	
	critic_loss		0.0943	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		404999	

	rollout/			
	ep_len_mean		152	
	ep_rew_mean		54.7	
	time/			
	episodes		2268	
	fps		111	

	time_elapsed		3681	
	total_timesteps		410357	
	train/			
	actor_loss		-22	
	critic_loss		0.0852	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		405356	

	rollout/			
	ep_len_mean		154	
	ep_rew_mean		54.8	
	time/			
	episodes		2272	
	fps		111	
	time_elapsed		3686	
	total_timesteps		410965	
	train/			
	actor_loss		-20.9	
	critic_loss		0.124	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		405964	

	rollout/			
	ep_len_mean		152	
	ep_rew_mean		54.7	
	time/			
	episodes		2276	
	fps		111	
	time_elapsed		3690	
	total_timesteps		411342	
	train/			
	actor_loss		-22.4	
	critic_loss		0.125	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		406341	

	rollout/			
	ep_len_mean		151	
	ep_rew_mean		54.7	
	time/			
	episodes		2280	
	fps		111	

	time_elapsed		3695	
	total_timesteps		411998	
	train/			
	actor_loss		-22.2	
	critic_loss		0.124	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		406997	

	rollout/			
	ep_len_mean		152	
	ep_rew_mean		54.8	
	time/			
	episodes		2284	
	fps		111	
	time_elapsed		3699	
	total_timesteps		412726	
	train/			
	actor_loss		-21.6	
	critic_loss		0.188	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		407725	

	rollout/			
	ep_len_mean		152	
	ep_rew_mean		54.7	
	time/			
	episodes		2288	
	fps		111	
	time_elapsed		3703	
	total_timesteps		413131	
	train/			
	actor_loss		-20.8	
	critic_loss		0.0712	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		408130	

	rollout/			
	ep_len_mean		152	
	ep_rew_mean		54.8	
	time/			
	episodes		2292	
	fps		111	

	time_elapsed		3707	
	total_timesteps		413637	
	train/			
	actor_loss		-21.2	
	critic_loss		0.0749	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		408636	

	rollout/			
	ep_len_mean		152	
	ep_rew_mean		54.7	
	time/			
	episodes		2296	
	fps		111	
	time_elapsed		3712	
	total_timesteps		414202	
	train/			
	actor_loss		-22.8	
	critic_loss		0.0711	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		409201	

	rollout/			
	ep_len_mean		153	
	ep_rew_mean		54.8	
	time/			
	episodes		2300	
	fps		111	
	time_elapsed		3716	
	total_timesteps		414658	
	train/			
	actor_loss		-21.9	
	critic_loss		0.137	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		409657	

Eval num_timesteps=415000, episode_reward=54.59 +/- 1.86
Episode length: 118.30 +/- 85.68

	eval/			
	mean_ep_length		118	
	mean_reward		54.6	
	time/			

	total_timesteps		415000	
	train/			
	actor_loss		-21.5	
	critic_loss		0.0528	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		409999	

	rollout/			
	ep_len_mean		152	
	ep_rew_mean		54.7	
	time/			
	episodes		2304	
	fps		111	
	time_elapsed		3723	
	total_timesteps		415433	
	train/			
	actor_loss		-21	
	critic_loss		0.1	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		410432	

	rollout/			
	ep_len_mean		145	
	ep_rew_mean		54.6	
	time/			
	episodes		2308	
	fps		111	
	time_elapsed		3726	
	total_timesteps		415755	
	train/			
	actor_loss		-21.6	
	critic_loss		0.0989	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		410754	

	rollout/			
	ep_len_mean		145	
	ep_rew_mean		54	
	time/			
	episodes		2312	
	fps		111	
	time_elapsed		3732	

	total_timesteps		416594	
	train/			
	actor_loss		-21	
	critic_loss		0.0673	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		411593	

	rollout/			
	ep_len_mean		145	
	ep_rew_mean		54	
	time/			
	episodes		2316	
	fps		111	
	time_elapsed		3736	
	total_timesteps		417064	
	train/			
	actor_loss		-21.6	
	critic_loss		0.0837	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		412063	

	rollout/			
	ep_len_mean		142	
	ep_rew_mean		54	
	time/			
	episodes		2320	
	fps		111	
	time_elapsed		3740	
	total_timesteps		417568	
	train/			
	actor_loss		-21.8	
	critic_loss		0.13	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		412567	

	rollout/			
	ep_len_mean		147	
	ep_rew_mean		54.1	
	time/			
	episodes		2324	
	fps		111	
	time_elapsed		3748	

	total_timesteps		418530	
	train/			
	actor_loss		-22.3	
	critic_loss		0.0961	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		413529	

	rollout/			
	ep_len_mean		147	
	ep_rew_mean		54.1	
	time/			
	episodes		2328	
	fps		111	
	time_elapsed		3754	
	total_timesteps		419144	
	train/			
	actor_loss		-21.2	
	critic_loss		0.0455	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		414143	

	rollout/			
	ep_len_mean		147	
	ep_rew_mean		54.1	
	time/			
	episodes		2332	
	fps		111	
	time_elapsed		3757	
	total_timesteps		419726	
	train/			
	actor_loss		-22.1	
	critic_loss		0.0496	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		414725	

Eval num_timesteps=420000, episode_reward=55.58 +/- 1.78
Episode length: 112.60 +/- 41.23

	eval/			
	mean_ep_length		113	
	mean_reward		55.6	
	time/			
	total_timesteps		420000	

train/		
actor_loss	-21.2	
critic_loss	0.0618	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	414999	

rollout/		
ep_len_mean	151	
ep_rew_mean	54.3	
time/		
episodes	2336	
fps	111	
time_elapsed	3764	
total_timesteps	420360	
train/		
actor_loss	-22	
critic_loss	0.0797	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	415359	

rollout/		
ep_len_mean	145	
ep_rew_mean	54.1	
time/		
episodes	2340	
fps	111	
time_elapsed	3766	
total_timesteps	420603	
train/		
actor_loss	-22.2	
critic_loss	0.0955	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	415602	

rollout/		
ep_len_mean	142	
ep_rew_mean	54.1	
time/		
episodes	2344	
fps	111	
time_elapsed	3771	
total_timesteps	421182	

train/		
actor_loss	-21.8	
critic_loss	0.0765	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	416181	

rollout/		
ep_len_mean	144	
ep_rew_mean	54.2	
time/		
episodes	2348	
fps	111	
time_elapsed	3775	
total_timesteps	421688	

train/		
actor_loss	-22	
critic_loss	0.12	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	416687	

rollout/		
ep_len_mean	141	
ep_rew_mean	54.2	
time/		
episodes	2352	
fps	111	
time_elapsed	3779	
total_timesteps	422187	

train/		
actor_loss	-22.5	
critic_loss	0.127	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	417186	

rollout/		
ep_len_mean	141	
ep_rew_mean	54.2	
time/		
episodes	2356	
fps	111	
time_elapsed	3785	
total_timesteps	422830	

train/		
actor_loss	-21.8	
critic_loss	0.101	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	417829	

rollout/		
ep_len_mean	142	
ep_rew_mean	54.2	
time/		
episodes	2360	
fps	111	
time_elapsed	3789	
total_timesteps	423449	
train/		
actor_loss	-21.5	
critic_loss	0.0937	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	418448	

rollout/		
ep_len_mean	150	
ep_rew_mean	53.7	
time/		
episodes	2364	
fps	111	
time_elapsed	3799	
total_timesteps	424614	
train/		
actor_loss	-21.5	
critic_loss	0.0513	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	419613	

Eval num_timesteps=425000, episode_reward=54.68 +/- 1.90
 Episode length: 102.20 +/- 51.79

eval/		
mean_ep_length	102	
mean_reward	54.7	
time/		
total_timesteps	425000	
train/		

	actor_loss		-22	
	critic_loss		0.117	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		419999	

	rollout/			
	ep_len_mean		148	
	ep_rew_mean		53.7	
	time/			
	episodes		2368	
	fps		111	
	time_elapsed		3804	
	total_timesteps		425138	
	train/			
	actor_loss		-21.9	
	critic_loss		0.055	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		420137	

	rollout/			
	ep_len_mean		144	
	ep_rew_mean		53.6	
	time/			
	episodes		2372	
	fps		111	
	time_elapsed		3806	
	total_timesteps		425376	
	train/			
	actor_loss		-21.3	
	critic_loss		0.051	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		420375	

	rollout/			
	ep_len_mean		145	
	ep_rew_mean		53.6	
	time/			
	episodes		2376	
	fps		111	
	time_elapsed		3810	
	total_timesteps		425820	
	train/			

	actor_loss		-21.9	
	critic_loss		0.0902	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		420819	

	rollout/			
	ep_len_mean		143	
	ep_rew_mean		53.6	
	time/			
	episodes		2380	
	fps		111	
	time_elapsed		3813	
	total_timesteps		426280	
	train/			
	actor_loss		-22.8	
	critic_loss		0.0605	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		421279	

	rollout/			
	ep_len_mean		140	
	ep_rew_mean		53.5	
	time/			
	episodes		2384	
	fps		111	
	time_elapsed		3817	
	total_timesteps		426707	
	train/			
	actor_loss		-22.3	
	critic_loss		0.0835	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		421706	

	rollout/			
	ep_len_mean		140	
	ep_rew_mean		53.6	
	time/			
	episodes		2388	
	fps		111	
	time_elapsed		3818	
	total_timesteps		427146	
	train/			

	actor_loss		-21.6	
	critic_loss		0.109	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		422145	

	rollout/			
	ep_len_mean		143	
	ep_rew_mean		53.7	
	time/			
	episodes		2392	
	fps		111	
	time_elapsed		3825	
	total_timesteps		427918	
	train/			
	actor_loss		-21.9	
	critic_loss		0.101	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		422917	

	rollout/			
	ep_len_mean		143	
	ep_rew_mean		53.7	
	time/			
	episodes		2396	
	fps		111	
	time_elapsed		3829	
	total_timesteps		428461	
	train/			
	actor_loss		-21.3	
	critic_loss		0.26	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		423460	

	rollout/			
	ep_len_mean		143	
	ep_rew_mean		53.7	
	time/			
	episodes		2400	
	fps		111	
	time_elapsed		3833	
	total_timesteps		428927	
	train/			

	actor_loss		-23.3	
	critic_loss		0.107	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		423926	

	rollout/			
	ep_len_mean		145	
	ep_rew_mean		53.1	
	time/			
	episodes		2404	
	fps		111	
	time_elapsed		3842	
	total_timesteps		429917	
	train/			
	actor_loss		-21.1	
	critic_loss		0.0466	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		424916	

Eval num_timesteps=430000, episode_reward=54.25 +/- 2.09
 Episode length: 142.80 +/- 122.44

	eval/			
	mean_ep_length		143	
	mean_reward		54.2	
	time/			
	total_timesteps		430000	
	train/			
	actor_loss		-22.6	
	critic_loss		0.065	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		424999	

	rollout/			
	ep_len_mean		147	
	ep_rew_mean		53.2	
	time/			
	episodes		2408	
	fps		111	
	time_elapsed		3846	
	total_timesteps		430441	
	train/			
	actor_loss		-21.7	

	critic_loss		0.0527	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		425440	

	rollout/			
	ep_len_mean		143	
	ep_rew_mean		53.7	
	time/			
	episodes		2412	
	fps		111	
	time_elapsed		3849	
	total_timesteps		430875	
	train/			
	actor_loss		-21.8	
	critic_loss		0.0545	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		425874	

	rollout/			
	ep_len_mean		149	
	ep_rew_mean		53.8	
	time/			
	episodes		2416	
	fps		111	
	time_elapsed		3858	
	total_timesteps		432008	
	train/			
	actor_loss		-22	
	critic_loss		0.0692	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		427007	

	rollout/			
	ep_len_mean		155	
	ep_rew_mean		53.4	
	time/			
	episodes		2420	
	fps		111	
	time_elapsed		3868	
	total_timesteps		433112	
	train/			
	actor_loss		-22.3	

	critic_loss		0.091	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		428111	

	rollout/			
	ep_len_mean		152	
	ep_rew_mean		53.3	
	time/			
	episodes		2424	
	fps		111	
	time_elapsed		3873	
	total_timesteps		433765	
	train/			
	actor_loss		-21.1	
	critic_loss		0.0733	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		428764	

	rollout/			
	ep_len_mean		151	
	ep_rew_mean		53.2	
	time/			
	episodes		2428	
	fps		111	
	time_elapsed		3877	
	total_timesteps		434197	
	train/			
	actor_loss		-22.1	
	critic_loss		0.0441	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		429196	

	rollout/			
	ep_len_mean		151	
	ep_rew_mean		53.3	
	time/			
	episodes		2432	
	fps		112	
	time_elapsed		3881	
	total_timesteps		434842	
	train/			
	actor_loss		-20.8	

	critic_loss		0.0489	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		429841	

Eval num_timesteps=435000, episode_reward=49.80 +/- 18.35
 Episode length: 175.50 +/- 129.19

	eval/			
	mean_ep_length		176	
	mean_reward		49.8	
	time/			
	total_timesteps		435000	
	train/			
	actor_loss		-21.6	
	critic_loss		0.0709	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		429999	

	rollout/			
	ep_len_mean		154	
	ep_rew_mean		53.2	
	time/			
	episodes		2436	
	fps		111	
	time_elapsed		3891	
	total_timesteps		435810	
	train/			
	actor_loss		-20.7	
	critic_loss		0.211	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		430809	

	rollout/			
	ep_len_mean		157	
	ep_rew_mean		53.3	
	time/			
	episodes		2440	
	fps		111	
	time_elapsed		3895	
	total_timesteps		436322	
	train/			
	actor_loss		-20.6	
	critic_loss		0.118	

	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		431321	

	rollout/			
	ep_len_mean		155	
	ep_rew_mean		53.3	
	time/			
	episodes		2444	
	fps		112	
	time_elapsed		3898	
	total_timesteps		436691	
	train/			
	actor_loss		-22.7	
	critic_loss		0.104	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		431690	

	rollout/			
	ep_len_mean		155	
	ep_rew_mean		53.2	
	time/			
	episodes		2448	
	fps		112	
	time_elapsed		3902	
	total_timesteps		437217	
	train/			
	actor_loss		-21.1	
	critic_loss		0.0638	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		432216	

	rollout/			
	ep_len_mean		154	
	ep_rew_mean		53.2	
	time/			
	episodes		2452	
	fps		112	
	time_elapsed		3905	
	total_timesteps		437617	
	train/			
	actor_loss		-21.7	
	critic_loss		0.06	

	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		432616	

	rollout/			
	ep_len_mean		153	
	ep_rew_mean		53.2	
	time/			
	episodes		2456	
	fps		112	
	time_elapsed		3907	
	total_timesteps		438140	
	train/			
	actor_loss		-21	
	critic_loss		0.0798	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		433139	

	rollout/			
	ep_len_mean		155	
	ep_rew_mean		53.3	
	time/			
	episodes		2460	
	fps		112	
	time_elapsed		3914	
	total_timesteps		438951	
	train/			
	actor_loss		-22.5	
	critic_loss		0.0923	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		433950	

	rollout/			
	ep_len_mean		151	
	ep_rew_mean		53.8	
	time/			
	episodes		2464	
	fps		112	
	time_elapsed		3920	
	total_timesteps		439708	
	train/			
	actor_loss		-21.8	
	critic_loss		0.0875	

	ent_coef	0.1	
	learning_rate	0.0003	
	n_updates	434707	

Eval num_timesteps=440000, episode_reward=56.29 +/- 2.00
Episode length: 152.80 +/- 55.65

	eval/		
	mean_ep_length	153	
	mean_reward	56.3	
	time/		
	total_timesteps	440000	
	train/		
	actor_loss	-22.5	
	critic_loss	0.0923	
	ent_coef	0.1	
	learning_rate	0.0003	
	n_updates	434999	

	rollout/		
	ep_len_mean	154	
	ep_rew_mean	53.2	
	time/		
	episodes	2468	
	fps	112	
	time_elapsed	3928	
	total_timesteps	440547	
	train/		
	actor_loss	-21.5	
	critic_loss	0.0695	
	ent_coef	0.1	
	learning_rate	0.0003	
	n_updates	435546	

	rollout/		
	ep_len_mean	155	
	ep_rew_mean	53.3	
	time/		
	episodes	2472	
	fps	112	
	time_elapsed	3931	
	total_timesteps	440884	
	train/		
	actor_loss	-22.6	
	critic_loss	0.0791	
	ent_coef	0.1	

learning_rate	0.0003
n_updates	435883

rollout/	
ep_len_mean	157
ep_rew_mean	53.3
time/	
episodes	2476
fps	112
time_elapsed	3936
total_timesteps	441495
train/	
actor_loss	-22.1
critic_loss	0.0408
ent_coef	0.1
learning_rate	0.0003
n_updates	436494

rollout/	
ep_len_mean	159
ep_rew_mean	53.3
time/	
episodes	2480
fps	112
time_elapsed	3940
total_timesteps	442161
train/	
actor_loss	-21.6
critic_loss	0.0881
ent_coef	0.1
learning_rate	0.0003
n_updates	437160

rollout/	
ep_len_mean	159
ep_rew_mean	53.3
time/	
episodes	2484
fps	112
time_elapsed	3944
total_timesteps	442614
train/	
actor_loss	-22.6
critic_loss	0.0773
ent_coef	0.1

learning_rate	0.0003
n_updates	437613

rollout/	
ep_len_mean	163
ep_rew_mean	53.3
time/	
episodes	2488
fps	112
time_elapsed	3951
total_timesteps	443431
train/	
actor_loss	-22.4
critic_loss	0.118
ent_coef	0.1
learning_rate	0.0003
n_updates	438430

rollout/	
ep_len_mean	159
ep_rew_mean	53.2
time/	
episodes	2492
fps	112
time_elapsed	3954
total_timesteps	443790
train/	
actor_loss	-21.9
critic_loss	0.088
ent_coef	0.1
learning_rate	0.0003
n_updates	438789

rollout/	
ep_len_mean	160
ep_rew_mean	53.2
time/	
episodes	2496
fps	112
time_elapsed	3960
total_timesteps	444438
train/	
actor_loss	-22.9
critic_loss	0.0584
ent_coef	0.1

	learning_rate		0.0003	
	n_updates		439437	

	rollout/			
	ep_len_mean		160	
	ep_rew_mean		53.2	
	time/			
	episodes		2500	
	fps		112	
	time_elapsed		3964	
	total_timesteps		444962	
	train/			
	actor_loss		-22.6	
	critic_loss		0.0754	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		439961	

Eval num_timesteps=445000, episode_reward=50.42 +/- 17.76
 Episode length: 167.70 +/- 118.06

	eval/			
	mean_ep_length		168	
	mean_reward		50.4	
	time/			
	total_timesteps		445000	
	train/			
	actor_loss		-21.4	
	critic_loss		0.0669	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		439999	

	rollout/			
	ep_len_mean		157	
	ep_rew_mean		53.8	
	time/			
	episodes		2504	
	fps		112	
	time_elapsed		3968	
	total_timesteps		445569	
	train/			
	actor_loss		-21.9	
	critic_loss		0.0928	
	ent_coef		0.1	
	learning_rate		0.0003	

	n_updates		440568	

	rollout/			
	ep_len_mean		157	
	ep_rew_mean		53.8	
	time/			
	episodes		2508	
	fps		112	
	time_elapsed		3973	
	total_timesteps		446176	
	train/			
	actor_loss		-23.1	
	critic_loss		0.112	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		441175	

	rollout/			
	ep_len_mean		160	
	ep_rew_mean		53.2	
	time/			
	episodes		2512	
	fps		112	
	time_elapsed		3979	
	total_timesteps		446872	
	train/			
	actor_loss		-21.7	
	critic_loss		0.0766	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		441871	

	rollout/			
	ep_len_mean		155	
	ep_rew_mean		53.2	
	time/			
	episodes		2516	
	fps		112	
	time_elapsed		3984	
	total_timesteps		447510	
	train/			
	actor_loss		-20.6	
	critic_loss		0.0761	
	ent_coef		0.1	
	learning_rate		0.0003	

	n_updates		442509	

	rollout/			
	ep_len_mean		148	
	ep_rew_mean		53.7	
	time/			
	episodes		2520	
	fps		112	
	time_elapsed		3988	
	total_timesteps		447926	
	train/			
	actor_loss		-20.8	
	critic_loss		0.0706	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		442925	

	rollout/			
	ep_len_mean		146	
	ep_rew_mean		53.7	
	time/			
	episodes		2524	
	fps		112	
	time_elapsed		3992	
	total_timesteps		448412	
	train/			
	actor_loss		-21	
	critic_loss		0.0563	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		443411	

	rollout/			
	ep_len_mean		150	
	ep_rew_mean		53.8	
	time/			
	episodes		2528	
	fps		112	
	time_elapsed		3997	
	total_timesteps		449181	
	train/			
	actor_loss		-22	
	critic_loss		0.248	
	ent_coef		0.1	
	learning_rate		0.0003	

	n_updates		444180	
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Eval num_timesteps=450000, episode_reward=55.60 +/- 1.41

Episode length: 123.30 +/- 34.07

	eval/			
	mean_ep_length		123	
	mean_reward		55.6	
	time/			
	total_timesteps		450000	
	train/			
	actor_loss		-21.9	
	critic_loss		0.089	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		444999	

	rollout/			
	ep_len_mean		152	
	ep_rew_mean		53.8	
	time/			
	episodes		2532	
	fps		112	
	time_elapsed		4005	
	total_timesteps		450022	
	train/			
	actor_loss		-21.9	
	critic_loss		0.0612	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		445021	

	rollout/			
	ep_len_mean		147	
	ep_rew_mean		53.8	
	time/			
	episodes		2536	
	fps		112	
	time_elapsed		4009	
	total_timesteps		450501	
	train/			
	actor_loss		-22.1	
	critic_loss		0.0585	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		445500	

rollout/		
ep_len_mean	149	
ep_rew_mean	53.8	
time/		
episodes	2540	
fps	112	
time_elapsed	4015	
total_timesteps	451236	
train/		
actor_loss	-22.2	
critic_loss	0.0803	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	446235	

rollout/		
ep_len_mean	153	
ep_rew_mean	53.9	
time/		
episodes	2544	
fps	112	
time_elapsed	4021	
total_timesteps	451972	
train/		
actor_loss	-21.6	
critic_loss	0.0526	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	446971	

rollout/		
ep_len_mean	155	
ep_rew_mean	53.9	
time/		
episodes	2548	
fps	112	
time_elapsed	4028	
total_timesteps	452749	
train/		
actor_loss	-22.8	
critic_loss	0.0901	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	447748	

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| rollout/          |          |
|   ep_len_mean    | 157      |
|   ep_rew_mean    | 53.9     |
| time/            |          |
|   episodes       | 2552     |
|   fps            | 112      |
|   time_elapsed   | 4031     |
|   total_timesteps | 453293   |
| train/           |          |
|   actor_loss     | -22.6    |
|   critic_loss    | 0.0734   |
|   ent_coef       | 0.1      |
|   learning_rate  | 0.0003   |
|   n_updates      | 448292   |
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| rollout/          |          |
|   ep_len_mean    | 158      |
|   ep_rew_mean    | 53.9     |
| time/            |          |
|   episodes       | 2556     |
|   fps            | 112      |
|   time_elapsed   | 4036     |
|   total_timesteps | 453895   |
| train/           |          |
|   actor_loss     | -21      |
|   critic_loss    | 0.0489   |
|   ent_coef       | 0.1      |
|   learning_rate  | 0.0003   |
|   n_updates      | 448894   |
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| rollout/          |          |
|   ep_len_mean    | 154      |
|   ep_rew_mean    | 53.8     |
| time/            |          |
|   episodes       | 2560     |
|   fps            | 112      |
|   time_elapsed   | 4040     |
|   total_timesteps | 454394   |
| train/           |          |
|   actor_loss     | -20.6    |
|   critic_loss    | 0.0989   |
|   ent_coef       | 0.1      |
|   learning_rate  | 0.0003   |
|   n_updates      | 449393   |
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rollout/		
ep_len_mean	151	
ep_rew_mean	53.7	
time/		
episodes	2564	
fps	112	
time_elapsed	4043	
total_timesteps	454786	
train/		
actor_loss	-22.7	
critic_loss	0.0567	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	449785	

Eval num_timesteps=455000, episode_reward=56.02 +/- 1.99
Episode length: 124.90 +/- 53.07

eval/		
mean_ep_length	125	
mean_reward	56	
time/		
total_timesteps	455000	
train/		
actor_loss	-21.4	
critic_loss	0.0459	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	449999	

rollout/		
ep_len_mean	148	
ep_rew_mean	54.2	
time/		
episodes	2568	
fps	112	
time_elapsed	4049	
total_timesteps	455358	
train/		
actor_loss	-22.9	
critic_loss	0.0752	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	450357	

rollout/		
ep_len_mean	151	
ep_rew_mean	54.3	
time/		
episodes	2572	
fps	112	
time_elapsed	4054	
total_timesteps	456018	
train/		
actor_loss	-22.3	
critic_loss	0.0741	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	451017	

rollout/		
ep_len_mean	150	
ep_rew_mean	54.3	
time/		
episodes	2576	
fps	112	
time_elapsed	4058	
total_timesteps	456510	
train/		
actor_loss	-21.1	
critic_loss	0.0581	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	451509	

rollout/		
ep_len_mean	153	
ep_rew_mean	54.4	
time/		
episodes	2580	
fps	112	
time_elapsed	4065	
total_timesteps	457510	
train/		
actor_loss	-23.1	
critic_loss	0.124	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	452509	

rollout/		
ep_len_mean	154	
ep_rew_mean	54.4	
time/		
episodes	2584	
fps	112	
time_elapsed	4069	
total_timesteps	458056	
train/		
actor_loss	-21.9	
critic_loss	0.0538	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	453055	

rollout/		
ep_len_mean	150	
ep_rew_mean	54.4	
time/		
episodes	2588	
fps	112	
time_elapsed	4072	
total_timesteps	458435	
train/		
actor_loss	-21.9	
critic_loss	0.0935	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	453434	

rollout/		
ep_len_mean	155	
ep_rew_mean	54.4	
time/		
episodes	2592	
fps	112	
time_elapsed	4079	
total_timesteps	459278	
train/		
actor_loss	-23.6	
critic_loss	0.134	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	454277	

rollout/		
ep_len_mean	155	
ep_rew_mean	54.4	
time/		
episodes	2596	
fps	112	
time_elapsed	4085	
total_timesteps	459925	
train/		
actor_loss	-23.1	
critic_loss	0.0843	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	454924	

Eval num_timesteps=460000, episode_reward=55.38 +/- 2.59

Episode length: 162.10 +/- 92.22

eval/		
mean_ep_length	162	
mean_reward	55.4	
time/		
total_timesteps	460000	
train/		
actor_loss	-22.4	
critic_loss	0.0802	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	454999	

rollout/		
ep_len_mean	154	
ep_rew_mean	54.4	
time/		
episodes	2600	
fps	112	
time_elapsed	4088	
total_timesteps	460350	
train/		
actor_loss	-23.3	
critic_loss	0.0644	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	455349	

rollout/		
ep_len_mean	155	
ep_rew_mean	54.4	
time/		
episodes	2604	
fps	112	
time_elapsed	4094	
total_timesteps	461082	
train/		
actor_loss	-21	
critic_loss	0.0768	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	456081	

rollout/		
ep_len_mean	155	
ep_rew_mean	54.4	
time/		
episodes	2608	
fps	112	
time_elapsed	4099	
total_timesteps	461641	
train/		
actor_loss	-22.2	
critic_loss	0.0479	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	456640	

rollout/		
ep_len_mean	153	
ep_rew_mean	54.9	
time/		
episodes	2612	
fps	112	
time_elapsed	4103	
total_timesteps	462134	
train/		
actor_loss	-21.9	
critic_loss	0.041	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	457133	

rollout/		
ep_len_mean	152	
ep_rew_mean	54.9	
time/		
episodes	2616	
fps	112	
time_elapsed	4108	
total_timesteps	462730	
train/		
actor_loss	-22	
critic_loss	0.0589	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	457729	

rollout/		
ep_len_mean	151	
ep_rew_mean	54.9	
time/		
episodes	2620	
fps	112	
time_elapsed	4110	
total_timesteps	463031	
train/		
actor_loss	-22.2	
critic_loss	0.106	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	458030	

rollout/		
ep_len_mean	157	
ep_rew_mean	55	
time/		
episodes	2624	
fps	112	
time_elapsed	4120	
total_timesteps	464124	
train/		
actor_loss	-22.1	
critic_loss	0.0886	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	459123	

rollout/		
ep_len_mean	153	
ep_rew_mean	54.9	
time/		
episodes	2628	
fps	112	
time_elapsed	4121	
total_timesteps	464521	
train/		
actor_loss	-22.7	
critic_loss	0.0723	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	459520	

rollout/		
ep_len_mean	149	
ep_rew_mean	54.8	
time/		
episodes	2632	
fps	112	
time_elapsed	4124	
total_timesteps	464881	
train/		
actor_loss	-22.8	
critic_loss	0.0618	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	459880	

Eval num_timesteps=465000, episode_reward=54.92 +/- 1.83
Episode length: 120.90 +/- 63.51

eval/		
mean_ep_length	121	
mean_reward	54.9	
time/		
total_timesteps	465000	
train/		
actor_loss	-21.1	
critic_loss	0.0546	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	459999	

rollout/		
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	ep_len_mean		148	
	ep_rew_mean		54.8	
	time/			
	episodes		2636	
	fps		112	
	time_elapsed		4128	
	total_timesteps		465273	
	train/			
	actor_loss		-22.6	
	critic_loss		0.14	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		460272	

	rollout/			
	ep_len_mean		145	
	ep_rew_mean		54.7	
	time/			
	episodes		2640	
	fps		112	
	time_elapsed		4133	
	total_timesteps		465769	
	train/			
	actor_loss		-21.6	
	critic_loss		0.0705	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		460768	

	rollout/			
	ep_len_mean		143	
	ep_rew_mean		54.7	
	time/			
	episodes		2644	
	fps		112	
	time_elapsed		4137	
	total_timesteps		466321	
	train/			
	actor_loss		-23.2	
	critic_loss		0.0704	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		461320	

	rollout/			
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	ep_len_mean		142	
	ep_rew_mean		54.7	
	time/			
	episodes		2648	
	fps		112	
	time_elapsed		4142	
	total_timesteps		466938	
	train/			
	actor_loss		-21.8	
	critic_loss		0.0873	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		461937	

	rollout/			
	ep_len_mean		143	
	ep_rew_mean		54.8	
	time/			
	episodes		2652	
	fps		112	
	time_elapsed		4148	
	total_timesteps		467630	
	train/			
	actor_loss		-22.2	
	critic_loss		0.0831	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		462629	

	rollout/			
	ep_len_mean		141	
	ep_rew_mean		54.8	
	time/			
	episodes		2656	
	fps		112	
	time_elapsed		4149	
	total_timesteps		468016	
	train/			
	actor_loss		-21.1	
	critic_loss		0.0734	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		463015	

	rollout/			
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	ep_len_mean		141	
	ep_rew_mean		54.8	
	time/			
	episodes		2660	
	fps		112	
	time_elapsed		4153	
	total_timesteps		468534	
	train/			
	actor_loss		-21.3	
	critic_loss		0.0517	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		463533	

	rollout/			
	ep_len_mean		144	
	ep_rew_mean		54.9	
	time/			
	episodes		2664	
	fps		112	
	time_elapsed		4159	
	total_timesteps		469215	
	train/			
	actor_loss		-21.6	
	critic_loss		0.0631	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		464214	

	rollout/			
	ep_len_mean		144	
	ep_rew_mean		54.9	
	time/			
	episodes		2668	
	fps		112	
	time_elapsed		4164	
	total_timesteps		469793	
	train/			
	actor_loss		-22.7	
	critic_loss		0.0687	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		464792	

Eval num_timesteps=470000, episode_reward=55.84 +/- 2.44
Episode length: 116.70 +/- 41.85

eval/		
mean_ep_length	117	
mean_reward	55.8	
time/		
total_timesteps	470000	
train/		
actor_loss	-22.7	
critic_loss	0.195	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	464999	

rollout/		
ep_len_mean	144	
ep_rew_mean	54.9	
time/		
episodes	2672	
fps	112	
time_elapsed	4170	
total_timesteps	470461	
train/		
actor_loss	-23.2	
critic_loss	0.0776	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	465460	

rollout/		
ep_len_mean	144	
ep_rew_mean	54.8	
time/		
episodes	2676	
fps	112	
time_elapsed	4174	
total_timesteps	470918	
train/		
actor_loss	-21.1	
critic_loss	0.0694	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	465917	

rollout/		
ep_len_mean	139	

	ep_rew_mean		54.7	
	time/			
	episodes		2680	
	fps		112	
	time_elapsed		4177	
	total_timesteps		471391	
	train/			
	actor_loss		-21.4	
	critic_loss		0.0582	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		466390	

	rollout/			
	ep_len_mean		141	
	ep_rew_mean		54.7	
	time/			
	episodes		2684	
	fps		112	
	time_elapsed		4182	
	total_timesteps		472204	
	train/			
	actor_loss		-21.2	
	critic_loss		0.0699	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		467203	

	rollout/			
	ep_len_mean		147	
	ep_rew_mean		54.9	
	time/			
	episodes		2688	
	fps		112	
	time_elapsed		4191	
	total_timesteps		473180	
	train/			
	actor_loss		-22.7	
	critic_loss		0.0576	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		468179	

	rollout/			
	ep_len_mean		144	

	ep_rew_mean		54.8	
	time/			
	episodes		2692	
	fps		112	
	time_elapsed		4195	
	total_timesteps		473660	
	train/			
	actor_loss		-22.2	
	critic_loss		0.147	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		468659	

	rollout/			
	ep_len_mean		140	
	ep_rew_mean		54.8	
	time/			
	episodes		2696	
	fps		112	
	time_elapsed		4197	
	total_timesteps		473970	
	train/			
	actor_loss		-21.7	
	critic_loss		0.0977	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		468969	

	rollout/			
	ep_len_mean		140	
	ep_rew_mean		54.7	
	time/			
	episodes		2700	
	fps		112	
	time_elapsed		4201	
	total_timesteps		474395	
	train/			
	actor_loss		-22.2	
	critic_loss		0.14	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		469394	

	rollout/			
	ep_len_mean		137	

	ep_rew_mean		54.7	
	time/			
	episodes		2704	
	fps		112	
	time_elapsed		4204	
	total_timesteps		474778	
	train/			
	actor_loss		-21.2	
	critic_loss		0.048	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		469777	

Eval num_timesteps=475000, episode_reward=54.64 +/- 1.71
Episode length: 98.10 +/- 47.19

	eval/			
	mean_ep_length		98.1	
	mean_reward		54.6	
	time/			
	total_timesteps		475000	
	train/			
	actor_loss		-22.4	
	critic_loss		0.178	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		469999	

	rollout/			
	ep_len_mean		140	
	ep_rew_mean		54.7	
	time/			
	episodes		2708	
	fps		112	
	time_elapsed		4209	
	total_timesteps		475609	
	train/			
	actor_loss		-23	
	critic_loss		0.0696	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		470608	

	rollout/			
	ep_len_mean		146	
	ep_rew_mean		54.1	

time/		
episodes	2712	
fps	113	
time_elapsed	4218	
total_timesteps	476736	
train/		
actor_loss	-21.7	
critic_loss	0.0792	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	471735	

rollout/		
ep_len_mean	146	
ep_rew_mean	54	
time/		
episodes	2716	
fps	113	
time_elapsed	4222	
total_timesteps	477281	
train/		
actor_loss	-22.1	
critic_loss	0.0651	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	472280	

rollout/		
ep_len_mean	152	
ep_rew_mean	54.1	
time/		
episodes	2720	
fps	113	
time_elapsed	4230	
total_timesteps	478207	
train/		
actor_loss	-22.6	
critic_loss	0.0484	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	473206	

rollout/		
ep_len_mean	145	
ep_rew_mean	54	

time/		
episodes	2724	
fps	113	
time_elapsed	4233	
total_timesteps	478650	
train/		
actor_loss	-20.9	
critic_loss	0.0995	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	473649	

rollout/		
ep_len_mean	149	
ep_rew_mean	54.1	
time/		
episodes	2728	
fps	113	
time_elapsed	4240	
total_timesteps	479446	
train/		
actor_loss	-20.3	
critic_loss	0.0669	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	474445	

Eval num_timesteps=480000, episode_reward=54.56 +/- 2.18
 Episode length: 109.90 +/- 81.42

eval/		
mean_ep_length	110	
mean_reward	54.6	
time/		
total_timesteps	480000	
train/		
actor_loss	-22.5	
critic_loss	0.0776	
ent_coef	0.1	
learning_rate	0.0003	
n_updates	474999	

rollout/		
ep_len_mean	151	
ep_rew_mean	54.1	
time/		

	episodes		2732	
	fps		113	
	time_elapsed		4243	
	total_timesteps		480008	
	train/			
	actor_loss		-22.1	
	critic_loss		0.06	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		475007	

	rollout/			
	ep_len_mean		154	
	ep_rew_mean		54.2	
	time/			
	episodes		2736	
	fps		113	
	time_elapsed		4248	
	total_timesteps		480646	
	train/			
	actor_loss		-21.8	
	critic_loss		0.0976	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		475645	

	rollout/			
	ep_len_mean		155	
	ep_rew_mean		54.2	
	time/			
	episodes		2740	
	fps		113	
	time_elapsed		4253	
	total_timesteps		481275	
	train/			
	actor_loss		-22.3	
	critic_loss		0.0485	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		476274	

	rollout/			
	ep_len_mean		158	
	ep_rew_mean		54.3	
	time/			

	episodes		2744	
	fps		113	
	time_elapsed		4260	
	total_timesteps		482115	
	train/			
	actor_loss		-22.3	
	critic_loss		0.101	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		477114	

	rollout/			
	ep_len_mean		158	
	ep_rew_mean		54.3	
	time/			
	episodes		2748	
	fps		113	
	time_elapsed		4266	
	total_timesteps		482770	
	train/			
	actor_loss		-22.2	
	critic_loss		0.0892	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		477769	

	rollout/			
	ep_len_mean		155	
	ep_rew_mean		54.2	
	time/			
	episodes		2752	
	fps		113	
	time_elapsed		4269	
	total_timesteps		483145	
	train/			
	actor_loss		-23.4	
	critic_loss		0.0588	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		478144	

	rollout/			
	ep_len_mean		155	
	ep_rew_mean		54.2	
	time/			

	episodes		2756	
	fps		113	
	time_elapsed		4270	
	total_timesteps		483551	
	train/			
	actor_loss		-21.9	
	critic_loss		0.113	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		478550	

	rollout/			
	ep_len_mean		155	
	ep_rew_mean		54.2	
	time/			
	episodes		2760	
	fps		113	
	time_elapsed		4274	
	total_timesteps		484009	
	train/			
	actor_loss		-22.3	
	critic_loss		0.0655	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		479008	

	rollout/			
	ep_len_mean		155	
	ep_rew_mean		54.1	
	time/			
	episodes		2764	
	fps		113	
	time_elapsed		4280	
	total_timesteps		484678	
	train/			
	actor_loss		-22.7	
	critic_loss		0.12	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		479677	

Eval num_timesteps=485000, episode_reward=54.37 +/- 2.18
Episode length: 97.30 +/- 58.83

	eval/			
	mean_ep_length		97.3	

	mean_reward		54.4	
	time/			
	total_timesteps		485000	
	train/			
	actor_loss		-22.2	
	critic_loss		0.0501	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		479999	

	rollout/			
	ep_len_mean		155	
	ep_rew_mean		54.2	
	time/			
	episodes		2768	
	fps		113	
	time_elapsed		4285	
	total_timesteps		485332	
	train/			
	actor_loss		-20.5	
	critic_loss		0.0717	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		480331	

	rollout/			
	ep_len_mean		153	
	ep_rew_mean		54.1	
	time/			
	episodes		2772	
	fps		113	
	time_elapsed		4288	
	total_timesteps		485729	
	train/			
	actor_loss		-22	
	critic_loss		0.0872	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		480728	

	rollout/			
	ep_len_mean		153	
	ep_rew_mean		54.1	
	time/			
	episodes		2776	

	fps		113	
	time_elapsed		4293	
	total_timesteps		486246	
	train/			
	actor_loss		-23.7	
	critic_loss		0.0656	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		481245	

	rollout/			
	ep_len_mean		152	
	ep_rew_mean		54.2	
	time/			
	episodes		2780	
	fps		113	
	time_elapsed		4296	
	total_timesteps		486617	
	train/			
	actor_loss		-23.7	
	critic_loss		0.0852	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		481616	

	rollout/			
	ep_len_mean		153	
	ep_rew_mean		54.2	
	time/			
	episodes		2784	
	fps		113	
	time_elapsed		4301	
	total_timesteps		487532	
	train/			
	actor_loss		-21	
	critic_loss		0.19	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		482531	

	rollout/			
	ep_len_mean		150	
	ep_rew_mean		54.1	
	time/			
	episodes		2788	

	fps		113	
	time_elapsed		4306	
	total_timesteps		488167	
	train/			
	actor_loss		-22.3	
	critic_loss		0.0555	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		483166	

	rollout/			
	ep_len_mean		149	
	ep_rew_mean		54.1	
	time/			
	episodes		2792	
	fps		113	
	time_elapsed		4309	
	total_timesteps		488527	
	train/			
	actor_loss		-22.5	
	critic_loss		0.0695	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		483526	

	rollout/			
	ep_len_mean		154	
	ep_rew_mean		54.2	
	time/			
	episodes		2796	
	fps		113	
	time_elapsed		4317	
	total_timesteps		489403	
	train/			
	actor_loss		-21.3	
	critic_loss		0.0544	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		484402	

	rollout/			
	ep_len_mean		155	
	ep_rew_mean		54.2	
	time/			
	episodes		2800	

	fps		113	
	time_elapsed		4320	
	total_timesteps		489849	
	train/			
	actor_loss		-22.3	
	critic_loss		0.0714	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		484848	

Eval num_timesteps=490000, episode_reward=56.16 +/- 2.38
Episode length: 177.50 +/- 84.78

	eval/			
	mean_ep_length		178	
	mean_reward		56.2	
	time/			
	total_timesteps		490000	
	train/			
	actor_loss		-22.7	
	critic_loss		0.0743	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		484999	

	rollout/			
	ep_len_mean		155	
	ep_rew_mean		54.2	
	time/			
	episodes		2804	
	fps		113	
	time_elapsed		4325	
	total_timesteps		490272	
	train/			
	actor_loss		-22.6	
	critic_loss		0.0737	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		485271	

	rollout/			
	ep_len_mean		153	
	ep_rew_mean		54.3	
	time/			
	episodes		2808	
	fps		113	

	time_elapsed		4330	
	total_timesteps		490888	
	train/			
	actor_loss		-20.9	
	critic_loss		0.158	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		485887	

	rollout/			
	ep_len_mean		145	
	ep_rew_mean		54.9	
	time/			
	episodes		2812	
	fps		113	
	time_elapsed		4331	
	total_timesteps		491228	
	train/			
	actor_loss		-23	
	critic_loss		0.1	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		486227	

	rollout/			
	ep_len_mean		145	
	ep_rew_mean		54.9	
	time/			
	episodes		2816	
	fps		113	
	time_elapsed		4336	
	total_timesteps		491784	
	train/			
	actor_loss		-22.8	
	critic_loss		0.0827	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		486783	

	rollout/			
	ep_len_mean		140	
	ep_rew_mean		54.8	
	time/			
	episodes		2820	
	fps		113	

	time_elapsed		4339	
	total_timesteps		492201	
	train/			
	actor_loss		-22.8	
	critic_loss		0.0461	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		487200	

	rollout/			
	ep_len_mean		143	
	ep_rew_mean		54.9	
	time/			
	episodes		2824	
	fps		113	
	time_elapsed		4345	
	total_timesteps		492910	
	train/			
	actor_loss		-22.2	
	critic_loss		0.116	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		487909	

	rollout/			
	ep_len_mean		143	
	ep_rew_mean		54.9	
	time/			
	episodes		2828	
	fps		113	
	time_elapsed		4351	
	total_timesteps		493751	
	train/			
	actor_loss		-22.1	
	critic_loss		0.134	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		488750	

	rollout/			
	ep_len_mean		142	
	ep_rew_mean		54.9	
	time/			
	episodes		2832	
	fps		113	

	time_elapsed		4355	
	total_timesteps		494242	
	train/			
	actor_loss		-23	
	critic_loss		0.0922	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		489241	

	rollout/			
	ep_len_mean		142	
	ep_rew_mean		54.9	
	time/			
	episodes		2836	
	fps		113	
	time_elapsed		4360	
	total_timesteps		494799	
	train/			
	actor_loss		-22.9	
	critic_loss		0.0613	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		489798	

Eval num_timesteps=495000, episode_reward=54.14 +/- 1.85
Episode length: 100.70 +/- 44.81

	eval/			
	mean_ep_length		101	
	mean_reward		54.1	
	time/			
	total_timesteps		495000	
	train/			
	actor_loss		-22.8	
	critic_loss		0.135	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		489999	

	rollout/			
	ep_len_mean		143	
	ep_rew_mean		54.9	
	time/			
	episodes		2840	
	fps		113	
	time_elapsed		4365	

	total_timesteps		495583	
	train/			
	actor_loss		-22.8	
	critic_loss		0.0813	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		490582	

	rollout/			
	ep_len_mean		140	
	ep_rew_mean		54.8	
	time/			
	episodes		2844	
	fps		113	
	time_elapsed		4369	
	total_timesteps		496099	
	train/			
	actor_loss		-22.6	
	critic_loss		0.14	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		491098	

	rollout/			
	ep_len_mean		141	
	ep_rew_mean		54.2	
	time/			
	episodes		2848	
	fps		113	
	time_elapsed		4376	
	total_timesteps		496887	
	train/			
	actor_loss		-22	
	critic_loss		0.155	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		491886	

	rollout/			
	ep_len_mean		143	
	ep_rew_mean		54.2	
	time/			
	episodes		2852	
	fps		113	
	time_elapsed		4381	

	total_timesteps		497424	
	train/			
	actor_loss		-22.6	
	critic_loss		0.0463	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		492423	

	rollout/			
	ep_len_mean		147	
	ep_rew_mean		54.2	
	time/			
	episodes		2856	
	fps		113	
	time_elapsed		4389	
	total_timesteps		498264	
	train/			
	actor_loss		-22.6	
	critic_loss		0.0728	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		493263	

	rollout/			
	ep_len_mean		146	
	ep_rew_mean		54.2	
	time/			
	episodes		2860	
	fps		113	
	time_elapsed		4392	
	total_timesteps		498644	
	train/			
	actor_loss		-23.1	
	critic_loss		0.0595	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		493643	

	rollout/			
	ep_len_mean		145	
	ep_rew_mean		54.3	
	time/			
	episodes		2864	
	fps		113	
	time_elapsed		4395	

	total_timesteps		499212	
	train/			
	actor_loss		-22.9	
	critic_loss		0.0796	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		494211	

	rollout/			
	ep_len_mean		144	
	ep_rew_mean		54.2	
	time/			
	episodes		2868	
	fps		113	
	time_elapsed		4399	
	total_timesteps		499725	
	train/			
	actor_loss		-22	
	critic_loss		0.0755	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		494724	

Eval num_timesteps=500000, episode_reward=56.78 +/- 1.68
 Episode length: 172.30 +/- 57.57

	eval/			
	mean_ep_length		172	
	mean_reward		56.8	
	time/			
	total_timesteps		500000	
	train/			
	actor_loss		-22.9	
	critic_loss		0.0729	
	ent_coef		0.1	
	learning_rate		0.0003	
	n_updates		494999	

New best mean reward!

Trening zakończony. Model zapisany do models/rolling_ball_final.zip

```
[3]: import matplotlib.pyplot as plt

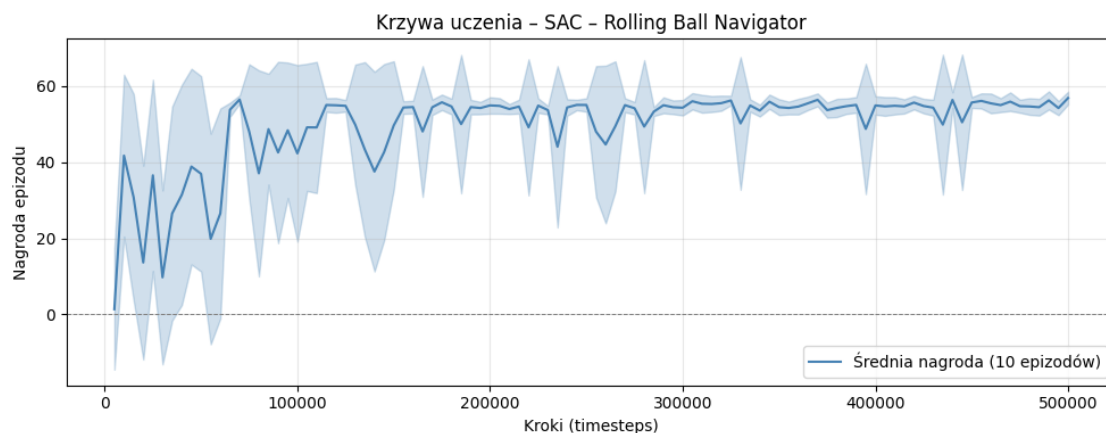
evaluations = np.load("logs/evaluations.npz")
timesteps = evaluations["timesteps"]
mean_rewards = evaluations["results"].mean(axis=1)
```

```

std_rewards = evaluations["results"].std(axis=1)

fig, ax = plt.subplots(figsize=(10, 4))
ax.plot(timesteps, mean_rewards, color="steelblue", label="Średnia nagroda (10_
    ↪epizodów)")
ax.fill_between(
    timesteps,
    mean_rewards - std_rewards,
    mean_rewards + std_rewards,
    alpha=0.25, color="steelblue",
)
ax.axhline(0, color="gray", linewidth=0.7, linestyle="--")
ax.set_xlabel("Kroki (timesteps)")
ax.set_ylabel("Nagroda epizodu")
ax.set_title("Krzywa uczenia - SAC - Rolling Ball Navigator")
ax.legend()
ax.grid(alpha=0.3)
plt.tight_layout()
plt.savefig("training_curve.png", dpi=150)
plt.show()

```



1.3 3. Ewaluacja

Wczytujemy najlepszy model zapisany przez EvalCallback i uruchamiamy go deterministycznie na zbiorze epizodów ewaluacyjnych. Mierzmy skuteczność (odsetek epizodów zakończonych sukcesem), średnią nagrodę i średnią liczbę kroków potrzebnych do dotarcia do celu.

```

[14]: from stable_baselines3 import SAC

model = SAC.load("models/best_model")

N_EVAL = 30

```

```

successes = 0
ep_rewards = []
ep_lengths = []

eval_env = RollingBallEnv()
for ep in range(N_EVAL):
    obs, _ = eval_env.reset(seed=ep)
    total_reward = 0.0
    while True:
        action, _ = model.predict(obs, deterministic=True)
        obs, reward, terminated, truncated, info = eval_env.step(action)
        total_reward += reward
        if terminated or truncated:
            if terminated:
                successes += 1
            ep_rewards.append(total_reward)
            ep_lengths.append(info["step"])
            break
eval_env.close()

print(f"Wyniki ewaluacji ({N_EVAL} epizodów):")
print(f"Skuteczność (dotarcie do celu): {successes / N_EVAL * 100:.1f}%")
print(f"Średnia nagroda: {np.mean(ep_rewards):.2f} ± {np.
    ↳std(ep_rewards):.2f}")
print(f"Średnia liczba kroków: {np.mean(ep_lengths):.1f} ± {np.
    ↳std(ep_lengths):.1f}")

```

Wyniki ewaluacji (30 epizodów):

Skuteczność (dotarcie do celu):	100.0%
Średnia nagroda:	54.90 ± 1.56
Średnia liczba kroków:	123.3 ± 76.2

```

[4]: import imageio.v2 as imageio
from stable_baselines3 import SAC

model = SAC.load("models/best_model")
rec_env = RollingBallEnv(render_mode="rgb_array")

frames = []
for ep in range(5):
    obs, _ = rec_env.reset(seed=None)
    frames.append(rec_env.render())
    while True:
        action, _ = model.predict(obs, deterministic=True)
        obs, _, terminated, truncated, _ = rec_env.step(action)
        frames.append(rec_env.render())
        if terminated or truncated:

```

```

        break
rec_env.close()

imageio.mimwrite("episode.mp4", frames, fps=30)
print(f"Zapisano episode.mp4 - 5 epizodów, {len(frames)} klatek ({len(frames) / 5} s @ 30 fps)")

```

IMAGEIO FFMPEG_WRITER WARNING: input image is not divisible by macro_block_size=16, resizing from (600, 600) to (608, 608) to ensure video compatibility with most codecs and players. To prevent resizing, make your input image divisible by the macro_block_size or set the macro_block_size to 1 (risking incompatibility).

Zapisano episode.mp4 - 5 epizodów, 646 klatek (21.5 s @ 30 fps)

```

[5]: from stable_baselines3 import SAC

model = SAC.load("models/best_model")
live_env = RollingBallEnv(render_mode="human")

for episode in range(10):
    obs, _ = live_env.reset()
    while True:
        action, _ = model.predict(obs, deterministic=True)
        obs, _, terminated, truncated, _ = live_env.step(action)
        if terminated or truncated:
            break

live_env.close()

```

1.4 4. Wnioski

1. **SAC i ciągła akcja** — algorytm dobrze pasuje do kulki sterowanej siłą w $[-1,1]^2$; deterministyczna ewaluacja (`best_model` z `EvalCallback`) daje powtarzalne metryki z komórki ewaluacyjnej.
2. **Gęste kształtowanie + rzadki bonus** — składnik $0.02 * (\text{prev_dist} - \text{dist})$ daje gradient w kierunku celu przy małej karze czasowej -0.01 , a $+50$ przy `terminated` utrwala zakończenie zadania. Bez tego (albo przy złym balansie wobec limitu kroków) agent nie ma sygnału do nauki.
3. **Normalizacja obserwacji** — wektor 13 liczb w `Box` o skalach $[-1,1]$ / $[0,1]$ jest zgodny z oczekiwaniami sieci MLP i ze stabilnym `critic_loss` (por. diagnoza w sprawozdaniu, sekcja 3.2).
4. **Projektowanie nagrody** — jawna kara za zderzenie (wcześniejsza wersja) mogła nagradzać beczynność; w finalnym kodzie wystarcza fizyka odbicia i nagroda za postęp w stronę celu.
5. **Entropia** — stałe `ent_coef=0.1` zamiast auto-tuningu ogranicza sprzężenie z niestabilnym krytykiem przy tych samych obserwacjach.

6. **Wynik** — przy 30 epizodach testowych pełna skuteczność i średnia długość epizodu znacznie poniżej `MAX_STEPS` wskazują, że polityka nie „gubi się” w eksploracji, lecz realizuje nawigację w oparciu o promień i wektor do celu.
7. **Grafika** — w `render` cel i kulka to sprite’y (`sprites/goal.png`, `sprites/agent.png`); logika kolizji i promieni (`GOAL_RADIUS`, `BALL_RADIUS`) jest niezależna od wyglądu bitmap.

Szczegółowe uzasadnienia eksperymentów po drodze (rzadka nagroda, skale wejść, kara za kolizje) są zebrane w `sprawozdanie.md` (sekcje 3–5).